

Counterexamples to the Hahn–Wilson conjecture at height two

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Abstract

We disprove the Hahn–Wilson conjecture at height two. For every sufficiently large prime p , we construct a connective p -complete spectrum X of exact fp-type two outside the ordinary thick subcategory generated by the specified standard form of $\mathrm{BP}\langle 2 \rangle_p^\wedge$. The same spectrum satisfies both localization comparisons $L_2^f X \simeq L_2 X$ and $L_{T(2)} X \simeq L_{K(2)} X$.

1 Introduction

Which infinite spectra can be built from truncated Brown–Peterson spectra using finitely many cofiber sequences and retracts? The Hahn–Wilson conjecture proposes a cohomological answer: finite presentation over the Steenrod algebra, together with a chromatic height bound, should characterize precisely these finite constructions. We construct counterexamples at height two. Their chromatic localizations retain strong finiteness properties; the obstruction lies in how the local pieces are joined.

1.1 The question and the theorem

Fix a prime p . Write $K(j)$ for Morava K -theory, with $K(0) = H\mathbb{Q}$. A nonzero finite p -local spectrum has *type* j if j is the least index for which its $K(j)$ -homology is nonzero. The thick subcategory theorem classifies thick subcategories of finite p -local spectra by type [15].

A bounded-below p -complete spectrum X is *fp* if $H^*(X; \mathbb{F}_p)$ is finitely presented over the mod- p Steenrod algebra. Mahowald and Rezk prove that this is equivalent to requiring $\pi_*(V \wedge X)$ to be finite in total for some nonzero finite p -local spectrum V [22, Proposition 3.2]. Here *finite in total* means finitely many nonzero homotopy groups, each a finite group. For an fp-spectrum, fp-type at most n means that this test holds for a finite spectrum of type $n + 1$, equivalently for every such spectrum. We include the zero spectrum. Exact fp-type two means fp-type at most two but not at most one.

For a p -complete spectrum B , let $\mathrm{Thick}(B)$ be its closure in $\mathrm{Sp}_p^\wedge$ under equivalences, finite sums, all integer shifts, cofiber sequences and retracts. These are the permitted operations; in particular this is ordinary thick closure. The conjecture is

$$\{X : X \text{ has fp-type at most } n\} = \mathrm{Thick}(\mathrm{BP}\langle n \rangle_p^\wedge). \quad (1.1)$$

Its written formulation appears in Lee–Pstrągowski [21, Conjecture 1.1 and Note 1], who report learning it from Dylan Wilson in 2021. A standard completed truncated Brown–Peterson spectrum has coefficients $\mathbb{Z}_p[v_1, \dots, v_n]$, where $|v_i| = 2(p^i - 1)$.

Let $L_n = L_{E(n)}$ denote Johnson–Wilson localization and L_n^f finite localization away from finite spectra of type greater than n . For $n > 0$, let $T(n)$ be a telescope of a periodic self-map of a finite type- n spectrum.

Theorem 1.1. *For every sufficiently large prime p , there is a connective p -complete spectrum X of exact fp -type two such that, for a standard form of $\mathrm{BP}\langle 2 \rangle_p^\wedge$,*

$$X \notin \mathrm{Thick}(\mathrm{BP}\langle 2 \rangle_p^\wedge).$$

The same spectrum satisfies

$$L_2^f X \simeq L_2 X, \quad L_{T(2)} X \simeq L_{K(2)} X.$$

The theorem refutes the universally quantified conjecture (1.1); its prime range is sufficiently large primes. We use a specific regular BP-module quotient throughout the proof. At these odd primes the underlying spectrum is independent of the standard quotient generators [20, Corollary 1.9]. This uniqueness uses the work of Angeltveit–Lind, with its corrected argument and the additional p -local result supplied by Lee [2, 20].

1.2 Global finiteness and the local classification

Mahowald–Rezk’s duality theory makes fp -spectra a natural class of infinite spectra with controlled chromatic behavior [22, Theorem 8.11]. Hahn–Wilson’s redshift theorem gives further examples: for the multiplicative forms in their theorem, $K(\mathrm{BP}\langle n \rangle_p^\wedge)_p^\wedge$ has fp -type $n + 1$ [13, Theorem B]. The cases of fp -type at most -1 and 0 are classical, and Lee–Pstrągowski prove the global generation conjecture at height one [21, Theorems 1.3 and 5.4].

Mahowald and Rezk asked whether the comparison $L_n^f X \rightarrow L_n X$ is an equivalence for fp -spectra [22, Conjecture 7.3]. Burklund, Hahn, Levy, and Schlank subsequently disproved the unrestricted telescope conjecture at every prime and every height at least two [10, Theorem A]. The finiteness distinction remains essential: Lee and Pstrągowski note that the relevant finite-smash homotopy of those counterexamples is bounded but not finite in total [21, Section 1.1, following Conjecture 1.2]. Thus those counterexamples do not supply the fp counterexample studied here.

There is also a positive classification at every height after Morava K -localization. A $K(n)$ -local spectrum Y is *locally fp* if $\pi_i(F \wedge Y)$ is finite for every integer i and every finite type- n spectrum F . Lee–Pstrągowski identify these objects with $\mathrm{Thick}(E_n)$ in the $K(n)$ -local category, equivalently with $\mathrm{Thick}(L_{K(n)}\mathrm{BP}\langle n \rangle)$ [21, Theorems 1.7 and 4.3]. Here E_n is Morava E -theory. The height-one local classification has an earlier source in Hahn–Mitchell’s Iwasawa theory [14, Theorem 8.1]. The local test requires finiteness in each degree; the global fp -type test above requires finiteness in total.

Set $X_K = L_{K(2)} X$. Then $X_K \in \mathrm{Thick}(E_2)$, and its cohomology $E_2^* X_K$ is finitely generated over the ring of operations $E_2^* E_2$. The natural map from X_K to its double dual, formed using the $K(2)$ -local sphere, is an equivalence. The original spectrum X has no global finite construction from $\mathrm{BP}\langle 2 \rangle_p^\wedge$. Lee–Pstrągowski suggest using a height decomposition and their local theorem to prove global generation for fp -spectra satisfying the telescope comparison [21, Remark 5.1]. Theorem 1.1 shows that this additional hypothesis does not suffice at height two. The tensor-ideal classification of locally dualizable spectra concerns different permitted constructions [5]; here we keep the single generator and ordinary thick closure fixed.

1.3 The construction problem

At large primes, chromatic algebraicity lets us build the example in a periodic derived category of sheaves on the stack of formal group laws. We choose Pstrągowski’s homology-compatible triangulated comparison, which at height two is available for $p > 8$ [27, Corollary 6.9]. Section 7

records the source qualifications and supplies the required local arguments for this comparison. This bound concerns the comparison; the operator constructed below requires larger primes. The comparison transports finite triangles and retracts. All completion, inverse-limit and gluing constructions are performed first in the algebraic category.

We use the weighted stack of p -typical formal group laws of height at most two. Work in characteristic p , extend scalars to $k = \mathbb{F}_{p^2}$, and quotient out the central strict automorphisms $1 + p\mathbb{Z}_p$; write $\mathcal{N}$ for the resulting stack. The coefficient v_i has weight $(p^i - 1)/(p - 1)$. Its Brown–Peterson coefficient chart is

$$f' : \mathcal{X}_k \longrightarrow \mathcal{N}, \quad \mathcal{X}_k = [(\mathrm{Spec} k[v_1, v_2] \setminus \{(0, 0)\})/\mathbb{G}_m].$$

The coordinate weights are $1, p + 1$; the map f' uses the law with all higher v_i equal to zero. Put $T' = f'_* \mathcal{O}_{\mathcal{X}_k}$. The weight line $\mathcal{O}(1)$ has the Cartier section $t = v_1$, which cuts out height two. The open $\mathcal{N}[1/t]$ is $\mathrm{Spec} k$; the completed chart retains a noncommutative algebra of operations.

Here is the precise shape of the algebraic problem. Put $D = 2(p - 1)$. The category $\mathbf{D}(\mathcal{N})$ consists of homological complexes with weight-twisted periodicity $C_{j+D} = C_j(1)$, where (1) is the weight-line twist.

For such a complex U , set

$$\pi_i U = [\mathcal{O}[i], U], \quad \widehat{U} = \mathrm{holim}_d \mathrm{cofib}(t^d : U[dD] \longrightarrow U).$$

Thus $\pi_i U$ denotes homotopy of derived global sections. Proposition 8.1 reduces the construction to finding U with

$$\begin{aligned} \pi_i U \text{ finite for every } i, \quad \widehat{U} \in \mathrm{Thick}(\widehat{T'}), \\ U \notin \mathrm{Thick}(T'). \end{aligned} \tag{1.2}$$

The transfer proof uses the first condition to obtain finite homotopy groups and establish the fp test for a connective cover. The second controls the change back from the reduced stack. The third supplies the obstruction to a global finite construction. The transfer proof uses nonsplit comparison triangles and nilpotence; it does not require these coefficient changes to have sections.

To produce U , begin with a cone of one operation on $\widehat{T'}$, and let Y be the sum of its shifts through one period. Write L for inversion of t . On the overlap, the operation acts on Laurent series. We arrange that it has many independent kernel vectors and cokernel classes. A triangular matching controls both kinds of defect and gives a closed image. They form the two summands of the overlap sections $\pi_i LY$.

In each degree we pair kernel and cokernel directions inside a discrete cocompact k -subspace of $\pi_i LY$. Since the open is $\mathrm{Spec} k$, these subspaces are realized by an open periodic complex V and a map $V \rightarrow LY$. Define U by the homotopy pullback

$$\begin{array}{ccc} U & \longrightarrow & Y \\ \downarrow & & \downarrow \\ V & \longrightarrow & LY. \end{array}$$

In each degree, the complete part maps to a compact open subspace of the overlap, with finite kernel. The chosen discrete subspace complements this image up to finite error.

Thus the joint map

$$\pi_i Y \oplus \pi_i V \longrightarrow \pi_i LY$$

has finite kernel and cokernel. The pullback long exact sequence makes $\pi_i U$ finite in every degree. Completion kills both open objects V and LY , so $\widehat{U} \simeq Y$. This proves the first two conditions in (1.2).

The last condition is the main obstruction. An independent estimate bounds certain joint operator kernels at Laurent precision N by $O(N^{1/3})$. Our paired defects grow at least as cN^β for a fixed $\beta > 1/3$ and every sufficiently large N . This discrepancy constrains every global map from U into a finite construction from T' : after a flat coefficient extension, vanishing on homology forces vanishing on the homology of the module dual. The dual condition removes the free part of each stage of a possible finite tower; the homology condition then passes to the preceding stage. Backward induction excludes every such tower and its retracts. We prove this categorical criterion before proving its growth hypothesis. The use of ghosts and finite extension towers has a general antecedent in Christensen [11]; the homology-to-dual-homology implication here is proved from the selected gluing data.

1.4 Reading the proof

The algebraic argument comes first. Section 2 defines the reduced setting and proves its chart geometry. Section 3 computes complete operations using compatible maps on finite reductions. It realizes perfect modules as actual complexes, so the calculation controls maps and cones as well as endomorphism groups. Sections 4 and 5 construct the Laurent operator and the gluing. Section 6 proves the finite-tower criterion and verifies its hypothesis for the glued object. Together, these arguments explain how complete operations control finite constructions and how the gluing obstructs them globally.

Sections 7 and 8 establish the comparison with spectra and prove the transfer proposition. The latter takes the connective cover of the resulting $E(2)$ -local spectrum, following the methods of Mahowald–Rezk [22, Proposition 3.5 and Section 7.2(2)]; in particular, it proves the total-finiteness test needed for the fp condition. Section 7 also gives the Brown–Peterson quotient calculation and proves a separate thick-transport principle for equivalences of underlying categories. Section 9 establishes exact type two and both localization comparisons, then compares the example with the positive local theory and proves the separate obstruction for the connective cover of Morava E -theory.

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2 Law charts and the height-two quotient

The counterexample will be built in a category of periodic complexes of sheaves on a stack of formal group laws. Our task here is to construct the stack and its chart sheaf, then establish tools for computing sections. We begin with the integral stack of laws of height at most two, pass to characteristic p , and quotient a central procyclic group of automorphisms. The quotient has a simple height-one open and explicit nilpotent neighborhoods of height two. The height-two stabilizer provides both the coefficient ring for the construction and the finite cohomological bounds needed for periodic complexes.

Throughout this section $p > 5$ is prime. Put

$$D = 2(p-1), \quad m_i = \frac{p^i - 1}{p-1}, \quad q = m_2 = p+1.$$

Additional large-prime conditions will be imposed only when constructing the operator used in the counterexample.

2.1 The integral law groupoid

We start over $\mathbb{Z}_{(p)}$ because the same flat chart will supply the characteristic- p construction and the later integral comparison. We use the classical Brown–Peterson Hopf algebroid

$$P = \mathbb{Z}_{(p)}[v_i : i \geq 1], \quad \Gamma_P = P[t_i : i \geq 1]$$

of strict p -typical formal group laws. Divide all its topological degrees by D , so that v_i and t_i have weight m_i . The formal-group and cooperation calculations used here are recalled in [29, Chapter 4]; for the Araki formulas see also [30, Appendix B.5.5–B.5.6, B.5.12, and B.5.15]. We choose Araki coordinates, for which the universal law F satisfies

$$[p]_F(T) = \sum_{i \geq 0}^F v_i T^{p^i}, \quad v_0 = p. \quad (2.1)$$

The superscript denotes addition in F , not ordinary addition. The following calculation recalls why this is an integral coordinate choice. Write $\log_F(T) = \sum_{i \geq 0} l_i T^{p^i}$ and successively subtract the lower terms of (2.1) from $[p]_F$ using formal-group subtraction. The logarithm shows that the next remainder starts at T^{p^n} ; its coefficient is integral, since all the preceding subtractions were integral. Comparison with the Hazewinkel recursion

$$pl_n = v_n^{\text{Haz}} + \sum_{0 < i < n} l_i (v_{n-i}^{\text{Haz}})^{p^i}$$

gives a triangular change of generators with diagonal $1 - p^{p^n-1}$, a unit in $\mathbb{Z}_{(p)}$.

A strict arrow from F_R to F_L has the form

$$\phi(T) = T +_{F_L} \sum_{i \geq 1}^F t_i T^{p^i}.$$

The coefficients of F_R are given by the right unit η_R . Commuting ϕ with the p -series gives

$$\eta_R v_i \equiv v_i \pmod{(p, v_1, \dots, v_{i-1})}. \quad (2.2)$$

In particular the ideal (p, v_1, v_2) defines an invariant closed locus.

Let $\mathcal{M}$ be the stack of this groupoid on $\mathrm{Spec} P \setminus V(p, v_1, v_2)$, also divided by $\mathbb{G}_m$ for the regraded weights. All stacks and sheaves in this section use fpqc descent. Define

$$\begin{aligned} B_2 &= \mathbb{Z}_{(p)}[v_1, v_2], \\ A_2 &= B_2[v_2^{-1}], & f : \mathcal{X} &\longrightarrow \mathcal{M}, \\ \mathcal{X} &= [(\mathrm{Spec} B_2 \setminus V(p, v_1, v_2))/\mathbb{G}_m], \end{aligned}$$

where f puts $v_{i>2} = 0$. We will also use the standard Johnson–Wilson coefficient chart

$$A_{\mathrm{Haz}} = \mathbb{Z}_{(p)}[v_1^{\mathrm{Haz}}, (v_2^{\mathrm{Haz}})^{\pm 1}],$$

with the higher Hazewinkel generators set equal to zero.

Proposition 2.1. *The map f is flat and affine. The maps $\mathrm{Spec} A_2 \rightarrow \mathcal{M}$ and $\mathrm{Spec} A_{\mathrm{Haz}} \rightarrow \mathcal{M}$ are flat affine covers. In graded coordinates, the cooperation Hopf algebroid of the latter chart has arrow ring*

$$A_{\mathrm{Haz}} \otimes_{P, \eta_L} \Gamma_P \otimes_{P, \eta_R} A_{\mathrm{Haz}}.$$

Proof. We verify flatness, affineness, and then coverage. The Landweber flatness criterion applies to each localization of B_2 inverting p , v_1 , or v_2 , and to A_{Haz} : the invariant regular sequence remains regular up to the point where its ideal becomes the unit ideal. The interpretation of this criterion as flatness of law charts is the standard equivalence between Landweber exact comodules and flat descent; see [18, Lemma 2.2, Corollary 2.3, and Theorem 2.5] and [23, Section 5, Theorem 26, and Proposition 27].

One can pass explicitly from exactness on graded comodules to the flatness assertion being used. Apply exact coinduction $\Gamma_P \otimes_P (-)$ to a sequence of graded P -modules, and then apply base change to the chart ring A . This tests flatness of $A \otimes_{P, \eta_L} \Gamma_P$ over P via η_R ; inversion of arrows interchanges the two units if necessary. Graded flatness here implies ordinary flatness. For an ungraded test module N , put a copy of N in every weight and let each homogeneous element act through its ordinary action into the correspondingly shifted copy. A fixed weight of the graded tensor product recovers the ungraded tensor product. Thus the graded exactness test also tests arbitrary modules.

After pulling back to an affine scheme of laws with its grading frame trivialized, the fiber of f is the affine scheme of strict arrows whose other end is a B_2 -law. The deleted locus cannot occur at that end by (2.2). This proves affineness of f . Fiber products of affine law charts are affine as well: in addition to the strict-arrow coordinates, they retain only an invertible grading parameter. In particular the relevant chart maps and diagonals are affine.

It remains to check that the indicated flat charts cover. This may be checked on geometric points. In characteristic zero the logarithm gives strict isomorphisms of the required p -typical form. In characteristic p , let $h = 1$ or 2 be the height, and compare two laws over an algebraically closed field with the same nonzero coefficient v_h . For a fixed left law, solve for an arrow by prescribing the right coefficients in order. After the preceding equations have been imposed, the equation at step $j \geq 1$ is

$$\eta_R v_{j+h} = v_h t_j^{p^h} - v_h^{p^j} t_j + \text{a polynomial already determined by } t_{<j}. \quad (2.3)$$

Indeed, compare degree p^{j+h} in $[p]_{F_L}(\phi(T)) = \phi([p]_{F_R}(T))$. All exponents in the left p -series are divisible by p^h , so only ϕ through degree p^j can enter there. The first occurrence of t_j is additive in degree p^j . On the right, apart from the displayed coefficient $\eta_R v_{j+h}$, only previously fixed

coefficients enter: the formal-sum coefficients needed at this step have total exponent at most p^j and hence, by homogeneity, use only generators of weight at most m_j . This gives (2.3). Its leading coefficient is nonzero, so it can be solved at every step. Both charts contain laws of the prescribed height and leading coefficient; for the Hazewinkel chart use the homogeneous triangular coordinate change above. This proves coverage. Finally the displayed arrow ring is the usual Landweber base-change calculation for cooperations, and is also the ring obtained directly by pulling back the two ends of the law groupoid. $\square$

We fix the chart sheaves

$$T_{\mathbb{Z}} = f_* \mathcal{O}_{\mathcal{X}}, \quad T_{\mathbb{Z}_p} = f_*(\mathcal{O}_{\mathcal{X}} \otimes_{\mathbb{Z}_{(p)}} \mathbb{Z}_p). \quad (2.4)$$

They are p -monic, meaning that multiplication by p is injective. This follows on $\mathcal{X}$ and is preserved by the affine pushforward. The later comparison will identify these particular sheaves with the algebraic images of Brown–Peterson spectra.

2.2 Removing the central procyclic group

The construction itself uses the characteristic- p law stack. On this stack a central copy of $\mathbb{Z}_p$ acts through strict automorphisms of every law. We remove this common inertia to obtain a height-two stabilizer whose invariants have cohomological dimension at most three.

Put $k = \mathbb{F}_{p^2}$. Write

$$i : \mathcal{M}_0 \longrightarrow \mathcal{M}, \quad e : \mathcal{M}_k \longrightarrow \mathcal{M}_0$$

for reduction to characteristic p and extension of scalars from $\mathbb{F}_p$ to k , respectively. Subscripts 0 and k on $\mathcal{X}$ and f mean the corresponding base changes. The proconstant group

$$Z = 1 + p\mathbb{Z}_p \simeq \mathbb{Z}_p$$

acts as central inertia by the strict automorphisms $[z]_F$. Here a proconstant group scheme is an inverse limit of finite constant group schemes; its functions over k are locally constant k -valued functions. Integer multiplication has the p -typical formal-sum expansion obtained from logarithms and successive subtraction as above. In characteristic p , the p -series has formal-variable order at least p , so these multiplications extend continuously to p -adic integers on every jet. They commute with all arrows and with the grading. The action is faithful on geometric fibers because the laws have finite height. Fix a topological generator c of Z .

We construct an affine groupoid presentation for the quotient and prove that the chart remains affine. These properties will make its pushforward accessible to the later coinduction calculation.

Lemma 2.2. *Let A be a k -algebra with a locally finite continuous action of Z , and let Z_m be the subgroup of index p^m . Suppose that*

$$A = \bigcup_{m \geq 0} A^{Z_m}$$

and that the action on $\mathrm{Spec} A$ is geometrically free. Then $\mathrm{Spec} A \rightarrow \mathrm{Spec}(A^Z)$ is an fpqc Z -torsor.

Proof. We first descend the free action to finite quotients. We then prove the finite torsor assertion directly and pass to the limit. Set $A_m = A^{Z_m}$. Every element has finite orbit, so A is integral over each A_m . Two geometric points with the same restriction to A_m differ by Z_m . To see this, restrict first to a subalgebra generated by finitely many finite orbits. For a finite group, distinct orbits of geometric points are separated by invariant functions: apply the Chinese remainder theorem and

take an invariant product of separating functions. Formation of finite-group invariants commutes with field extension, since it is a finite kernel and a field extension is flat. The finite-level assertions then give nonempty closed sets of possible elements of Z_m ; compactness supplies a compatible element for the whole algebra.

Integrality lifts every geometric point of A_m to one of A . The induced Z/Z_m -action on $\mathrm{Spec} A_m$ is free. In fact, if a coset fixes such a point, lift it to a point of A . The translate of that lift and the original lift differ by Z_m by the preceding paragraph. Geometric freeness upstairs forces the original coset to be trivial.

For completeness, a finite constant group K acting geometrically freely on an affine scheme gives a torsor over the invariant ring even when $|K|$ is not invertible. At a lift of a geometric point of the invariant spectrum, choose elements $b_1, \dots, b_{|K|}$ whose evaluations on its orbit form an invertible matrix. Such elements exist by the Chinese remainder theorem after field extension, and hence a determinant of elements from the original algebra is nonzero there. If $D_b = \det(g(b_j))_{g \in K, j}$, then D_b^2 is invariant. After inverting it, solve for the coefficients of any element in the basis b_j using this matrix. Uniqueness makes the coefficients invariant. The same matrix gives the tensor torsor isomorphism. These neighborhoods cover the invariant spectrum by integrality.

Applying this argument to A_m shows that $\mathrm{Spec} A_m$ is a finite faithfully flat Z/Z_m -torsor over $\mathrm{Spec} A^Z$. Passing to the filtered union gives faithful flatness and the torsor identity for A . The resulting affine map is quasi-compact, hence fpqc. $\square$

Proposition 2.3. *There is a flat affine groupoid stack $\mathcal{N}$ and a map $r : \mathcal{M}_k \rightarrow \mathcal{N}$ that quotients the vertical central Z -action on arrows. Its pullback is exact and identifies sheaves on $\mathcal{N}$ with sheaves on $\mathcal{M}_k$ having trivial vertical Z -action. Moreover*

$$f' = r f_k : \mathcal{X}_k \longrightarrow \mathcal{N}$$

is flat and affine. The weight line descends to $\mathcal{N}$, and $t = v_1$ is an invariant Cartier section of $\mathcal{O}(1)$ there.

Proof. We construct the quotient on an affine presentation, and then check the chart and the weight line. Take $S = \mathrm{Spec}(A_2 \otimes k)$ and quotient its affine arrow scheme $S \times_{\mathcal{M}_k} S$ by vertical composition with Z . The hypotheses of Lemma 2.2 hold. The action is geometrically free by the leading orders of finite-height p -series, and it is locally finite on the arrow coefficients by uniform continuity on each formal-variable jet. Thus the quotient is affine and the original arrow scheme is an fpqc torsor over it. Composition descends by centrality: after locally lifting two composable arrows, changing either lift changes their composite by the corresponding vertical element. Identity and inverse descend as well. Source and target remain flat by faithfully flat descent. This defines the asserted groupoid and $\mathcal{N}$. Descent of a sheaf's arrow isomorphism through the torsor gives the claimed description of r^* and its exactness.

Similarly, $S \times_{\mathcal{N}} \mathcal{X}_k$ is the quotient of $S \times_{\mathcal{M}_k} \mathcal{X}_k$ by vertical composition. An isomorphism in the quotient has local lifts whose choices form a Z -torsor, as is visible on the arrow chart. There is no residual automorphism from $\mathcal{X}_k$ in this vertical subgroup: an automorphism from $\mathcal{X}_k$ is a grading element, and its grading parameter must be identity to lie in the strict subgroup. This assertion holds fpqc locally over arbitrary bases, not only on geometric points. The quotient is therefore represented by the invariant affine of Lemma 2.2. Its map to S is flat, since the map upstairs is flat and the torsor is faithfully flat. This proves the assertion about f' .

The torsor descriptions commute with base change. The central action is trivial on the weight line, so that line descends. Equation (2.2) in characteristic p makes v_1 invariant. It is a non-zero-divisor on the law charts, proving the Cartier assertion. $\square$

Use the same notation t on $\mathcal{M}_0$ and $\mathcal{M}_k$, and set

$$T_0 = f_{0*}\mathcal{O}_{\mathcal{X}_0}, \quad T_k = f_{k*}\mathcal{O}_{\mathcal{X}_k}, \quad T' = f'_*\mathcal{O}_{\mathcal{X}_k}. \quad (2.5)$$

The settings and their chart objects are arranged as follows:

$$\begin{array}{ccccccc} \mathcal{N} & \xleftarrow{r} & \mathcal{M}_k & \xrightarrow{e} & \mathcal{M}_0 & \xrightarrow{i} & \mathcal{M} \\ T' & & T_k & & T_0 & & T_{\mathbb{Z}}, T_{\mathbb{Z}_p}. \end{array}$$

The upper row records maps of stacks: r quotients the central automorphisms, e extends scalars from $\mathbb{F}_p$ to k , and i is the characteristic- p inclusion. The lower row records the chart objects in those settings. Sections 3–6 carry out the algebraic construction on $\mathcal{N}$ in the periodic category defined next, with T' as its chart object. Sections 7–8 return to the integral setting and transfer the construction to spectra.

The Cartier section t determines the height-one open and the height-two thickenings. We next specify the category and the section groups in which the construction is measured.

2.3 Periodic complexes and sections

Later we will assemble quotients by powers of t using inverse limits, so the category must allow arbitrary quasi-coherent sheaves as well as finite constructions. We fix its conventions before describing the charts on which those quotients are computed. The weight twist of a graded module is

$$W(s)_a = W_{s+a}.$$

For a stack $\mathcal{S}$ with a flat affine groupoid presentation and weight line $\mathcal{O}(1)$, let $\mathbf{D}(\mathcal{S})$ be the derived category of twisted periodic homological complexes of quasi-coherent sheaves:

$$C_{j+D} = C_j(1), \quad d : C_j \longrightarrow C_{j-1}. \quad (2.6)$$

The differential respects the periodic identifications, and we invert quasi-isomorphisms. We retain the associated stable homotopy theory when forming homotopy limits or colimits. Write $C[h] = \Sigma^h C$, with $(C[h])_j = C_{j-h}$ and the usual sign on the differential. A sheaf W denotes the pure complex with $C_0 = W$ and zero terms outside the residue class 0 (mod D), extended by the twists.

For the integral stack $\mathcal{M}$, Section 7 identifies this periodic category, by flat descent and an explicit degree regrading, with the differential E_*E -comodule model used in the chromatic comparison, where $E = E(2)$.

For any of these categories set

$$\pi_j C = [\mathcal{O}[j], C]_{\mathbf{D}(\mathcal{S})}. \quad (2.7)$$

This is sections homotopy, not underlying sheaf homology. For example, a sheaf map $W \rightarrow W'(s)$ extends to a map $W[sD] \rightarrow W'$ of pure complexes. Such sheaf maps need not account for all derived maps. After the chart calculations, we will construct the model structure and finite resolutions that justify our computations of these derived groups.

2.4 The height-one open and height-two thickenings

We now describe the two pieces determined by t . The open has no remaining stabilizer after the central quotient. On each nilpotent thickening, the stabilizer acts on one explicit coordinate ring; this will turn the reductions of T' into concrete coefficient modules.

Proposition 2.4. *There are equivalences*

$$\mathcal{M}_0[1/t] \simeq BZ, \quad \mathcal{N}[1/t] \simeq \operatorname{Spec} k.$$

The first is split by the law with $v_1 = 1$ and $v_{i>1} = 0$.

Proof. Normalize $v_1 = 1$ by its weight-one grading. We will show that every normalized law is locally isomorphic to the stated law and then determine its stabilizer. An arrow from the stated law into an arbitrary normalized law is obtained by imposing $\eta_R v_{j+1} = 0$ successively. Modulo the preceding equations,

$$\eta_R v_{j+1} = t_j^p - t_j + \text{a polynomial in } t_{<j}.$$

This is the height-one case of the coefficient comparison in (2.3); equivalently, only ϕ^p on the left and $\phi(T^p)$ on the right can contribute the new t_j . The equations are successively finite étale, so their limit is faithfully flat and supplies the cover.

On the splitting law the automorphism equations reduce recursively to $t_j^p = t_j$. Their group of solutions is exactly Z : $[1 + ap^j](T) -_F T$ supplies any prescribed next digit in $\mathbb{F}_p$, and successive correction and completeness identify all strict automorphisms. There is no remaining grading stabilizer, since strict arrows preserve the normalized coefficient $v_1 = 1$. The first equivalence follows. Base change to k and quotient by the central group give the second. $\square$

Let

$$\Lambda = \mu_q(k), \quad R_d = k[x]/(x^d) \quad (d \geq 1), \quad R = k[[x]].$$

The group Λ has order q , prime to p . The rings R_d will be the coordinate rings of the thickenings, and R records their compatible coefficients. Let F_0 be the Honda law with $v_2 = 1$ and $v_i = 0$ for $i \neq 2$, $i \geq 1$. Denote its proconstant group of strict automorphisms by G^1 , and put

$$G_{\text{up}} = G^1 \rtimes \Lambda, \quad H = G^1/Z, \quad G = H \rtimes \Lambda.$$

The action of Λ on strict automorphisms is the weight action.

Proposition 2.5. *The chart law F_x with $v_1 = x$, $v_2 = 1$, and $v_{i>2} = 0$ gives compatible presentations*

$$\mathcal{M}_k/(t^d) = [\operatorname{Spec} R_d/G_{\text{up}}], \quad \mathcal{N}/(t^d) = [\operatorname{Spec} R_d/G]. \quad (2.8)$$

Under these presentations, $\mathcal{X}_k/(t^d)$ is $[\operatorname{Spec} R_d/\Lambda]$. Sheaves are smooth, that is, discrete continuous, semilinear representations over R_d . The compatible coefficient action on R satisfies

$$g(x) \in x + x^2 R \quad (g \in H), \quad \lambda(x) = \lambda^{-1} x \quad (\lambda \in \Lambda).$$

If u is the chart basis of the weight line with $t = xu$, then

$$g(u) = \frac{x}{g(x)} u \quad (g \in H), \quad \lambda(u) = \lambda u \quad (\lambda \in \Lambda). \quad (2.9)$$

For every strict label representing an element $g \in H$, there is a formal isomorphism $F_{g(x)} \rightarrow F_x$ over R reducing to that label.

Proof. We normalize v_2 and then identify arrows by lifting their special-fiber labels. Uniqueness of the lifts will also identify composition over every nilpotent base.

On $V(t^d)$ the coordinate v_2 is invertible. Normalize it to 1 by adjoining a q -th root of the grading parameter. This gives a flat cover by the stated chart: the map obtained by adjoining a grading parameter λ has coordinates $(v_1, v_2) = (\lambda x, \lambda^q)$ and is the composite of the original flat cover with the grading action. On $\mathcal{X}_k$ the remaining grading group is Λ , giving the asserted chart there; its ambient $v_2 \neq 0$ open is $[\operatorname{Spec} k[x]/\Lambda]$.

We identify all arrows and their composition, not just their geometric fibers. If ρ is the grading parameter and t_j are the strict-arrow parameters with left end x , the right chart point is $y = x/\rho$, since $\eta_R v_1 = x$. The remaining equations are

$$\eta_R v_2 = \rho^q, \quad \eta_R v_{j+2} = 0 \quad (j \geq 1). \quad (2.10)$$

Modulo x the first right unit is 1. Modulo the preceding equations the subsequent ones are $t_j^{p^2} - t_j = 0$. This follows by comparing $\phi(T)^{p^2}$ with $\phi(T^{p^2})$ for the Honda law; its coefficients lie in $\mathbb{F}_p$, and all preceding digits are fixed by the square of Frobenius. Thus the special-fiber arrow scheme is proconstant with labels

$$(\rho_0, t_1^0, t_2^0, \dots) \in \Lambda \times k^{\mathbb{N}_{>0}}.$$

Labels with $\rho_0 = 1$ give G^1 , and grading-only arrows split off Λ .

For a prescribed left end x , each label lifts uniquely through the nilpotent layers. At a square-zero layer $(x^b)/(x^{b+1})$, only the special-fiber Jacobian acts. Its first diagonal entry is a unit because q is prime to p ; the other diagonal entries are -1 . After adding combinations of earlier rows, the system is lower triangular. Each equation is a finite polynomial, so the linear correction system on the infinite product of coefficient modules is solved recursively from its first row. This proves existence and uniqueness on every layer and over every chart algebra. Proconstant labels are unchanged by a nilpotent thickening, so the construction is functorial. In particular composition is determined by composition of the special-fiber labels.

Write the g -arrow as an arrow from the right end y to the left end x , and define $g(x) = y$. If an h -arrow is composed on its source side, its left end is y , and the composite label is gh . The resulting coefficient pullback is $g \circ h$. This convention therefore gives left semilinear actions: the moving source coordinate $g(x)$ supplies their coefficient action. Strict labels are tangent to identity, and the grading labels act as stated. Central $[z]$ fixes x and has trivial grading: its unique lift is $[z]_{F_x}$ itself. Dividing by it gives the second presentation in (2.8).

For the last assertion, a strict-label lift has $\rho \in 1 + xR$. There is a unique $\chi \in 1 + xR$ with $\chi^{1-p} = \rho$, since $1 - p$ is a unit. The linear coordinate change $T \mapsto \chi T$ sends

$$v_i \mapsto \chi^{1-p^i} v_i = \rho^{m_i} v_i.$$

It carries the law at $y = x/\rho$ to the source of the strict arrow. Composing gives the required formal isomorphism over R . Finally invariance of $t = xu$ gives (2.9) over the domain R . Reducing this identity gives the formula on each R_d ; thus no division by a nilpotent coordinate is involved. $\square$

Proposition 2.6. *The sheaves T' and T_k are t -monic. Set $T'_d = T'/t^d$ and $T_{k,d} = T_k/t^d$; their ordinary quotients also represent their derived multiplication cofibers. On the chart of Proposition 2.5,*

$$T'_d = C_{\text{lc}}(G/\Lambda, R_d) = C_{\text{lc}}(H, R_d),$$

with action

$$(gF)(s) = g(F(g^{-1}s)) \quad (g \in G, s \in G/\Lambda).$$

The inner value is acted on through the coefficient action on R_d . The analogous description of $T_{k,d}$ uses G_{up}/Λ .

Proof. The maps from $\mathcal{X}_k/(t^d)$ in Proposition 2.5 are the subgroup maps for Λ , so their pushforward is coinduction. To check the semilinear formula explicitly, let M be a semilinear G -module. An R_d -linear Λ -equivariant functional $\epsilon : M \rightarrow R_d$ has adjoint

$$m \mapsto (s \mapsto s(\epsilon(s^{-1}m))).$$

The function is constant on cosets, is locally constant by smoothness, and is equivariant for the stated action. Evaluation at 1Λ is the inverse adjunction. This proves the formula. Multiplication by t is injective on $\mathcal{O}_{\mathcal{X}_k}$. Affineness and the projection formula show it remains injective after either chart pushforward, and identify the quotients with the pushforwards from $\mathcal{X}_k/(t^d)$. Monicity makes these ordinary quotients the multiplication cofibers of pure complexes. $\square$

2.5 The reduced stabilizer and its operation ring

To describe the completed coefficient ring, we now study H and its action on R . The formal-group digits give ordered coordinates on H and a polynomial associated graded for its completed skew group ring. This description also gives the cohomological dimension bound directly, without a uniform-group hypothesis on the whole of H .

For $i \geq 1$, let $G^1(i)$ consist of strict Honda automorphisms whose digits t_j vanish for $1 \leq j < i$, and let $H(i)$ be its image in H . Set

$$\Gamma_s = H(2s+1) \quad (s \geq 0).$$

Lemma 2.7. *The filtration is complete and separated, and*

$$H(i)/H(i+1) = \begin{cases} (k, +), & i \text{ odd}, \\ (k/\mathbb{F}_p, +), & i \text{ even}, \end{cases} \quad [H(i), H(j)] \subset H(i+j). \quad (2.11)$$

The p -th-power map carries each graded digit to the identical digit two indices later. Choose g_1, g_2 with first digits an $\mathbb{F}_p$ -basis of k , and choose $g_3 = \gamma$ with only nonzero digit $t_2 = \theta \notin \mathbb{F}_p$. Then

$$(a_1, a_2, a_3) \mapsto g_1^{a_1} g_2^{a_2} g_3^{a_3}$$

gives unique coordinates $\mathbb{Z}_p^3 \rightarrow H$. Modulo Γ_s , the coordinates are taken modulo p^s . In particular Γ_s consists of coordinates divisible by p^s , and $H(2)$ consists of coordinates whose first two entries are divisible by p . The element γ commutes with Λ .

Proof. The special-fiber labels, with composition truncated on each jet, make G^1 a compact Hausdorff inverse limit of finite groups. The central subgroup Z is compact and hence closed, so H is compact Hausdorff. Each $H(i)$ is closed. Nested cosets in this compact group show that $H \rightarrow \varprojlim_i H/H(i)$ is surjective. To see that its kernel is trivial, represent an element in every $H(i)$ by $g \in G^1$. The nonempty closed sets

$$\{z \in Z : gz^{-1} \in G^1(i)\}$$

are nested. Their intersection is nonempty, and any element in it shows that $g \in Z$, since the digit filtration of G^1 is separated. This proves completeness and separatedness of the quotient filtration.

Work first in the ring of actual endomorphisms of F_0 , with its formal-group addition and composition. An automorphism beginning in digit i is $1 + h$, where h has order at least p^i . Composition orders give the commutator inclusion. In the binomial expansion of $(1 + h)^p - 1$, the leading term at order p^{i+2} is ph : here $[p]_{F_0}(T) = T^{p^2}$, and for $p > 3$ the remaining terms have larger order. Its coefficient is the original digit, since that digit lies in $k = \mathbb{F}_{p^2}$. All digits occur by the arrow calculation above. The central subgroup has its first nonzero digit in $\mathbb{F}_p$ at an even index, as follows from $[p^a]_{F_0}(T) = T^{p^{2a}}$. Quotienting gives (2.11) and the power assertion.

For the coordinates, proceed from s to $s + 1$. The p^s -th powers of g_1, g_2 correct the two independent digits of $H(2s + 1)/H(2s + 2)$. They do so also when inserted in their ordered positions, because the resulting commutators lie in the next filtration step. The p^s -th power of g_3 then corrects the remaining digit of $H(2s + 2)/H(2s + 3)$. These corrections are unique. Completeness and separatedness of the digit filtration give the coordinates on the inverse limit and all the asserted subgroup descriptions. The weight $m_2 = q$ is trivial on Λ and the other digits of γ vanish, proving its commutation assertion. $\square$

Define the completed rings, with coefficients on the left, by

$$\Omega = k[[H]], \quad \Delta = R[[H]].$$

The second is a skew group ring for the action on R : it is the inverse limit of $R_d \# (H/\Gamma_s)$, with s large enough to detect the action on R_d . Thus $ga = g(a)g$ for $a \in R$. Put $U_i = g_i - 1$. The filtration below is an auxiliary filtration on these coefficient rings; it is distinct from both x -adic order and the homological grading of periodic complexes.

Proposition 2.8. *The ring Δ has a complete separated decreasing filtration given by the ordered weights*

$$\text{wt}(x) = 4, \quad \text{wt}(U_1) = \text{wt}(U_2) = 1, \quad \text{wt}(U_3) = 3.$$

Its associated graded is the commutative polynomial ring

$$\text{gr } \Delta = k[\sigma x, A, B, C]. \quad (2.12)$$

The element x is normal, $\Omega = \Delta/(x)$, and the induced filtration gives $\text{gr } \Omega = k[A, B, C]$. Here A, B, C are the initial symbols of U_1, U_2, U_3 .

Proof. The coordinates of Lemma 2.7 show that ordered monomials in U_1, U_2, U_3 form a topological k -basis of Ω . At level H/Γ_s , the ordered group elements and the ordered difference monomials with exponents less than p^s are related by an invertible binomial change of basis; larger p^s -th powers vanish in characteristic p . Including powers of x gives the corresponding basis of Δ .

Initially filter by the closed spans of all words of weight at least a given integer. These spaces tend to zero. Indeed at each finite quotient the action preserves the ideal (x) , so x is normal there. It is also nilpotent, and modulo x the ideal generated by the remaining letters is the augmentation ideal of a finite p -group algebra. Thus the ideal generated by all the letters is nilpotent at each finite quotient. Conversely, the word-filtration spaces contain deep open neighborhoods by the ordered topological basis.

There is a strict gain in weight whenever letters are interchanged. For U_1, U_2 , the commutator lies in $H(2)$, whose first two coordinates are divisible by p . Characteristic- p binomial expansion

therefore gives weight at least 3 for its difference from 1, exceeding the original weight 2. Commuting U_1 or U_2 with U_3 gives an element of Γ_1 , so the difference has weight at least $p > 4$. Finally

$$U_i x - x U_i = (g_i(x) - x) g_i$$

has weight at least 8, since $g_i(x) - x \in x^2 R$. This is larger than either $\text{wt}(U_i) + 4$.

We can consequently reorder every word modulo strictly higher weight, and iterate to all weights by completeness. Every element of a word-filtration stage expands in ordered monomials of at least that weight. Uniqueness in the topological basis proves the reverse inclusion and shows that there are no further associated-graded relations. This proves (2.12).

Normality of x follows from $gx = x(g(x)/x)g$. Weighting each group coefficient by the continuous unit $g(x)/x$, or by its inverse, defines the requisite continuous change of coefficients even in the completion: at precision x^d choose a group quotient detecting the action modulo x^{d+1} . Thus the ideal (x) is two-sided and quotienting simply removes the x coordinate and its weight. $\square$

2.6 Derived sections and cohomological bounds

We now turn the chart calculations into bounds for derived sections. First we justify the model and resolution tools for the periodic categories. The polynomial associated graded of $k[[H]]$ will yield a compact resolution of k of length three. Applying continuous Hom computes derived H -invariants; exact Λ -invariants and the forgetful functor from semilinear R_d -modules then give the bound on the nilpotent charts. Passing through the open, the central quotient, and the integral base adds the degrees recorded below. The bounds are uniform for arbitrary sheaves, so they also control unbounded periodic complexes.

Lemma 2.9. *For each flat affine groupoid stack used here, twisted periodic complexes admit the injective model structure with quasi-isomorphisms as weak equivalences and monomorphisms as cofibrations. It is stable, its fibrant objects have injective sheaf terms, and maps into a fibrant object are computed up to compatible chain homotopy. If a sheaf W satisfies $\text{Ext}^{>a}(W, -) = 0$ for some finite a , maps from its pure shifts can be computed by applying $\text{Hom}(W, -)$ termwise to any quasi-isomorphic replacement whose terms are $\text{Hom}(W, -)$ -acyclic.*

Proof. The sheaf categories in question are Grothendieck categories. Colimits and AB5 follow from flat affine descent. For a generating set, choose an infinite cardinal κ dominating the sizes of the base and arrow rings. Starting with a bounded-size family of elements of a comodule, repeatedly adjoin components in finite expressions for their coactions and close under the module operations. Flatness injects the tensor of the resulting submodule into that of the original module, so the coaction identities restrict. The resulting subcomodules have bounded cardinality and form a generating set. For $\mathcal{X}$ one may use its finitely many grading-trivialized principal charts; their disjoint union is affine and the same argument applies. The other stacks have the affine presentations constructed above.

Twisted periodic complexes are Grothendieck as well. Disk objects formed from sheaf generators in adjacent indices, with identity differential and then repeated with the twists, give generators. We use Smith's recognition theorem in the form of [8, Theorem 1.7]; the solution-set condition follows from accessibility by [8, Proposition 1.15]. Quasi-isomorphisms satisfy the two-out-of-three property and are closed under retracts, as is seen on homology. They are accessibly embedded and accessible. For an explicit size argument, enlarge κ to include the twist data. In an arrow of complexes start with boundedly many elements, close under coactions and differentials, and

then adjoin preimages for cycles in its cone. Iteration expresses any quasi-isomorphism as a κ^+ -directed union of quasi-isomorphic subarrows of size at most κ . Objects and arrows of that size are κ^+ -presentable: their module presentations, coaction identities, and finitely many periodic differential identities can all be tested at that size. These tests can be performed on the flat affine charts.

Acyclic monomorphisms are preserved by pushout and transfinite composition, by AB5. Monomorphisms are cofibrantly generated: for a generator G_0 , adjoin successively the images of maps from G_0 by pushing out inclusions of its subobjects, taking the inverse image of the subobject already attained at each step. There is a set of these subobject inclusions. A map right orthogonal to all monomorphisms is a split epimorphism with injective kernel. An injective object in the category of complexes is contractible: its monomorphism into the cone of its identity splits, making it a retract of a contractible complex. Such an epimorphism is therefore a quasi-isomorphism. This verifies the remaining recognition condition.

Ordinary chain mapping cylinders are compatible with the twists; they identify ordinary mapping cones with homotopy cofibers. The chain shift is an equivalence, so the model structure is stable. Testing fibrancy against disk inclusions shows that fibrant complexes have injective terms. Since all objects are cofibrant, the chain cylinder also gives the claimed computation of maps by chain homotopy.

For the last assertion, a fibrant replacement computes the derived maps. A quasi-isomorphism between complexes with $\text{Hom}(W, -)$ -acyclic terms remains a quasi-isomorphism after applying $\text{Hom}(W, -)$. Indeed its cone is acyclic with such terms. Dimension shifting along the short exact sequences of cycles shifts the possible obstruction to exactness past degree a , where it vanishes. The argument requires only the uniform bound on $\text{Ext}(W, -)$ and therefore works for the unbounded periodic unfolding. $\square$

Lemma 2.10. *The right derived functors of sections in the sheaf categories above commute with filtered colimits and are computed by the flat affine coinduced resolution. The resolution is compatible with closed base change. If $\text{Ext}^{>a}(\mathcal{O}, -) = 0$, its finite truncation gives a sections-acyclic resolution that computes (2.7) on periodic complexes. In particular π_* then commutes with filtered colimits and telescopes.*

Proof. Choose an affine fpqc chart $l : S \rightarrow \mathcal{S}$ with S affine. The functor $I = l_* l^*$ is exact, its unit is monic, and its unit cokernel J is exact. Its values are sections-acyclic by flat affine adjunction. Iterating I and J gives the usual coinduced resolution. On the chart all these functors, including ordinary sections of the coinduced terms, commute with filtered colimits; AB5 therefore proves the same for their cohomology. After restricting to a closed substack, the base-changed chart gives exactly the corresponding resolution. This proves the assertion about sheaves pushed forward from closed loci without requiring derived closed pushforward to be fully faithful.

Under the dimension bound, the resolution can end in J^a , which is sections-acyclic by dimension shifting. These functors commute with weight twists. The finite totalization thus computes sections on periodic complexes, by Lemma 2.9; no infinite totalization is needed. The colimit assertion follows termwise. $\square$

We use exact pullback and pushforward functors below also on periodic complexes. An exact pullback left adjoint is left Quillen for these injective structures and is adjoint to the right derived pushforward. These adjunctions let the preceding chart calculations control both sheaves and the periodic complexes built from them.

Lemma 2.11. *Invariants of smooth k -linear H -modules have cohomological dimension at most 3. The same is true for G and for sections on $\mathcal{N}/(t^d)$, for every $d \geq 1$.*

Proof. The key step is to lift the three-step Koszul resolution from the associated graded ring and use it to compute continuous invariants. Lift the graded Koszul resolution of k over $\text{gr } \Omega = k[A, B, C]$ to a resolution of length three by finite free compact left Ω -modules. Here is the lifting argument. At each step the actual kernel is complete. Lift the homogeneous columns generating its associated-graded kernel to elements of that kernel and map in a free module with the corresponding shifted filtration. Successive subtraction of initial symbols converges and makes this a strict surjection. Its complete kernel has associated graded equal to the next Koszul kernel. At the final step that graded kernel vanishes, so separatedness makes the actual kernel zero. The resulting differentials compose to zero because their columns were lifted into the actual kernels.

For a smooth discrete module M , continuous $\text{Hom}_\Omega(-, M)$ applied to this resolution has terms finite sums of M , and hence gives a delta functor. It is acyclic in positive degrees on $C_{\text{lc}}(H, M_{\text{vec}})$, coinduced from the underlying vector space: evaluation and translation identify the resulting complex with continuous k -linear Hom from the compact resolution into M_{vec} . The adjunction is continuous because a map from a compact free module to a discrete vector space kills a sufficiently deep stable filtration stage. Exact sequences of compact k -vector spaces split continuously, as is seen by dualizing to discrete vector spaces; thus this Hom complex is exact in positive degrees. Every smooth module embeds in such a coinduction. Effaceability identifies the delta functor with derived invariants and proves the bound.

For G , take H -invariants and then exact Λ -invariants. Restriction of injectives from G to H preserves injectives, because finite-index induction is exact. Finally, forgetting the R_d -module structure of a semilinear representation preserves injectives: it has the exact left adjoint given by the free R_d -module with diagonal action. Applying the quotient-chart description proves the assertion about $\mathcal{N}/(t^d)$. $\square$

Proposition 2.12. *Projective dimensions in the Ext sense satisfy*

$$\text{pd}_{\mathcal{N}}(\mathcal{O}) \leq 4, \quad \text{pd}_{\mathcal{M}_k}(\mathcal{O}), \text{pd}_{\mathcal{M}_0}(\mathcal{O}) \leq 5, \quad \text{pd}_{\mathcal{M}}(\mathcal{O}) \leq 6. \quad (2.13)$$

For sheaves on $\mathcal{M}_k$, the right adjoint of r^ has*

$$r_*M = \ker(c - 1 : M \rightarrow M), \quad R^1r_*M = \text{coker}(c - 1 : M \rightarrow M), \quad R^{>1}r_*M = 0, \quad (2.14)$$

where the kernel and cokernel descend to $\mathcal{N}$.

Proof. We first pass from the nilpotent charts to $\mathcal{N}$, then restore the central group, and finally pass from characteristic p to the integral stack.

A t^∞ -torsion sheaf on $\mathcal{N}$ is a filtered union of sheaves annihilated by powers of t , with the evident weight twists in the kernel maps. Lemmas 2.10 and 2.11 show that its sections cohomology vanishes above degree 3. The open of Proposition 2.4 has exact sections. For an arbitrary sheaf M , the kernel and cokernel of its localization map $M \rightarrow M[1/t]$ are t^∞ -torsion. The open is affine over the stack, so localization is exact open restriction followed by affine pushforward. The two short exact sequences associated to the kernel and cokernel give the extra degree and prove the bound 4.

The right adjoint r_* is vertical invariants. Smoothness and centrality ensure that the kernel and cokernel of $c - 1$ have trivial vertical action and descend; they give a two-term delta functor. To prove that it computes the derived functor, efface its degree-one term by embedding a sheaf

into $I = l_* l^*$ for an affine law chart $l : S \rightarrow \mathcal{M}_k$. On the arrow chart, the action is translation on the arrow algebra tensored over the other endpoint with the module coefficients. The map $c - 1$ is surjective there. Indeed the arrow scheme is a Z -torsor over its quotient, and fpqc locally it is split. A locally constant function is represented at a finite central period. Increase that period by a factor p ; the sum of the repeated old period is zero in characteristic p , so the function can be integrated as a discrete difference on the larger cycle. This proves surjectivity locally, hence on the sheaf. Effaceability proves (2.14).

Because r^* is exact, r_* preserves injectives. Derived sections composed with (2.14) therefore give the bound 5 on $\mathcal{M}_k$. Finite flat extension from $\mathbb{F}_p$ to k commutes with the coinduced chart resolution of sections and detects vanishing, giving the same bound on $\mathcal{M}_0$.

For $\mathcal{M}$, first consider p -power-torsion sheaves. A sheaf annihilated by p^a has a finite filtration with characteristic- p subquotients; a general p^∞ -torsion sheaf is the filtered union of its kernels of powers of p . The preceding bound and Lemma 2.10 give vanishing above degree 5. On the rational open the logarithm trivializes the strict law groupoid, leaving only $B\mathbb{G}_m$, whose sections are exact. Apply the same kernel-cokernel argument to $M \rightarrow M[1/p]$ to obtain the last bound in (2.13). $\square$

2.7 The central comparison triangle

The finite resolutions above justify derived sections on periodic complexes. For the reverse transfer, we now construct the central pushforward itself from exact functors and retain the natural comparison triangle, including its possibly nonzero connecting map.

Proposition 2.13. *Choose an affine law chart $l : S \rightarrow \mathcal{M}_k$, set $I = l_* l^*$, and let J be the cokernel of its unit. The two-term totalization*

$$Q = [r_* I \longrightarrow r_* J], \quad \text{in homological degrees } 0, -1, \quad (2.15)$$

computes Rr_ on twisted periodic complexes and preserves quasi-isomorphisms. There are natural identifications*

$$QT_k \simeq T', \quad \pi_*(r^* U) \cong \pi_*(Qr^* U) \quad (U \in \mathbf{D}(\mathcal{N})),$$

and natural triangles

$$U \longrightarrow Qr^* U \longrightarrow U[-1] \longrightarrow U[1]. \quad (2.16)$$

Proof. The composite $rl : S \rightarrow \mathcal{N}$ is an affine chart, so I has r_* -acyclic values by affine adjunction. Equation (2.14) and the sequence $0 \rightarrow M \rightarrow IM \rightarrow JM \rightarrow 0$ show that J also has r_* -acyclic values. Both $r_* I$ and $r_* J$ are exact: I and J are exact, and the acyclicity of their values removes the connecting obstruction after applying r_* . Hence the finite totalization (2.15) preserves quasi-isomorphisms. On injective-term fibrant replacements it agrees up to quasi-isomorphism with ordinary r_* , by the same two-term acyclic resolution. It therefore computes Rr_* on periodic complexes, not just on sheaves. The twist line is pulled back from downstairs, so all these functors and identifications respect the periodicity.

Flat affine adjunction along f_k and f' identifies $Rr_* T_k$ with T' . Derived adjunction for r^* , with $r^* \mathcal{O} = \mathcal{O}$, gives the equality of sections homotopy. For a sheaf pulled back from $\mathcal{N}$, the vertical kernel and cokernel in (2.14) are both the original sheaf. Thus there is a natural four-term exact sequence

$$0 \longrightarrow U \longrightarrow r_* I r^* U \longrightarrow r_* J r^* U \longrightarrow U \longrightarrow 0.$$

It applies termwise also to periodic complexes. Its two connecting maps give (2.16), with the consistent homological signs. In particular the triangle is natural, but no splitting of its final connecting map is asserted. $\square$

The next sections carry out the algebraic construction in $\mathbf{D}(\mathcal{N})$. The open is $\mathrm{Spec} k$, the thickenings carry the semilinear G -action, and the reductions of T' are coinduced modules. Section 3 uses these descriptions and the finite bounds to assemble the reductions by homotopy limits.

3 Completion, operations, and the overlap

We now describe how a twisted periodic complex is recovered from its restriction to the height-one open and its completion along the height-two stratum. The fracture square supplies this reconstruction. To use it for the construction of our object, we also need an explicit description of finite constructions from the completed chart sheaf. We obtain one by representing every operation by compatible ordinary maps on all nilpotent thickenings. Matrices of these maps then form actual complexes, whose sections and localized sections can be computed. The last part identifies an intrinsic topology on the overlap when the original object has finite sections homotopy in every degree, and proves continuity for every map between such completed objects.

We work at a prime $p > 5$, as in the geometric construction. Throughout this section the categories are the characteristic- p categories $\mathbf{D}(\mathcal{M}_0)$, $\mathbf{D}(\mathcal{M}_k)$, and $\mathbf{D}(\mathcal{N})$ of Section 2.3. The line section $t = v_1$ has degree $D = 2(p - 1)$. Recall that the homological and weight conventions are $C_{j+D} = C_j(1)$, $d : C_j \rightarrow C_{j-1}$, and $C[b]_j = C_{j-b}$, with the usual sign on the differential of a shift. We write $\pi_i C = [\mathcal{O}[i], C]$ for sections homotopy; this is distinct from the sheaf homology of C . We retain $R = k[[x]]$, $R_d = k[x]/(x^d)$, $\Lambda = \mu_q(k)$, and $\Delta = R[[H]]$ from the nilpotent charts. All limits and colimits of objects in the derived categories are homotopy limits and colimits.

3.1 The fracture square

For $d \geq 1$ define

$$P/t^d = \mathrm{cofib}(P[dD] \xrightarrow{t^d} P), \quad \widehat{P} = \mathrm{holim}_d P/t^d, \quad LP = P[1/t]. \quad (3.1)$$

The multiplication maps, their cones, and the transition maps in these towers are chosen naturally at the level of twisted complexes. We call an object t -invertible if multiplication $t : P[D] \rightarrow P$ is an equivalence, and *complete* if $P \rightarrow \widehat{P}$ is an equivalence.

For the general fracture description relating completion and localization, compare Barthel–Heard–Valenzuela [6, Corollaries 2.26 and 2.28]. We prove the required form directly for arbitrary twisted periodic complexes in these categories.

Proposition 3.1 (Fracture). *The functor L is the flat affine restriction and pushforward for the open $t \neq 0$, whose derived category it embeds fully faithfully. Completion is exact and idempotent. Its values are right orthogonal, in every shift, to t -invertible objects. For every P , the fiber of $P \rightarrow \widehat{P}$ is t -invertible, and there are natural equivalences*

$$P/t^d \simeq \widehat{P}/t^d \quad (d \geq 1).$$

The following square is a homotopy pullback:

$$\begin{array}{ccc} P & \longrightarrow & \widehat{P} \\ \downarrow & & \downarrow \\ LP & \longrightarrow & L\widehat{P}. \end{array} \quad (3.2)$$

Thus the original object is recovered by matching its open and complete parts over the localized completion. Moreover,

$$\pi_i LP = \operatorname{colim}_{s \geq 0} \pi_{i+sD} P. \quad (3.3)$$

Proof. The localization telescope is $P \rightarrow P[-D] \rightarrow P[-2D] \rightarrow \cdots$, with transition maps induced by t . Exactness of filtered colimits represents this telescope by the ordinary termwise colimit. On the affine charts this is localization at the principal open defined by t , so it is the asserted open push-pull functor. Flat affine adjunction identifies the open category with its essential image.

Taking the inverse limit of the multiplication-cofiber triangles shows that the fiber of $P \rightarrow \widehat{P}$ is the limit of the tower

$$\cdots \xrightarrow{t} P[(d+1)D] \xrightarrow{t} P[dD] \xrightarrow{t} \cdots.$$

Shifting this tower by one position does not change its limit. Consequently t is invertible on this fiber. On the other hand, t^d acts nullhomotopically on P/t^d , by the standard homotopy on its multiplication cone. If A is t -invertible, naturality of t therefore gives $[A[j], P/t^d] = 0$ for all j : multiplication by t^d is simultaneously invertible from the source and zero on the target. The same assertion for mapping spectra, followed by the inverse limit, gives $[A[j], \widehat{P}] = 0$.

Inverse limits are exact in the stable category, so completion is exact. If Q is right orthogonal to the t -invertible objects, then the fiber of $Q \rightarrow \widehat{Q}$ is both t -invertible and right orthogonal to the t -invertible objects. It is therefore zero. This proves idempotence and characterizes the complete objects by that orthogonality. The t -invertible fiber of $P \rightarrow \widehat{P}$ has zero quotient by every t^d , giving the asserted equivalence of reductions. Localizing this same fiber triangle leaves its fiber unchanged, which proves the pullback assertion in (3.2).

Finally, the finite cohomological bounds of Proposition 2.12, together with exact filtered colimits, allow sections-homotopy to commute with this telescope. Equivalently, compute derived sections with the bounded cohomological range established there and commute the filtered colimit through the resulting calculation. This gives (3.3). $\square$

The fracture square reduces reconstruction to a map on the overlap. The following description of the open categories specifies which maps can be chosen there and records a vanishing used in the transfer step.

Lemma 3.2 (Objects on the open). *The category $\mathbf{D}(\mathcal{N}[1/t])$ is the category of D -periodic complexes of k -vector spaces. Every object is the sum, over the D residue classes, of its pure homology objects, and every graded periodic map on homology is realizable.*

The category $\mathbf{D}(\mathcal{M}_0[1/t])$ is the category of D -periodic complexes of smooth Z -representations over $\mathbb{F}_p$. Every object also splits as the sum of its pure homology objects. Maps can have an Ext^1 component, but a pure object has no self-map to its 2-shift when $D > 3$.

Proof. The chart identifications give $\mathcal{N}[1/t] = \operatorname{Spec} k$ and $\mathcal{M}_0[1/t] = BZ$, and the nowhere-vanishing section t trivializes the periodicity line. The assertion for vector spaces follows by splitting cycles and boundaries.

Fix the chosen topological generator c of Z and write $s = c - 1$. Smooth representations identify with locally s -nilpotent $\mathbb{F}_p[s]$ -modules: the identity $(1 + s)^{p^a} = 1 + s^{p^a}$ identifies annihilation by a power of s with invariance under a sufficiently small open subgroup. This torsion category is hereditary. Indeed, embed a torsion module in a divisible $\mathbb{F}_p[s]$ -module and take its s -power-torsion submodule. That submodule is still divisible: division by s can be performed within the torsion submodule, and every polynomial prime to s acts invertibly on it. It is thus injective. Its quotient

by the original module is again divisible and torsion, giving an injective resolution of length one in the torsion category.

Represent a periodic complex by a fibrant complex I with injective terms. Write Z_j , B_j , and H_j for its cycles, boundaries, and homology. From $0 \rightarrow Z_{j+1} \rightarrow I_{j+1} \rightarrow B_j \rightarrow 0$ and heredity we obtain

$$\mathrm{Ext}^1(H_j, B_j) = \mathrm{Ext}^2(H_j, Z_{j+1}) = 0.$$

The identity of H_j therefore lifts to a map $H_j \rightarrow Z_j$. Choose these lifts for one representative of each period residue and extend them periodically. They give a quasi-isomorphism from the finite sum of shifted pure H_j to I . Finally, periodizing a length-one injective resolution of a pure object computes its maps; when $D > 3$, neither its resolution degree 0 nor its resolution degree 1 can contribute to a self-map with shift 2. $\square$

3.2 The complete operation algebra

Recall from Proposition 2.6 that $T' = f'_* \mathcal{O}$ is t -monic and that

$$T'_d = T' / t^d = f'_* \mathcal{O}_{\mathcal{X}_k / (t^d)}$$

is a pure sheaf. On the nilpotent chart it is $C_{\mathrm{lc}}(H, R_d)$, coinduced from Λ , with diagonal semilinear action $(gF)(h) = g(F(g^{-1}h))$ in coset coordinates. The chart basis u of the weight line satisfies $t = xu$ and $h(u) = x/h(x)u$. These formulas will identify a line twist with a fractional coefficient lattice.

We use $\Delta[x^{-1}]$ for the two-sided localization at the powers of x . Here is the precise reason this localization contains Δ . With coefficients on the left, the skew relation gives

$$hx = x \frac{h(x)}{x} h, \quad xh = \frac{x}{h(x)} h x \quad (h \in H).$$

Both displayed fractions are units of R . As in the proof of Proposition 2.8, at coefficient precision x^d we may refine the group quotient until it detects these units modulo x^d , using the coefficient action modulo x^{d+1} . The two inverse continuous changes of coefficients give $\Delta x = x\Delta$. Left multiplication by x is injective on the compatible coefficient systems defining Δ , and the unit change gives right injectivity as well. Thus x is normal and regular, so its powers admit this two-sided localization. This coefficient element is generally noncentral.

We shall compute the graded algebra

$$\mathcal{E}_n = [\widehat{T}'[n], \widehat{T}'].$$

The product convention is composition: ab means first apply b and then apply the appropriately shifted a . With our periodicity convention, an ordinary sheaf map $T'_d \rightarrow T'_d(s)$ gives a map $T'_d[sD] \rightarrow T'_d$. The calculation must retain these ordinary maps: their strict compositions will make matrices of operations into complexes.

Proposition 3.3 (Complete operations and their representatives). *There is an identification of graded algebras*

$$\mathcal{E}_{sD} = (x^{-s}\Delta)^\Lambda \quad (s \in \mathbb{Z}), \quad \mathcal{E}_n = 0 \quad (n \notin D\mathbb{Z}). \quad (3.4)$$

Products use left-coefficient skew convolution in $\Delta[x^{-1}]$, with degrees added. Every element of $\mathcal{E}_{sD}$ has a unique compatible system of ordinary sheaf-map representatives $T'_d \rightarrow T'_d(s)$, and these

representatives realize multiplication strictly. Under (3.4), the operation t of degree D is the coefficient $1 \in x^{-1}\Delta$; it is central in $\mathcal{E}$.

Put

$$A_{\text{op}} = (\Delta[x^{-1}])^\Lambda.$$

With a formal t recording degree D , localization identifies

$$\mathcal{E}[t^{-1}] = A_{\text{op}}[t^{\pm 1}]. \quad (3.5)$$

Proof. We first exclude the extension and inverse-limit terms so that all operations have ordinary compatible representatives. We then normalize the line twists and compute composition at every finite precision.

The fiber of $T' \rightarrow \widehat{T}'$ is t -invertible, so Proposition 3.1 identifies maps into $\widehat{T}'$ out of T' or out of $\widehat{T}'$. Maps into the limit can consequently be computed from the tower of targets T'_d .

Flat affine adjunction along f' computes $[T'[n], T'_d]$ on $\mathcal{X}_k$, with target $\mathcal{O}_{\mathcal{X}_k/(t^d)}$. This target is pushed forward from the open $v_2 \neq 0$, so we may work on $[\text{Spec } k[x]/\Lambda]$. Let M be the restriction of $(f')^*T'$ to this chart. It is x -monic and

$$M/x^d M = C_{\text{lc}}(H, R_d).$$

As an R_d -module the latter is the scalar extension of $C_{\text{lc}}(H, k)$, hence is free. Since $|\Lambda| = q$ is invertible in k , averaging a lift over Λ proves equivariant projectivity as well.

Choose an equivariant projective resolution of M of length one over $k[x]$. Start with a surjection from a free module over the skew group ring $k[x] \# \Lambda$; its underlying $k[x]$ -module is free. Since $k[x]$ is hereditary, its kernel is projective as a $k[x]$ -module; averaging again promotes this to equivariant projectivity. Reduction modulo x^d preserves exactness, since M is x -monic. Applying Hom into $R_d u^s$ factors through this reduced resolution. Its positive cohomology vanishes by the projectivity of $M/x^d M$ just proved. Periodize the length-one resolution, with the specified line twists. Its bounded resolution length and projective terms make it valid for computing maps to twisted periodic complexes: mapping from it takes acyclics to acyclics. Since $D > 2$, the augmentation degree and the relation degree cannot coincide modulo D . Thus the only possible groups are the ordinary maps in degrees sD and their first extensions; the latter vanish. In particular,

$$[T'[sD], T'_d] = \text{Hom}(T', T'_d(s)), \quad [T'[n], T'_d] = 0 \quad (n \notin D\mathbb{Z}). \quad (3.6)$$

These ordinary Hom groups have surjective transition maps. Indeed, a map at level d can be viewed as a functional on $M/x^{d+1}M$ with values in $R_d u^s$; equivariant projectivity of $M/x^{d+1}M$ lifts it through $R_{d+1} u^s \twoheadrightarrow R_d u^s$. The Milnor exact sequence therefore has zero derived-limit term. Each ordinary map in (3.6) kills t^d and factors uniquely through T'_d . Compatibility in d gives actual maps of towers of pure complexes, and composition is their ordinary composition. This factorization is a statement about sheaf maps; it does not assert full faithfulness of closed pushforward on derived categories.

We next identify these representatives and their product. Evaluation at the identity coset in coinduction identifies a map with a Λ -equivariant R_d -linear functional

$$C_{\text{lc}}(H, R_d) \longrightarrow R_d u^s.$$

Without the twist, dualizing the functions on each finite group quotient gives an R_d -valued measure, with masses added under passage to a smaller quotient. Taking both group and coefficient limits gives Δ . To include the twist, use

$$R_d u^s \cong x^{-s} R / x^{d-s} R, \quad u^s \longmapsto x^{-s}. \quad (3.7)$$

This is compatible with the group action: the formula $g(u) = (x/g(x))u$ of Proposition 2.5 becomes the ordinary action on the fractional coefficient x^{-s} . For the tame action, $\lambda(u) = \lambda u$ follows from the invariance of $t = xu$, and agrees with $\lambda(x^{-s}) = \lambda^s x^{-s}$. Thus the normalization is equivariant for every positive or negative s . Equivariance of the functional is therefore invariance of its measure under the coefficient and conjugation action of Λ . This gives the groups in (3.4).

For clarity, write integrals in left-coefficient notation. They are finite sums after passage to a sufficiently deep coefficient and group quotient, and their values are compatible as those quotients are refined. If b is the measure describing a map and F its input function, equivariance reconstructs its value at $h \in H$ as

$$h \left(\int_z b(z)(h^{-1}F)(z) \right) = \int_z h(b(z))F(hz). \quad (3.8)$$

The same formula applies to fractional coefficient lattices using (3.7). In our diagonal semilinear coinduction coordinates the projection formula is the pointwise map

$$C_{lc}(H, R_d) \otimes L_j \longrightarrow C_{lc}(H, L_j), \quad F \otimes l \longmapsto (h \longmapsto F(h)l), \quad L_j = R_d u^j.$$

It is equivariant: applying g to either side gives $g(F(g^{-1}h))g(l)$ at h . In particular the second factor here is l , not $h(l)$. Tensoring both sides of a map by the j th line twist therefore multiplies both its input function and its output at the identity by x^{-j} . Its normalized measure is unchanged. This is the R -linear extension of the evaluation functional to the shifted lattice; it is not a claim that x commutes with the resulting skew operator on sections.

Apply the map b first and then the map a . Evaluation of their composite, using (3.8), is

$$\int_h \int_z a(h)h(b(z))F(hz),$$

which is precisely left-coefficient skew convolution ab . All these identities hold at each finite level, including for negative twists. For example, if the degrees of a and b are sD and jD , changing their representatives respectively by coefficients in $x^{d-s}R$ and $x^{d-j}R$ changes the product only by $x^{d-s-j}R$. This follows because every h preserves x -adic order. Group quotients may always be refined to detect the relevant units $h(x)/x$ and their inverses to the required coefficient order. Thus composition and reduction are strictly compatible, as asserted.

On invariant sections the input function is the orbit function of its value f at the identity. The action consequently becomes

$$a(f) = \int_H a(h)h(f). \quad (3.9)$$

Multiplication by $t = xu$ normalizes to the coefficient 1, proving the assertion about t . In particular it is central in the graded operation algebra, although x need not be central in Δ . Finally, inverting this degree- D inclusion allows arbitrarily large fractional coefficient lattices in every fixed degree. Their union is $\Delta[x^{-1}]$, and taking Λ -invariants commutes with this union. This proves (3.5). $\square$

The same finite-level calculation gives the vanishing of self-maps to the 2-shift for all three characteristic- p chart sheaves. The assertion includes every shift of each sheaf.

Corollary 3.4. *For each $j \in \mathbb{Z}$ there are no self-maps to the 2-shift of $\widehat{T}'[j]$, $\widehat{T}_k[j]$, or $\widehat{T}_0[j]$.*

Proof. For $\widehat{T}'$ this is immediate from (3.4). The finite-level calculation (3.6) applies equally to the full-center chart sheaf T_k , replacing the functions on H by the functions for G_{up} . They are again free over R_d and equivariantly projective for the tame subgroup Λ . Hence the same extension and Milnor terms vanish.

For T_0 , use extension of scalars $e^* : \mathbf{D}(\mathcal{M}_0) \rightarrow \mathbf{D}(\mathcal{M}_k)$. Since $k/\mathbb{F}_p$ is finite separable, trace duality identifies this extension functor also as a right adjoint to restriction. It preserves homotopy limits, and therefore commutes with completion. Moreover the unit $F \rightarrow e_*e^*F$ is naturally split as an $\mathbb{F}_p$ -linear map, so extension of scalars detects zero maps. The asserted vanishing follows from that for $\widehat{T}_k$. $\square$

3.3 Finite constructions from the complete generator

The operations just computed act strictly on the entire quotient tower. We now turn this action into a description of all finite constructions from $\widehat{T}'$. The important point is that the relations in a differential matrix will hold as equalities of maps at every level, so its cones and sections can be calculated from the same operators.

Give $\mathcal{E}$ its homological grading and zero differential. A *finite-cell right dg $\mathcal{E}$ -module* is a finite iterated extension of shifts of the free right module $\mathcal{E}$. It can be represented by a finite graded-free module with a filtration whose successive quotients are shifted free modules. A perfect right module is a retract of a finite-cell module in the derived category; their category is $\text{Perf}(\mathcal{E})$.

The description of an algebraic thick subcategory by perfect modules over a dg endomorphism algebra is a general form of dg Morita theory; compare Keller [19, Section 3.6 and Theorem 3.8(b)]. Here we use the computed graded algebra $\mathcal{E}$ itself with zero differential. Its strict action on the finite reductions, proved in Proposition 3.3, is what permits this choice.

Proposition 3.5 (Strict realization). *There is an exact fully faithful functor*

$$\mathcal{R} : \text{Perf}(\mathcal{E}) \longrightarrow \mathbf{D}(\mathcal{N})$$

from perfect right dg modules which sends $\mathcal{E}$ to $\widehat{T}'$ and identifies its image with $\text{Thick}(\widehat{T}')$. For a finite-cell module $\mathbf{P}$, its image is the homotopy limit of an actual compatible tower Y_d of complexes whose terms are finite sums of shifts of T_d' .

Proof. First let $\mathbf{P}$ be a finite-cell module. Choose a homogeneous basis of $\mathbf{P}$ compatible with its cell filtration. Replace a basis vector e_b of degree b by the shifted pure complex $T_d'[b]$ at every level d . A degree- j right-module matrix entry

$$e_b \longmapsto e_{b'} a, \quad a \in \mathcal{E}_{b+j-b'},$$

is represented in chain degree m by

$$a : (T_d')_{m-b} \longrightarrow (T_d')_{m+j-b'}. \quad (3.10)$$

Use the unique representatives of Proposition 3.3 for all entries, including the differential entries, where $j = -1$. The source and target use the same line-twist periodicity identifications in every degree and at every level d . On the normalized charts these are the powers of the descent unit for u already used in (3.7).

Strict composition makes the matrix relation $d^2 = 0$ an equality of actual maps. All nonzero degrees of $\mathcal{E}$ are even, so no coefficient sign enters this relation. For a degree- j matrix f , the Hom differential is

$$d_Q f - (-1)^j f d_P.$$

Formula (3.10) respects this differential, and therefore respects chain maps, chain homotopies, the signed shift, and ordinary mapping cones. It also makes all these complexes and maps compatible

under the transitions $d + 1 \rightarrow d$. Every matrix entry uses its compatible tower map, so its representative is fixed at all levels at once.

The order of multiplication agrees with the right-module convention. An endomorphism of the free right module determined by $1 \mapsto a$ sends $z \mapsto az$, by left multiplication. Composing the endomorphism for b and then the one for a gives ab , the same order as the operators in Proposition 3.3. Tensoring a right module with a left module consequently uses the left action of these same operators.

Take the homotopy limit of the constructed tower. Limits are additive and exact in the stable category, so the realization of a levelwise cone is the cone of the realized map. Compatible chain homotopies induce equal maps. A change of homogeneous basis gives isomorphic towers through its change-of-basis matrices. We obtain an exact functor on the homotopy category of finite cells. Mapping a shifted free module to an acyclic module gives an acyclic Hom complex, and induction along a finite free-cell filtration preserves this property. Finite cells are therefore homotopically projective, so their homotopy maps compute their derived maps.

On the free generator and every shift, the functor induces precisely the isomorphisms of Proposition 3.3. Fix a free source and use the long exact sequences of maps along a cell filtration of the target to obtain the isomorphism for every finite-cell target. Then apply the long exact sequences along a cell filtration of the source. This proves full faithfulness for all finite cells. The target derived category is idempotent complete, so the functor extends fully faithfully to retracts. Its image is closed under shifts, finite sums, cones, and retracts and contains $\hat{T}'$. Conversely, every object in its image is constructed by those operations from $\hat{T}'$. Its image is therefore exactly $\text{Thick}(\hat{T}')$. $\square$

For example, if $a \in \mathcal{E}_0$, the two-cell module $\text{cofib}(a : \mathcal{E} \rightarrow \mathcal{E})$ realizes $\text{cofib}(a : \hat{T}' \rightarrow \hat{T}')$. The module map is left multiplication on the free right module. We next compute the sections of this cone using the same left operator on a coefficient lattice; its kernel and cokernel are thus sections data of an actual completed object.

3.4 Sections and finite reductions

Let $\mathbf{L}$ be the graded left $\mathcal{E}$ -module

$$\mathbf{L}_{sD} = (x^{-s}R)^\Lambda \quad (s \in \mathbb{Z}), \quad \mathbf{L}_n = 0 \quad (n \notin D\mathbb{Z}), \quad (3.11)$$

with zero differential and action (3.9). Its finite reductions are the graded left modules

$$(\mathbf{L}_d)_{sD} = (x^{-s}R/x^{d-s}R)^\Lambda \quad (s \in \mathbb{Z}), \quad (\mathbf{L}_d)_n = 0 \quad (n \notin D\mathbb{Z}),$$

and $\mathbf{L} = \varprojlim_d \mathbf{L}_d$ degree by degree. Set

$$\Phi = (k((x)))^\Lambda = k((x^q)).$$

Here t acts on normalized sections by the inclusion of one fractional lattice in the next. In particular $\mathbf{L}[t^{\pm 1}] = \Phi[t^{\pm 1}]$, with t recording degree D . For a complete object Y write $W(Y) = \pi_*LY$.

Proposition 3.6 (Section formulas). *If $Y = \mathcal{R}(\mathbf{P})$ for a perfect right dg $\mathcal{E}$ -module $\mathbf{P}$, then naturally*

$$\begin{aligned} \pi_*Y &= H_*(\mathbf{P} \otimes_{\mathcal{E}}^{\mathbf{L}} \mathbf{L}), \\ W(Y) &= H_*(\mathbf{P} \otimes_{\mathcal{E}}^{\mathbf{L}} \Phi[t^{\pm 1}]). \end{aligned} \quad (3.12)$$

For a finite-cell model the ordinary cellular tensor computes each derived tensor. The actions in these formulas are the left actions of (3.9).

Proof. Suppose first that $\mathbf{P}$ is a finite-cell module and write Y_d for its realization tower. Each term of Y_d is a finite sum of twists of T'_d . Those twists are sections-acyclic: flat affine adjunction reduces sections to the nilpotent chart of $\mathcal{X}_k$, where taking Λ -invariants is exact. The bounded derived-sections calculation from Proposition 2.12 therefore computes $\pi_* Y_d$ by the termwise sections complex. By (3.9), that complex is $\mathbf{P} \otimes_{\mathcal{E}} \mathbf{L}_d$. This identifies differentials and maps, as well as the underlying groups.

Write $C_d = \mathbf{P} \otimes_{\mathcal{E}} \mathbf{L}_d$. In each fixed degree its terms, cycles, boundaries, and homology are finite groups: k is finite, the fractional lattices modulo x^d are finite dimensional, and there are only finitely many cells. Each of these towers therefore satisfies the Mittag-Leffler condition, so its first derived inverse limit vanishes. Taking inverse limits in the exact sequences defining cycles, boundaries, and homology shows that $H_*(\varprojlim_d C_d) = \varprojlim_d H_*(C_d)$. The Milnor sequence for $Y = \operatorname{holim}_d Y_d$ and the finite graded-free underlying module of $\mathbf{P}$ now give

$$\pi_* Y = H_* \left(\varprojlim_d \mathbf{P} \otimes_{\mathcal{E}} \mathbf{L}_d \right) = H_*(\mathbf{P} \otimes_{\mathcal{E}} \mathbf{L}).$$

The tensor-limit interchange is degreewise and involves only finitely many summands. Finite-cell right modules are homotopically flat: after tensoring their cell filtration with an acyclic left module, every successive quotient is a shift of that acyclic module, so induction makes the full tensor acyclic. Thus the ordinary cellular tensor computes the derived tensor in the first formula.

Now take the t -colimit. Proposition 3.1 identifies it with the localized sections. Filtered colimits are exact, and $\mathbb{L}[t^{-1}] = \Phi[t^{\pm 1}]$, which gives the second formula with the same homotopical flatness justification. The identifications are natural for finite-cell maps. Applying them to idempotent summands proves both formulas for all perfect modules. $\square$

The topology of the overlap vector spaces will use these complete section groups. To make the topology on $\pi_i Y$ intrinsic, we identify the tower just constructed with the reductions of the complete object.

Lemma 3.7 (The tower is the reduction tower). *For a finite-cell realization $Y = \operatorname{holim}_e Y_e$ there are natural equivalences*

$$Y/t^d \simeq Y_d \quad (d \geq 1).$$

Under these equivalences, the tower projection is the intrinsic reduction map. The profinite topology on $\pi_i Y$ obtained from the finite-level sections calculation is consequently its inverse-limit topology from $\pi_i(Y/t^d)$.

If Y' is a retract of Y , let e_d be the idempotent of Y_d in the derived category induced by the idempotent defining this retract. Then Y'/t^d is the summand of Y_d split by e_d , and

$$\pi_i Y' = \varprojlim_d \operatorname{im}(\pi_i e_d)$$

with its intrinsic profinite topology.

Proof. Multiplication by t^d is strictly zero on Y_d , because it is strictly zero on every shifted T'_d in its terms. For $e \geq d$, map the ordinary cone of $t^d : Y_e[dD] \rightarrow Y_e$ to Y_d by the tower transition on the unshifted summand and zero on the shifted summand. Strict annihilation makes this a chain map. These maps are compatible in e and natural for the matrix maps defining our finite-cell constructions. Exactness of homotopy limits gives a map $Y/t^d \rightarrow Y_d$.

On the free generator this is the equivalence $\widehat{T}'/t^d \simeq T'/t^d = T'_d$ of Proposition 3.1, represented by the ordinary quotient. The comparison respects mapping cones and shifts, with the usual sign

isomorphism for a shifted cone. Induction on the finite cell filtration proves the equivalence for Y and identifies the tower projection with its intrinsic reduction map.

The groups $\pi_i Y_d$ are finite by the section calculation, and their first derived inverse limits vanish. Hence $\pi_i Y = \varprojlim_d \pi_i Y_d$ is profinite, and the natural equivalences just proved identify this topology with the intrinsic one. For a retract Y' of Y , naturality sends its defining derived idempotent to an idempotent e_d on every Y_d . Reduction preserves the retract, so Y'/t^d is the summand split by e_d ; this statement takes place in the derived category and requires no strict idempotent representative. On the finite group $\pi_i Y_d$, the map $\pi_i e_d$ is an ordinary idempotent. Taking its image commutes with the inverse limit, because the associated image and kernel decompositions are compatible with the transition maps. This gives the displayed formula for $\pi_i Y'$ and its intrinsic reduction topology. $\square$

3.5 Compact and discrete parts of the overlap

We have described the complete objects built from $\hat{T}'$ by finite algebraic constructions. We now study the topology of the overlap vector spaces. When an object has finite sections homotopy in every degree before completion, its open image occupies a discrete cocompact subgroup of the overlap. For completions of objects with this finiteness, the topology is intrinsic, and every map between them induces continuous maps on the overlap vector spaces.

Proposition 3.8 (Intrinsic overlap topology). *Every $P \in \text{Thick}(T')$ has finite $\pi_i P$ for every integer i . Now let $P \in \mathbf{D}(\mathcal{N})$ have finite $\pi_i P$ for every integer i . For each integer i , the group $\pi_i \hat{P}$ is naturally profinite, with the intrinsic inverse-limit topology from the finite groups $\pi_i(\hat{P}/t^d)$. The vector space $W_i(\hat{P}) = \pi_i L\hat{P}$ has a locally compact Hausdorff k -vector-space topology obtained as the increasing union of the compact open images of $\pi_{i+sD}\hat{P}$, for $s \geq 0$. The image of $\pi_i LP$ is discrete and cocompact in this space.*

If $Q \in \mathbf{D}(\mathcal{N})$ also has finite $\pi_j Q$ for every integer j , then every map $f : \hat{P} \rightarrow \hat{Q}$ induces continuous maps $\pi_i \hat{P} \rightarrow \pi_i \hat{Q}$ and $W_i(\hat{P}) \rightarrow W_i(\hat{Q})$ for every integer i , in the topologies just described. No global map $P \rightarrow Q$ inducing f is required. If $\hat{P} \simeq \mathcal{R}(\mathbf{P})$ for a perfect right dg $\mathcal{E}$ -module $\mathbf{P}$, then under this identification the natural identifications of Proposition 3.6 and Lemma 3.7 match the profinite topology on $\pi_i \hat{P}$ with the one obtained from the realization's finite-level sections and intrinsic reductions. Consequently they match the compact images and the increasing-union topology on $W_i(\hat{P})$.

Proof. For T' , flat affine adjunction identifies sections-homotopy with the cohomology of twists on $\mathcal{X}_k = \mathcal{P}_k(1, q)$. The two-variable principal-open Čech complex computes it, also in the twisted periodic category. Weight by weight, its H^0 is spanned by nonnegative monomials of that weight and its H^1 by Laurent monomials with both powers negative. Both sets are finite in a fixed weight because the two weights 1 and q are positive. Thus every $\pi_i T'$ is finite. Degreewise finiteness is preserved by shifts, finite sums, triangles, and retracts, proving the first assertion.

Now let P satisfy the stated degreewise finiteness hypothesis. The cofiber sequence for t^d makes every $\pi_i(P/t^d)$ finite. The Milnor sequence therefore gives

$$\pi_i \hat{P} = \varprojlim_d \pi_i(P/t^d), \quad (3.13)$$

with zero first derived-limit term. Give this group its compact Hausdorff inverse-limit topology. The equivalence $P/t^d \simeq \hat{P}/t^d$ is compatible with the projections: both maps $\hat{P} \rightarrow P/t^d$ agree after precomposition by $P \rightarrow \hat{P}$, and this determines a map uniquely because its fiber is t -invertible whereas the target is complete. This proves that (3.13) describes the intrinsic reduction topology.

The fracture square gives a long exact sequence with the map

$$\pi_i \hat{P} \oplus \pi_i LP \longrightarrow W_i(\hat{P}). \quad (3.14)$$

Its kernel is a quotient of $\pi_i P$ and its cokernel embeds in $\pi_{i-1} P$. In particular both are finite. The same holds in every shifted degree. Write $C_s = \pi_{i+sD} \hat{P}$ and let K_s be its image in $W_i(\hat{P})$ under the localization identifications. The kernel $C_s \rightarrow K_s$ is finite, by applying (3.14) in degree $i + sD$. Give K_s its compact Hausdorff quotient topology.

The maps $C_s \rightarrow C_{s+1}$ are continuous by naturality of the intrinsic reduction maps. Their cokernels are finite, by the multiplication-cofiber sequence and $\hat{P}/t \simeq P/t$. Thus the resulting inclusions $K_s \subset K_{s+1}$ are continuous and of finite index. A continuous injection from a compact space to a Hausdorff space is a closed embedding; here finite index also makes it open. By (3.3), the union of these compact open subgroups is exactly $W_i(\hat{P})$. Give the union its corresponding topology. This is a locally compact Hausdorff topological k -vector space.

Let B be the image of $\pi_i LP$. Its intersection with K_0 is finite, because a common image determines a pair in the finite kernel of (3.14). In the Hausdorff group K_0 , choose a neighborhood of zero avoiding the nonzero points of this finite intersection. It meets B only in zero, so B is discrete. A discrete subgroup of a Hausdorff topological group is closed. The quotient by $K_0 + B$ is finite, by the finite cokernel of (3.14); hence $W_i(\hat{P})/B$ is covered by finitely many translates of the compact image of K_0 . Thus B is cocompact.

Finally, let Q and $f : \hat{P} \rightarrow \hat{Q}$ be as in the statement. The map f induces maps of the intrinsic finite reduction groups, so it is continuous on the profinite homotopy groups. After localization its restriction to each compact stage is continuous, so it is continuous on the unions just constructed. No global map $P \rightarrow Q$ inducing f is needed. The finite-level section calculation of Proposition 3.6 and the natural reduction equivalences of Lemma 3.7 identify the profinite topologies for the perfect realization of $\hat{P}$ in the statement. These identifications commute with t , so they identify each compact image and the union topology after localization. This includes the case where P is a retract of a finite-cell module. $\square$

4 A complete operator with many defects

We now choose one endomorphism of $\hat{T}'$ from the operation algebra of Proposition 3.3, using finite matrix blocks on Laurent series to control its action on the overlap. The construction also keeps a prescribed reduction on the closed chart, which the obstruction argument will use after gluing. Integral interpolation measures from the group action let us control successive matrix depths; a choice of base- p digits then determines which rows and columns remain unmatched.

Retain the chart action of Proposition 2.5, the digit filtration of Lemma 2.7, and the complete operation algebra of Proposition 3.3. Thus

$$k = \mathbb{F}_{p^2}, \quad R = k[[x]], \quad \Delta = R[[H]], \quad \mathcal{E}_0 = \Delta^\Lambda.$$

The initial symbol of $U_3 = \delta_\gamma - \delta_1$ in the weight filtration of Proposition 2.8 is C ; the weights of x and U_3 are respectively 4 and 3. Here δ_g denotes a Dirac measure, and 1 denotes the identity of H .

Put

$$K_0 = k((x)), \quad w = x^{-q}, \quad \Phi = K_0^\Lambda = k((w^{-1})).$$

The subscript in K_0 refers to this Laurent field, not to Morava K -theory.

Give Φ its Laurent-series topology, with standard compact open subspace $k[[w^{-1}]]$. We seek one operator $a \in \mathcal{E}_0$ with closed image and two fixed families: nonzero kernel series in $k[[w^{-1}]]$

with distinct leading positions, and classes of positive monomials w^j . The leading position of a nonzero series is the greatest exponent of w occurring in it. For fixed constants $c > 0$ and $\beta > 1/3$, which may depend on p , and every sufficiently large integer N , at least cN^β leading positions from the first family should lie in $[-N, 0]$, while at least cN^β of the selected classes with $1 \leq j \leq N$ should be independent in

$$\Phi / (a\Phi + k[[w^{-1}]]) .$$

The first family lies in the compact subspace, while the second remains independent after quotienting the cokernel by the image of that subspace. Proposition 4.6 supplies matrix conditions, and Section 5 proves that they yield these analytic properties.

For $g \in H$ define

$$z_g = \frac{g(x)}{x}, \quad h_g = \frac{g(w)}{w} = z_g^{-q}, \quad u_g = h_g - 1. \quad (4.1)$$

Both z_g and h_g belong to $1 + xR$. Since q is prime to p ,

$$\ell(g) := \text{ord}_x(u_g) = \text{ord}_x(z_g - 1),$$

where the value ∞ is allowed. For $l \geq 1$ let

$$H_l = \{g \in H : \ell(g) \geq l\}, \quad N_l = [H : H_l].$$

These are the normal open congruence subgroups given by the action on R/x^{l+1} . Since the action preserves valuation, the distance between $z_g, z_{g'}$, and also between $u_g, u_{g'}$, has valuation $\ell(g^{-1}g')$.

4.1 Displacement and interpolation measures

The next bound relates the digit filtration of H to the rate at which distinct orbit points approach one another. It has two uses. Here it bounds the denominators of divided differences, so that interpolation produces measures in Δ . In the obstruction argument, the same estimate will bound joint operator kernels.

Lemma 4.1 (Displacement bound). *If $g \in H(i) \setminus H(i+1)$, then*

$$\ell(g) \leq r_i := \sum_{j=1}^i p^{\lfloor j/2 \rfloor}. \quad (4.2)$$

The action of H on R is faithful. With $r_0 = 0$, one has

$$\kappa := \sum_{l \geq 1} \frac{1}{N_l} \leq \frac{p}{p-1}. \quad (4.3)$$

Moreover, for every $s \geq 1$,

$$\Gamma_s \subset H_{p^s} \subset H(2s-1), \quad N_{p^s} \geq p^{3(s-1)}. \quad (4.4)$$

Proof. We first bound the displacement of an element from its first nonidentity digit. The comparison is made on torsion points of the deforming formal group, where the digit gives an exact valuation and the displacement gives a lower bound for the same expression.

Choose a strict representative of g whose first nonidentity digit is at index i , with coefficient $b \neq 0$. For even i , choose $b \notin \mathbb{F}_p$, using the quotient by the central subgroup. The chart construction gives a formal isomorphism

$$\tilde{g} : F_y \longrightarrow F_x, \quad y = g(x), \quad [p]_{F_x}(T) = xT^p + {}_{F_x}T^{p^2}.$$

Its coefficients lie in R : the grading change is effected by a $(p-1)$ -st root congruent to 1, followed by the compatible strict arrow. The reduction of $\tilde{g}(T) - {}_{F_x}T$ begins with bT^{p^i} .

Extend the valuation $v(x) = 1$ to a completed algebraic closure of K_0 . Formal group series over R can be evaluated on its positive-valuation ball. The zeros of $[p^d]_{F_x}$ in this ball form a cyclic group of order p^d , and every zero has multiplicity p^d . Indeed, Weierstrass preparation, using reduction $T^{p^{2d}}$, gives a polynomial of degree p^{2d} all of whose zeros reduce to zero; its unit factor evaluates to a unit. The order at zero over K_0 is p^d . Translation by a torsion point is an invertible formal coordinate change and preserves multiplicity, so the group of zeros has order p^d . Its subgroup killed by p has order p , which proves cyclicity.

A primitive level- d point ζ has valuation

$$\lambda_d = \frac{1}{(p-1)p^{2d-1}}.$$

For $d = 1$, the two summands of the p -series must have equal valuation. Inductively, to have $v([p]\zeta) = \lambda_{d-1}$, one must have $v(\zeta) < \lambda_1$; in that range the p^2 -power summand dominates. This proves the formula.

Fix i and take d sufficiently large. The residue of

$$\frac{\tilde{g}(\zeta) - {}_{F_x}\zeta}{\zeta^{p^i}}$$

is b : coefficients of lower powers of T lie in xR , and their valuations exceed $p^i \lambda_d$ for large d . The differences $\tau - {}_{F_x}\zeta$, as τ ranges over the level- d torsion points, are the multiples $[a]_{F_x}\zeta$ for $a \bmod p^d$. If $s = \nu_p(a) < d$, their valuation is $p^{2s} \lambda_d$; the zero multiple has infinite valuation. When $2s = i$, the residue after division by ζ^{p^i} is $a/p^s \bmod p \in \mathbb{F}_p^\times$, since $[p^s]_{F_0}(T) = T^{p^{2s}}$. It cannot cancel b .

Ordinary and formal differences have the same valuation, because $T -_{F_x} S$ is $T - S$ times a unit. Consequently

$$v(\tilde{g}(\zeta) - \tau) = p^{\min(i, 2s)} \lambda_d,$$

with $s = \infty$ for $\tau = \zeta$.

There are $(1 - 1/p)p^{d-s}$ multiples of exact order parameter $s < d$. Counting these factors with multiplicity p^d in the Weierstrass factorization gives

$$\begin{aligned} v([p^d]_{F_x}(\tilde{g}(\zeta))) &= p^{2d} \lambda_d \left((1 - 1/p) \sum_{0 \leq s < \lceil i/2 \rceil} p^s + p^{i - \lceil i/2 \rceil} \right) \\ &= \frac{p^{\lceil i/2 \rceil} + p^{\lfloor i/2 \rfloor + 1} - 1}{p - 1} = r_i + \frac{p}{p - 1}. \end{aligned} \quad (4.5)$$

On the other hand, substitution gives $F_y \equiv F_x \bmod x^{\ell(g)+1}$. Since $[p^d]_{F_x}(\zeta) = 0$, the value $[p^d]_{F_y}(\zeta)$ has valuation strictly greater than $\ell(g) + 1$ when $\ell(g)$ is finite. Its image under $\tilde{g}$ has the same valuation and equals $[p^d]_{F_x}(\tilde{g}(\zeta))$. Thus $\ell(g) < r_i + 1/(p - 1)$, which proves (4.2). If $y = x$, the latter value would instead be zero, contradicting (4.5). Since the digit filtration is separated, the action is faithful.

For $r_i < l \leq r_{i+1}$, the displacement bound implies

$$N_l \geq [H : H(i + 1)] = p^{2i - \lfloor i/2 \rfloor}.$$

The length of this interval is $p^{\lfloor (i+1)/2 \rfloor}$, so its contribution to κ is at most p^{-i} . Summing proves (4.3).

The inequality $r_{2s-2} < p^s$ proves the second inclusion in (4.4), and the digit indices give its index bound. For the first inclusion, every tangent group operator $g - 1$ raises valuation on R by at least one. In characteristic p , $g^{p^s} - 1 = (g - 1)^{p^s}$, so $g^{p^s}(x) \equiv x \bmod x^{p^s+1}$. Apply this to the ordered coordinates of Γ_s . $\square$

The balanced-node interpolation below adapts Bhargava's P -ordering and generalized-factorial framework; see [9, §4]. The particular orbit enumeration, divided-difference measures, and subsequent operator and ghost arguments are developed here.

To use the displacement estimate for interpolation, choose elements $b_j^H \in H$, $j \geq 0$, such that modulo every H_l their indices give compatible repeated enumerations of the N_l cosets: two indices give the same coset exactly when they are congruent modulo N_l . Such enumerations are obtained by refinement through the nested finite quotients and then compactness. We call this sequence *balanced*. Put

$$u_j = u_{b_j^H}, \quad r'_m = \sum_{l \geq 1} \left\lfloor \frac{m}{N_l} \right\rfloor \leq \kappa m. \quad (4.6)$$

Faithfulness implies $N_l \rightarrow \infty$, so each sum is finite and the points $u_0, \dots, u_m$ are distinct.

The interpolation below is a form of Amice's method of very well-distributed sequences [1, Sections 2.4 and 6.2]; we give the required estimates directly for this group action.

Lemma 4.2 (Integral divided differences). *For $m \geq 0$, the finite measure*

$$D_m = \sum_{i=0}^m \frac{x^{r'_m}}{\prod_{0 \leq j \leq m, j \neq i} (u_i - u_j)} \delta_{b_i^H} \quad (4.7)$$

belongs to Δ . Its moments satisfy

$$\int_H D_m(g) u_g^n = \begin{cases} 0, & 0 \leq n < m, \\ x^{r'_m}, & n = m, \end{cases} \quad \int_H D_m(g) u_g^n \in x^{r'_m+n-m} R \quad (n > m).$$

Proof. For a fixed $i \in \{0, \dots, m\}$, at most $\lfloor m/N_L \rfloor$ other indices are congruent to i modulo N_L . The distance formula following (4.1) therefore bounds the denominator's valuation by r'_m , proving integrality. By Lagrange interpolation, the divided difference of Z^n at $u_0, \dots, u_m$ is the coefficient of Z^m in its remainder modulo $\prod_{i=0}^m (Z - u_i)$. It is zero for $n < m$ and one for $n = m$. For $n > m$ it is a homogeneous polynomial of degree $n - m$ in the u_i . Since each $u_i \in xR$, the last bound follows. $\square$

Thus D_m kills the first m moments, prescribes the next one, and makes every later moment increasingly divisible by x . The bound $r'_m \leq \kappa m$ controls the cost of this prescription. We now use these measures to change one depth of the Laurent matrix at a time.

4.2 Control of triangular coefficients

We now impose the quantitative conditions on the prime. Choose an odd integer M such that

$$M \geq 81, \quad p > 16M^2, \quad 2\lfloor M/8 \rfloor > p^{1/3}. \quad (4.8)$$

These conditions hold simultaneously at every sufficiently large prime: take an odd integer M asymptotic to $p^{2/5}$. Indeed $M^2/p \rightarrow 0$ and $M/p^{1/3} \rightarrow \infty$. They are compatible with every earlier large-prime hypothesis. No particular numerical lower bound for p will be needed.

Put $\alpha = -1/2 \in \mathbb{Z}_p$ and $\delta = (p-1)/2$; every base- p digit of α is δ . We construct a by weighting a measure $\nu \in \Delta^\Lambda$ pointwise by h_g^α . Powers with a p -adic exponent are defined by the convergent binomial series in $1 + xR$. Weighting is Λ -equivariant because w is invariant. As throughout the operation calculus, measures and weighted integrals are evaluated at finite coefficient and group levels and then passed to the limit. For coefficient precision x^e , $e \geq 1$, choose $s \geq 1$ with $p^s \geq e$. Lemma 4.1 gives $\Gamma_s \subset H_{p^s} \subset H_e$, so the distance formula and the binomial series make $h_g^z \bmod x^e$ constant on cosets of Γ_s for every $z \in \mathbb{Z}_p$. Weighting therefore commutes with mass addition within this cofinal family of group quotients; coarser quotients receive the resulting pushforward. These finite values are also compatible under coefficient reduction. The resulting operator is continuous on Φ and preserves its pole filtration. It satisfies

$$a(w^j) = w^j \int_H \nu(g) h_g^{\alpha+j}.$$

The integral lies in $R^\Lambda = k[[x^q]]$. Hence the matrix is lower-going: a column indexed by j has nonzero rows only at indices at most j . We call $b = j - l$ the depth of the entry in column j and row l . For $b \geq 0$, the coefficient in column w^j and row w^{j-b} is

$$f_b(\alpha + j), \quad f_b(z) = [x^{qb}] \int_H \nu(g) h_g^z. \quad (4.9)$$

Write z_i for the base- p digits of $z \in \mathbb{Z}_p$. If $X = p^r \leq qb < pX$, then f_b and all smaller-depth coefficient functions depend only on $z \bmod pX$, since $h_g^{pX} \equiv 1 \bmod x^{qb+1}$.

Lemma 4.3 (Coefficient control). *Suppose (4.8) holds, $X = p^r \leq qb < pX$, and $b \geq X/(2M)$. By adding a measure in $(x^{X+1}\Delta)^\Lambda$ to ν , one may change $f_b(z)$ by any polynomial of degree at most M in z_r , independently for every lower label $z \bmod X$, without changing any coefficient function of smaller depth. Successive such corrections over increasing depths converge in Δ and do not change reduction modulo x .*

Proof. For $0 \leq m < (M+1)X$, use a scalar multiple over k of $x^{qb-r'_m}D_m$ and average under Λ . This averaging is defined because $|\Lambda| = q = p+1$ is invertible in k . By (4.3) and (4.8),

$$qb - r'_m \geq \left(\frac{q}{2M} - \kappa(M+1) \right) X > X. \quad (4.10)$$

The measure is therefore in $x^{X+1}\Delta$. This ideal is Λ -stable, so the averaged measure lies in $(x^{X+1}\Delta)^\Lambda$. Expand $h_g^z = (1+u_g)^z$. Lemma 4.2 shows that the correction vanishes in all coefficient orders below qb and changes the coefficient of x^{qb} by $\binom{z}{m} \bmod p$. Averaging preserves this coefficient, since x^{qb} is invariant and $h_{\lambda g \lambda^{-1}} = \lambda(h_g)$.

The characteristic- p identity

$$(1+v)^z = \prod_{i \geq 0} (1+v^{p^i})^{z_i}$$

expresses these binomial functions as products of digit binomials. For digits below r , their binomial bases span all functions. At digit r the indices $0, \dots, M$ span the polynomials of degree at most $M < p$. Every product of these bases occurs with $0 \leq m < (M+1)X$, proving the asserted independence. Each stage has only finitely many depths and lower labels, so its total correction lies in $(x^{X+1}\Delta)^\Lambda$, even when every depth is used. For a fixed $e \geq 1$, all stages $X \geq e$ vanish in every finite quotient $R_e \# (H/\Gamma_s)$ for which the action is defined. Thus the partial measures stabilize in every such quotient, uniformly in s at fixed e , and converge in Δ^Λ . The smaller-depth vanishing above separately makes every finite matrix block constant once its largest depth has been processed. Each correction vanishes modulo x . $\square$

The coefficient-control lemma leaves the reduction of the measure unchanged. We choose that reduction now and also make the early matrix depths vanish. This gives a common starting point for the later pivot decisions.

Lemma 4.4 (Initialization). *There is a measure $\nu \in \Delta^\Lambda$ with reduction $\delta_\gamma - \delta_1$ such that*

$$\int_H \nu(g) u_g^n = 0 \quad (0 \leq n < p^2).$$

For this initial measure, $\int_H \nu(g) h_g^z \in x^{p^2}R$ for every $z \in \mathbb{Z}_p$. After any convergent sequence of the corrections in Lemma 4.3, the weighted operator a satisfies

$$a \bmod x = \delta_\gamma - \delta_1 = U_3, \quad \sigma(a) = C. \quad (4.11)$$

Proof. Lemma 4.1 gives $N_2 \geq p^2$. The residues of u_g/x lie in k , and distinct classes in H/H_2 give distinct residues. Hence also $N_2 \leq |k| = p^2$. The first p^2 balanced points u_j/x consequently have distinct residues. Moreover u_γ/x has zero residue: γ commutes with Λ , so h_γ is invariant, and its positive x -exponents are divisible by $q > 1$.

The measure $\delta_\gamma - \delta_1$ integrates every power $(u/x)^n$, $0 \leq n < p^2$, to zero modulo x . The Vandermonde matrix at the first p^2 points u_j/x is invertible over R . Solving this finite system gives

corrections at those points with coefficients in xR that kill these moments exactly. Average the resulting measure under Λ . Vanishing of each moment is preserved, as is the reduction because γ commutes with Λ . Expanding h_g^z now proves the coefficient bound. The later corrections preserve the reduction, and weighting by $h_g^\alpha \equiv 1 \pmod{x}$ does too. Finally, x has weight 4, whereas U_3 has weight 3 and initial symbol C ; this proves (4.11). $\square$

We have obtained an actual complete operation with fixed initial symbol, together with freedom to change later matrix coefficients. It remains to spend that freedom on the two rank properties needed for gluing. The next lemma describes exactly when a new coefficient creates a pivot; the digit construction following it will leave many rows and columns unmatched at all large scales.

4.3 Ranks and pivot decisions

Let $A = (A_{l,j})_{l,j \in \mathbb{Z}}$ be a lower-going triangular matrix, so $A_{l,j} = 0$ for $l > j$. For $l \leq j$, write $R(l, j)$ for the rank of its finite block on row and column indices from l to j , and put $R(l, j) = 0$ when $l > j$. Equivalently, this is the rank on rows at least l and columns at most j : triangularity makes all remaining entries in that range zero.

Lemma 4.5 (Rank matching). *For such a matrix, the numbers*

$$R(l, j) - R(l, j - 1) - R(l + 1, j) + R(l + 1, j - 1) \quad (4.12)$$

are 0 or 1 and define a partial matching of rows with columns. The rank $R(l, j)$ counts the matched pairs in its block. When entries are fixed in order of increasing depth $b = j - l$, a new match at (l, j) is possible exactly when both its row and column are still unmatched and the corner entry avoids one specified scalar. That scalar is determined by smaller-depth entries. Decisions at a fixed depth are independent of one another.

Proof. For fixed j , put $d_j(l) = R(l, j) - R(l, j - 1)$. Adding one column changes the rank by 0 or 1. If column j is independent of the earlier columns on rows at least $l + 1$, it remains independent after row l is added: the newly admitted earlier column l is zero on the old rows. Thus $d_j(l) \geq d_j(l + 1)$. The difference $d_j(l) - d_j(l + 1)$ is therefore 0 or 1, with at most one nonzero value in each column. The analogous argument for row increments gives at most one in each row. Telescoping over the finite block proves that $R(l, j)$ counts its matched pairs.

Both corner indices are still unmatched precisely when

$$R(l, j - 1) = R(l + 1, j - 1) = R(l + 1, j).$$

To see the exceptional corner value explicitly, permute the corner column to the first position and write the block as

$$\begin{pmatrix} c & r \\ u & B \end{pmatrix}.$$

The displayed rank equalities say that $r = \rho B$ and $u = B\eta$ for some row vector ρ and column vector η . Row and column operations reduce the block to B together with the single entry $c - \rho B\eta$. The value $\rho B\eta$ is independent of the choices: changing ρ by a vector annihilating B , or changing η by a vector annihilated by B , leaves it fixed. Exactly this value of c keeps the old rank; every other value creates a new match. All entries of r, u, B have smaller depth, which also proves the independence of decisions at the current depth. $\square$

At a depth b with $X = p^r \leq qb < pX$, every coefficient of the matrix in (4.9) used in the current rank tests depends only on the column label modulo pX . Translating both indices by pX therefore preserves each finite block used in a test. All eligibility tests and their exceptional corner values are periodic in j with that period. Fixing a lower label $(\alpha + j) \bmod X$ leaves p replicas, indexed by the next digit z_r . At any allowed depth, Lemma 4.3 therefore permits the following rule:

Prescribe no pivot at any chosen set of at most M eligible next digits, and prescribe a pivot at every other eligible next digit.

Indeed, choose a polynomial correction of degree at most M whose value at each chosen digit is the exceptional corner value minus the current coefficient. Because at most M values have been prescribed, the solution space has positive dimension. At any other eligible digit, equality of the corrected entry with its exceptional corner value cuts out a proper affine hyperplane in this space: a polynomial vanishing at the prescribed digits need not vanish at the additional one. There are at most p such hyperplanes, each containing at most a fraction $1/p^2$ of the space, so their union cannot cover it. The rule is applied independently at each lower label; it does not distinguish replicas at still higher digits.

Define center digits and a central set of their indices by

$$\begin{aligned} d_i &= \left\lfloor \frac{(i + 1/2)p}{M} \right\rfloor \quad (0 \leq i < M), \\ s_0 &= (M - 1)/2, \quad d_{s_0} = \delta, \quad R_0 = \lfloor M/8 \rfloor, \\ I &= \{s_0 - R_0 + 1, \dots, s_0 + R_0\}, \quad B_0 = |I| = 2R_0, \\ \beta &= \log_p B_0 > 1/3. \end{aligned} \tag{4.13}$$

The cyclic positive gaps between the d_i are $p/M \pm 1$. The set I meets neither endpoint $0, M - 1$. For an index set S , write $d(S) = \{d_i : i \in S\}$; in particular, $I - 1 = \{i - 1 : i \in I\}$ is again a set of center indices.

The central digits serve two related purposes. Their repeated choices will give B_0 surviving possibilities at each new digit, which produces the exponent β . At sufficiently large stages, the digit δ will be retained for every possible pivot crossing a fixed pole bound. The next proposition proves these assertions for the resulting matrix, including the uniform conclusion for every large cutoff.

Proposition 4.6 (Operator with persistent defects). *Under (4.8), there is $a \in \mathcal{E}_0$ satisfying (4.11) whose triangular matrix on Φ has the following properties.*

1. *For every fixed $J \in \mathbb{Z}$, only finitely many matched pairs (l, j) satisfy $l \leq J < j$.*
2. *There are constants $c > 0$ and N_0 such that for every integer $N \geq N_0$, at least cN^β columns in $[-N, 0]$ and at least cN^β rows in $(0, N]$ are forever unmatched.*

Here the matching is that of Lemma 4.5; the constants may depend on p .

Proof. We begin at one initial depth and then proceed by stages. At each subsequent stage, adjacent clusters of lower labels are paired, and only specified central clusters retain next-digit replicas. We will record the surviving labels exactly. Once the stages have defined the operator, that record will both exclude late crossings of a fixed bound and count the unmatched labels at every large scale.

Give every row and column index j the label $z = \alpha + j$. Begin with

$$d = \lfloor p/M \rfloor.$$

All entries of smaller depth are zero by initialization, since $qd < p^2$. This depth belongs to the $X = p$ stage and satisfies $d \geq p/(2M)$. Match every column except those with

$$z_0 = \delta, \quad z_1 \in \{d_0, \dots, d_{M-1}\}.$$

The pivot rule permits this: at the one specified lower label protect M digits, and at all other lower labels protect none. The corresponding surviving rows have

$$z_0 = \delta - d, \quad z_1 \in \{d_0, \dots, d_{M-1}\},$$

without a borrow in subtraction. These statuses persist until the end of $qb < p^2$. Surviving lower labels could meet again only at a depth congruent to d modulo p , and no second such depth occurs in this range.

For the induction, take $X = p^r$, $r \geq 2$, and suppose the surviving labels modulo X are exactly

	z_0	$z_j \ (1 \leq j \leq r-2)$	z_{r-1}	
columns	δ	$d(I)$	$d_i \ (0 \leq i < M)$	(4.14)
rows	$\delta - d$	$d(I-1)$	$d_i \ (0 \leq i < M)$.	

All choices displayed are independent, and each has all p values of the new digit z_r . Group these lower labels into clusters indexed by i , according to their last digit d_i .

The first possible band of edges in the range $X \leq qb < pX$ consists of all pairings from column cluster i to row cluster $i-1$, with cluster indices taken cyclically. Their depths lie in

$$X(1/M - 2/p) \leq b \leq X(1/M + 2/p), \quad b \geq X/(2M). \quad (4.15)$$

To verify this, the allowed depth interval is $[X/q, X - X/q)$ because $q = p + 1$. The portions of a pair of lower labels below digit $r-1$ differ by at most

$$(p/4 + 3)X/p^2.$$

For $r = 2$ this is immediate. For $r > 2$, their digits in position $r-2$ have center indices in $I \cup (I-1)$, of diameter at most $2R_0$, while all lower contributions have absolute value less than X/p^2 . Thus same-cluster differences and their cyclic complements fall outside the allowed depth interval. For distinct clusters separated by k_1 cyclic steps, $1 \leq k_1 < M$, the center difference is $k_1 p/M \pm 1$. Adding the lower portions puts the depth in $X(k_1/M \pm 2/p)$. The bounds on p separate the $k_1 = 1$ band from all later bands and give (4.15). Every pair of lower labels in the adjacent clusters occurs there.

Process this first band by increasing depth. Call a lower label *full* if all its p next-digit replicas remain. When an edge has two full endpoints, match them at every next digit, except that for column cluster $i \in I$ retain the M digits

$$z_r \in \{d_0, \dots, d_{M-1}\}. \quad (4.16)$$

In this case there is no cyclic wrap to the row cluster: the depth is the positive difference of the ordinary lower representatives. Consequently the retained next digits on the two sides agree. Mark both endpoints no longer full. Whenever either endpoint is not full, there are at most M eligible

next digits, since the next-digit replicas of the two endpoints correspond bijectively at a fixed depth. Require no pivot at all of them. These are precisely the choices allowed by the pivot rule. At one depth, different column lower labels have different row lower labels, so the prescriptions are compatible.

Every full label is paired full-to-full once. An edge with a nonfull endpoint creates no pivots, so it cannot remove any replica from an endpoint that is still full. Full status can only be lost, never regained. For each pair of adjacent clusters, the possible edges form a complete bipartite graph whose sides both have B_0^{r-2} lower labels. If a full label were left over on one side, equal numbers removed from the two sides would leave one on the other side too. Their edge would already have been processed while both endpoints were still full, a contradiction. After this band, every label has at most M surviving next-digit replicas. At every later depth in the stage, require no further eligible pivots. Thus precisely the column cluster indices I and the row cluster indices $I - 1$ remain, with the full center set at the next digit. This is (4.14) for $r + 1$.

Figure 1 records these two features of the induction: possible pairings run from a column cluster to the preceding row cluster, and central clusters preserve the center digits. The coefficient corrections converge by Lemma 4.3, so the stages define an actual operator a satisfying (4.11).

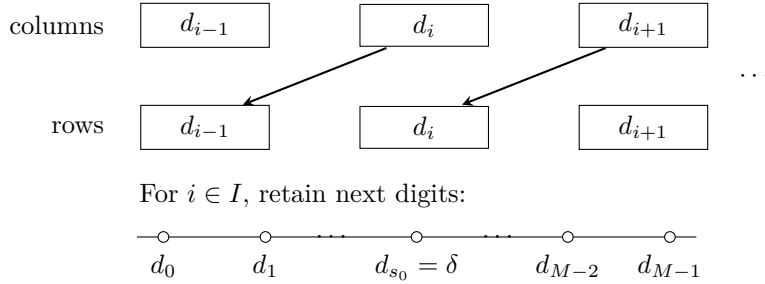


Figure 1: One matching stage. Arrows show the first band of possible pairings from a column cluster to the preceding row cluster; each indicated cluster pair contains all lower-label pairs. In a retained central cluster $i \in I$, the marked next digits have no pivot. The spacings are schematic. The fixed-cut argument below uses the protected digit $\delta = d_{s_0}$ to rule out crossings at sufficiently large stages.

Fix now $J \in \mathbb{Z}$. At a sufficiently large stage, a possible matched pair crossing J would have column index

$$J < j \leq J + X(1/M + 2/p),$$

since new matches occur only in the first band. Its lower label is

$$(\alpha + j) \bmod X = (X - 1)/2 + j$$

without wrap, and its next digit is δ . If the lower label is eligible, its cluster index lies in I . In fact its representative divided by X differs from $1/2$ by at most $1/M + 2/p + o(1)$; if its last digit is d_i , the same representative differs from $(i + 1/2)/M$ by at most $1/p$. Hence i is within three places of s_0 for large X , which is inside I since $M \geq 81$. But the next digit $\delta = d_{s_0}$ was explicitly retained there. Such a pair cannot be matched. The remaining finitely many depths each have only finitely many pairs crossing J . This proves the first assertion.

For the second assertion, use labels with the prescribed units digit and all later digits in $d(I)$ for columns or $d(I - 1)$ for rows, eventually all equal to δ . These survive at every stage. Changing finitely many digits of α gives an integer offset, so each such label is $\alpha + j$ for a genuine integer

row or column index j . At a highest modified position $r \geq 1$, choose a center digit strictly below δ for a column or strictly above it for a row. Both sets of center digits contain such choices. The change at position r has absolute value at least p^r , while all lower changes together have absolute value less than p^r , so this determines the appropriate sign of the offset. The remaining $r - 1$ intermediate digits can be chosen freely, giving at least B_0^{r-1} distinct offsets of absolute value less than p^{r+1} . For $p^{r+1} \leq N < p^{r+2}$, these offsets lie in the required intervals and

$$B_0^{r-1} = p^{\beta(r-1)} \geq p^{-3\beta} N^\beta.$$

Reducing the constant if necessary to work on both sides proves the assertion for every sufficiently large integer N , including cutoffs between successive powers of p . $\square$

5 From pivots to a gluing lattice

Fix the operator a of Proposition 4.6. We first pass from its finite-block pivots to actual kernel and cokernel vectors in Φ . The finite-crossing property makes the resulting quotient Hausdorff and controls its compact open subspaces. We then pair the two sorts of defects in a discrete lattice and glue it to the cone of a . The resulting object will satisfy the first two conditions of the transfer criterion; exclusion from $\text{Thick}(T')$ will be the next section's task.

For $J \in \mathbb{Z}$, put

$$F_J \Phi = w^J k[[w^{-1}]], \quad \mathcal{Q} = \Phi / a\Phi, \quad \mathcal{K} = \ker(a : \Phi \longrightarrow \Phi). \quad (5.1)$$

The $F_J \Phi$ are compact open subspaces for the Laurent-series topology, with $\Phi = \bigcup_J F_J \Phi$ and $\bigcap_J F_J \Phi = 0$. Give $\mathcal{K}$ the subspace topology and $\mathcal{Q}$ the quotient topology. The *leading position* of a nonzero series is its greatest exponent of w .

The operator a is continuous and preserves every $F_J \Phi$. Consequently, for $L \leq J$, its matrix block on indices $L, \dots, J$ is the induced map on the finite-dimensional space $F_J \Phi / F_{L-1} \Phi$. This remains true for inputs with infinitely many nonzero coefficients below L : their tails contribute nothing to the displayed rows. The rank of this block is the $R(L, J)$ used in Lemma 4.5.

Lemma 5.1 (Pivots and the pole topology). *An integer j is an unmatched column of a if and only if there is a vector in $\mathcal{K}$ with leading term w^j . An integer l is a matched row if and only if it is the leading position of a nonzero vector in $a\Phi$. Moreover, for every $J \in \mathbb{Z}$,*

$$\dim_k \frac{a\Phi \cap F_J \Phi}{a(F_J \Phi)} = \#\{(l, j) \text{ matched} : l \leq J < j\} < \infty. \quad (5.2)$$

The image $a\Phi$ is closed. Thus $\mathcal{Q}$ and $\mathcal{K}$ are Hausdorff locally compact k -vector spaces, with compact open subspaces given respectively by the images of $F_J \Phi$ and by $\mathcal{K} \cap F_J \Phi$.

Proof. Suppose column j is unmatched. For every lower cutoff $L \leq j$, its rank increment in the block $[L, j]$ is zero. Hence

$$S_L = \{f \in w^j + F_{j-1} \Phi : a(f) \in F_{L-1} \Phi\}$$

is nonempty: a solution of the finite coefficient equations extends by a zero tail. Continuity of a makes S_L closed in the compact space $w^j + F_{j-1} \Phi$. Lowering L adds equations, so these sets are nested. Compactness gives a point in their intersection, and its image lies in $\bigcap_L F_{L-1} \Phi = 0$.

This is an actual kernel vector with leading term w^j . Conversely, such a vector makes column j dependent in every finite block, so the column is unmatched. In particular,

$$\dim_k \frac{a(F_j\Phi)}{a(F_{j-1}\Phi)} = \begin{cases} 1, & \text{if column } j \text{ is matched,} \\ 0, & \text{if column } j \text{ is unmatched.} \end{cases} \quad (5.3)$$

Indeed, $F_j\Phi/F_{j-1}\Phi$ is generated by w^j , and $a(w^j) \in a(F_{j-1}\Phi)$ is equivalent to the existence of the kernel vector just constructed.

If row l is matched, then at some finite column cutoff its addition increases rank. A finite column combination therefore has nonzero coefficient in that row and zero coefficients above it. The combination is an actual Laurent polynomial input, and triangularity makes all rows above the finite block vanish too. Conversely, suppose $a(f)$ has leading position l . Write $f = f_{\geq l} + f_{< l}$, where $f_{\geq l}$ is a Laurent polynomial and $f_{< l} \in F_{l-1}\Phi$. Since a preserves $F_{l-1}\Phi$, the infinite tail contributes nothing on rows at least l . The finite polynomial $f_{\geq l}$ therefore exhibits the same row increment in a finite block, so row l is matched.

For $J' > J$, equation (5.3) shows that the dimension of $a(F_{J'}\Phi)/a(F_J\Phi)$ is the number of matched columns in $(J, J']$. The map from this quotient to $F_{J'}\Phi/F_J\Phi$ induced by projection has rank $R(J+1, J')$: the images of input terms at or below J lie in $F_J\Phi$ and vanish under the projection. The kernel of this induced map is

$$\frac{a(F_{J'}\Phi) \cap F_J\Phi}{a(F_J\Phi)}.$$

Subtracting the projected rank therefore counts exactly the matches with $l \leq J < j \leq J'$. Every Laurent input belongs to some $F_{J'}\Phi$, so taking the increasing union over J' proves (5.2). This calculation precedes any closedness assertion. The crossing count is finite by Proposition 4.6.

The image $a(F_J\Phi)$ is compact by continuity. The quotient in (5.2) is finite-dimensional over the finite field k , hence is a finite set. Thus $a\Phi \cap F_J\Phi$ is a finite union of translates of $a(F_J\Phi)$, and is compact and closed. The $F_J\Phi$ form an open cover of Φ , so $a\Phi$ is closed in Φ . The quotient is Hausdorff, and the quotient map is open, proving the asserted compact open subspaces of $\mathcal{Q}$. The kernel is closed by continuity, and its intersections with the compact open stages give the assertion for $\mathcal{K}$. $\square$

5.1 The complete cone and its overlap

In the category of complete objects, or equivalently in the perfect $\mathcal{E}$ -module calculus of Proposition 3.5, define

$$m = \text{cofib}(a : \widehat{T}' \longrightarrow \widehat{T}'), \quad Y = \bigoplus_{r=0}^{D-1} m[r]. \quad (5.4)$$

Recall that $W_i(Y) = \pi_i LY$ denotes the normalized overlap. The finite sum over all residues ensures that both kinds of defect occur in every degree.

Lemma 5.2 (Compact stages of the cone). *For every degree i , the normalized overlap has an identification*

$$W_i(Y) = \mathcal{Q} \oplus \mathcal{K}, \quad (5.5)$$

compatible with the t -period identifications. The map $\pi_i Y \longrightarrow W_i(Y)$ has finite kernel and compact open image. Each such image is commensurable with

$$C_0^{\text{lat}} = \text{im}_{\mathcal{Q}}(F_0\Phi) \oplus (\mathcal{K} \cap F_0\Phi). \quad (5.6)$$

The topology of $\pi_i Y$ is the intrinsic compact topology from its finite reductions, and the induced topology on the overlap is the locally compact topology in (5.5). Every fixed compact stage of this overlap is contained in the image of one sufficiently late t -translated group $\pi_{i+sD} Y$.

Proof. The overlap of $\hat{T}'$ is supported in degrees divisible by D , with normalized coefficient space Φ and the action of a used above. Since $D > 2$, the cofiber calculation in each summand has separate cokernel and kernel residues. In degree i , the cokernel component comes from $m[r]$ with $i \equiv r \pmod{D}$, by the cofiber inclusion from $\hat{T}'[r]$. The kernel component comes from $m[r']$ with $i \equiv r' + 1 \pmod{D}$, by the connecting map into $\hat{T}'[r' + 1]$. Summing gives (5.5) and its periodic compatibility.

For complete homotopy, Proposition 3.6 replaces Φ by the compact coefficient stages

$$(x^{-s}R)^\Lambda = F_{[s/q]} \Phi.$$

The two components may use different values of s and hence of J ; both increase without bound under positive t -translation. On a cokernel component, the kernel of the map from the stage cokernel to $\mathcal{Q}$ is exactly $(a\Phi \cap F_J\Phi)/a(F_J\Phi)$, finite by Lemma 5.1; its image is the compact open image of $F_J\Phi$. On a kernel component, the stage gives simply $\mathcal{K} \cap F_J\Phi$, injecting into $\mathcal{K}$. This proves the finite-kernel and compact-open claims. Any two compact open subspaces here are commensurable: their intersection is open in each compact space, so the two quotient spaces are finite.

By Lemma 3.7, the kernel and cokernel calculations at a fixed compact stage are inverse limits of the corresponding finite-level calculations. These inverse systems of finite groups satisfy the Mittag-Leffler condition: at each fixed level, their decreasing sequences of images stabilize. Their inverse limits are therefore exact, and the resulting compact topology is the intrinsic topology from reductions. Taking the t -colimit recovers the same locally compact union topology on $\mathcal{Q} \oplus \mathcal{K}$. Finally the coefficient stages increase without bound under positive t -translation. Choosing a single sufficiently large stage for both components proves the last assertion. $\square$

5.2 A lattice pairing the defects

We choose the open part by constructing, in each degree, a discrete complement to the compact open subspace $C_0^{\text{lat}} \subset \mathcal{Q} \oplus \mathcal{K}$. Such a complement can be obtained by lifting an algebraic basis of $(\mathcal{Q} \oplus \mathcal{K})/C_0^{\text{lat}}$ and taking its span. Adding a vector of $\mathcal{K} \cap F_0\Phi$ to the kernel coordinate of a lift does not change its quotient class. We use this freedom to attach the distinguished compact kernel vectors to independent polar cokernel classes.

For each forever unmatched positive row j , let $q_j \in \mathcal{Q}$ be the class of w^j . For each forever unmatched nonpositive column l , choose $k_l \in \mathcal{K}$ with leading term w^l , as supplied by Lemma 5.1. In particular $k_l \in \mathcal{K} \cap F_0\Phi \subset R$. Pair these lists in order of increasing j and increasing $|l|$; write (j, l) for a resulting pair of an unmatched row and an unmatched column. Since $(0, k_l) \in C_0^{\text{lat}}$, the pair (q_j, k_l) and the vector $(q_j, 0)$ have the same image in the quotient. The next lemma verifies the independence of these quotient classes and records simultaneous bounds on their leading positions.

Lemma 5.3 (Simultaneous defect pairs). *The vectors $(q_j, k_l) \in \mathcal{Q} \oplus \mathcal{K}$, indexed by the paired lists, are linearly independent modulo C_0^{lat} . For every sufficiently large integer N , at least cN^β pairs, for a fixed $c > 0$, simultaneously satisfy $j \leq N$ and $|l| \leq N$. For these pairs, their kernel coordinates are linearly independent modulo $x^{qN+1}R$.*

Proof. The q_j are independent modulo $\text{im}_{\mathcal{Q}}(F_0\Phi)$. Otherwise a finite nontrivial relation, after subtracting a vector in $F_0\Phi$, would be an actual image of a whose leading position is an unmatched positive row. This contradicts Lemma 5.1. It also proves independence of the pairs modulo C_0^{lat} .

Let $n_+(N)$ count the positive unmatched rows at most N , and let $n_-(N)$ count the nonpositive unmatched columns of absolute value at most N . In the ordered pairing, the first condition holds precisely for the first $n_+(N)$ pairs and the second for the first $n_-(N)$ pairs. Thus exactly $\min\{n_+(N), n_-(N)\}$ pairs satisfy both conditions. Proposition 4.6 bounds each count below by a constant times N^β for every sufficiently large N , giving the simultaneous bound. The leading terms of their kernel vectors are the distinct powers $w^l = x^{q|l|}$, with $q|l| \leq qN$. A nonzero combination has one of these as its lowest x -power, so cannot lie in $x^{qN+1}R$. $\square$

In every period residue, extend the independent images of the pairs to an algebraic basis of $(\mathcal{Q} \oplus \mathcal{K})/C_0^{\text{lat}}$. Choose lifts retaining the specified pairs, and let $\mathcal{L}_i$ be their span. The projection of this algebraic span to the quotient is an isomorphism. It therefore meets C_0^{lat} trivially and, together with C_0^{lat} , fills the whole space. Since C_0^{lat} is open, the span is discrete. Each open coset of C_0^{lat} contains exactly one point of the span, so the span is also closed. Finally, the compact space C_0^{lat} maps onto the quotient by the span, proving cocompactness. Make the same choice along each t -period.

By Lemma 3.2, the t -open category is the split category of D -periodic complexes of k -vector spaces. There is consequently an open object V , also regarded as pushed forward to $\mathbf{D}(\mathcal{N})$, and a map $V \rightarrow LY$ whose maps on homotopy are precisely these lattice inclusions. Define

$$U = Y \times_{LY} V \quad \text{in } \mathbf{D}(\mathcal{N}), \quad (5.7)$$

where the pullback is a homotopy pullback.

Proposition 5.4 (The glued object). *The object U of (5.7) satisfies*

$$\pi_i U \text{ is finite for every } i, \quad \widehat{U} \simeq Y \in \text{Thick}(\widehat{T}'), \quad LU \simeq V.$$

Under these equivalences the structure map $LU \rightarrow L\widehat{U}$ is the chosen lattice map.

Proof. By Lemma 5.2, the map from $\pi_i Y$ to $\pi_i LY$ has finite kernel and compact open image; call this image C_i . It is commensurable with C_0^{lat} , and $\pi_i V = \mathcal{L}_i$ is a discrete complement to C_0^{lat} . Consider the joint map

$$\pi_i Y \oplus \pi_i V \rightarrow \pi_i LY.$$

After dividing its kernel by the finite kernel of $\pi_i Y \rightarrow C_i$, one obtains $C_i \cap \mathcal{L}_i$. This intersection injects into $C_i/(C_i \cap C_0^{\text{lat}})$, which is finite. The cokernel of the joint map is a quotient of $C_0^{\text{lat}}/(C_i \cap C_0^{\text{lat}})$, also finite. Thus the joint map has finite kernel and cokernel in every degree. The pullback long exact sequence expresses $\pi_i U$ as an extension of the kernel in degree i by the cokernel in degree $i+1$. Both are finite, so $\pi_i U$ is finite.

Both V and LY are t -invertible and hence have zero completion. Exactness of completion and completeness of Y give $\widehat{U} \simeq Y$. Localizing the pullback instead gives $LU \simeq V$, since the localization of $Y \rightarrow LY$ is an equivalence. These identifications preserve the structure map. The object Y is a finite sum of shifts of a two-cell cone on $\widehat{T}'$, so lies in $\text{Thick}(\widehat{T}')$. $\square$

The construction has now supplied the finite homotopy and complete thick-membership parts of Proposition 8.1. The following consequence of the chosen lattice is the restriction that will obstruct global thick membership.

Lemma 5.5 (Constraint on global maps). *For every map $g : U \rightarrow P$ with $P \in \text{Thick}(T')$, the induced map*

$$W_i(\widehat{g}) : \mathcal{Q} \oplus \mathcal{K} \longrightarrow W_i(\widehat{P})$$

is continuous and sends each paired vector (q_j, k_l) into the discrete lattice $\text{im}(\pi_i LP \rightarrow W_i(\widehat{P}))$.

Proof. Objects of $\text{Thick}(T')$ have finite homotopy in each degree. The same is true of U by Proposition 5.4. Proposition 3.8 therefore makes every map between their completions continuous on overlap homotopy. Under Lemma 5.2, the source topology is precisely the locally compact topology on $\mathcal{Q} \oplus \mathcal{K}$ used here.

The additional lattice assertion uses that $\widehat{g}$ comes from the global map g . Naturality of the fracture square gives a commutative square from $LU \rightarrow L\widehat{U}$ to $LP \rightarrow L\widehat{P}$. Each paired vector is represented in $\pi_i LU = \pi_i V = \mathcal{L}_i$. Applying $\pi_i Lg$ and then the target structure map places its image in $\text{im}(\pi_i LP \rightarrow W_i(\widehat{P}))$, which is the discrete lattice provided by Proposition 3.8. This argument uses the representative in $\pi_i LU$; an individual pair need not lift to $\pi_i U$. $\square$

To see how the pairing will be used, choose a compact open target subspace containing the images of the kernel coordinates. Modulo this subspace, each paired image is its polar image. Since all paired images lie in a discrete lattice, the part of their span inside the compact subspace has finite dimension. Thus a rank bound for the polar images controls the rank of the paired images up to a fixed finite error. The next section obtains the required bound under its homology-vanishing hypothesis.

6 The obstruction to a global finite construction

The object U of Proposition 5.4 has finite homotopy groups in every degree, and its completion is the finite construction

$$Y = \bigoplus_{r=0}^{D-1} m[r], \quad m = \text{cofib}(a : \widehat{T}' \longrightarrow \widehat{T}').$$

The remaining condition in the transfer criterion is $U \notin \text{Thick}(T')$. We must rule out every finite tower of shifts of T' and every retract of such a tower. Ruling out a retract of a finite sum of shifts would still leave longer towers untreated.

We first isolate the finite-tower argument. Its input is an exact functor to perfect modules over a periodic algebra over a domain with principal left and right ideals. The criterion uses ordinary homology and the homology of module duals to pass backward through any finite tower: the dual test removes the free part of a tower tail, and the ordinary test then uses torsion to force vanishing. We next construct a flat extension of the operation coefficients for which Y has nonzero torsion homology. The final part verifies the relation between the two tests for maps that come from global objects.

6.1 A criterion for arbitrary finite constructions

We state the two tests abstractly before constructing the coefficient extension that will make them apply to the operator a and the gluing lattice. Torsion homology by itself cannot be the obstruction: the cone of $2 : \mathbb{Z} \rightarrow \mathbb{Z}$ is a two-cell object with homology $\mathbb{Z}/2$. The additional information will restrict the maps that exist in the original category.

Let $\mathcal{B}_0 = S_0[t^{\pm 1}]$, where S_0 is a domain with principal left and right ideals, t is central of positive even homological degree D_0 , and the differential is zero. Write $\text{Perf}(\mathcal{B}_0)$ for the full subcategory of the derived category consisting of perfect right dg $\mathcal{B}_0$ -modules. For a right module over a domain, we call an element *torsion* if it is annihilated on the right by a nonzero ring element. In the principal domains used below the Ore property makes these elements a submodule. A torsion module has no nonzero map to a free module: the image of a torsion element would be annihilated by a nonzero element of the domain.

For $M \in \text{Perf}(\mathcal{B}_0)$ its module dual is the left dg module

$$M^* = R\text{Hom}_{\mathcal{B}_0}(M, \mathcal{B}_0).$$

A map $f : M \rightarrow N$ induces $H(f) : H(M) \rightarrow H(N)$ and $H(f^*) : H(N^*) \rightarrow H(M^*)$. The dual test has the opposite direction, and in both tests vanishing means zero in every degree.

Testing finite constructions by maps invisible to homology is part of the theory of ghosts and projective classes developed by Christensen [11, Theorem 3.5 and Section 8]. The following criterion adds a relation between the two tests for maps in the original category. Its proof is included.

Theorem 6.1 (A finite-tower criterion). *Let $\mathcal{C}$ be an idempotent-complete triangulated category, let $T_0, U_0 \in \mathcal{C}$, and let*

$$\Psi : \mathcal{C} \longrightarrow \text{Perf}(\mathcal{B}_0)$$

be an exact functor with $\Psi(T_0) \simeq \mathcal{B}_0$. Assume that $H(\Psi(U_0))$ is torsion as a right S_0 -module in every degree and is nonzero in at least one degree. Assume also that, for every $P \in \text{Thick}_{\mathcal{C}}(T_0)$ and every map $g : U_0 \rightarrow P$ in $\mathcal{C}$,

$$H(\Psi(g)) = 0 \implies H((\Psi(g))^*) = 0. \quad (6.1)$$

Then $U_0 \notin \text{Thick}_{\mathcal{C}}(T_0)$.

Here $\text{Thick}_{\mathcal{C}}(T_0)$ is the ordinary thick closure, using isomorphisms, finite sums, all integer shifts, triangles and retracts. The quantifier in (6.1) concerns maps in $\mathcal{C}$. In our application, their global origin is precisely what imposes a constraint on the gluing lattice.

The proof uses two algebraic facts. The first separates free homology from torsion homology over the coefficient ring.

Lemma 6.2 (Periodic splitting). *Let S_0 be a domain whose left and right ideals are principal, and let D_0 be a positive even integer. Every finitely generated right S_0 -module is a direct sum of a torsion module and a finite free module and has a finite free resolution of length at most one. Every perfect right dg module over $S_0[t^{\pm 1}]$, with $|t| = D_0$ and zero differential, splits in its derived category into the D_0 -periodizations of its shifted homology modules. In particular it splits as a finite sum of free shifts and a perfect module with torsion homology.*

Proof. First, S_0 is noetherian on both sides, since every one-sided ideal is principal. Nonzero right ideals dS_0 and eS_0 must intersect. Otherwise the right ideals $e^i dS_0$, $i \geq 0$, would form an infinite internal direct sum in S_0 : in a finite relation, its term of least e -power belongs both to dS_0 and to eS_0 after canceling that power, and hence vanishes; continue inductively. This contradicts noetherianity. The same argument on the left gives the two-sided Ore division ring of fractions Q_0 . Clearing common denominators gives exact localization of modules. Consequently, for a finitely generated right module M , the kernel

$$M_{\text{tor}} = \ker(M \longrightarrow M \otimes_{S_0} Q_0)$$

consists exactly of the elements right-annihilated by a nonzero element of S_0 .

The quotient M/M_{tor} embeds in the finite-dimensional right Q_0 -vector space $M \otimes_{S_0} Q_0$. Choose coordinates in that space and finitely many generators of the embedded module. In each coordinate, clear their common *left* denominators. Left multiplication in a coordinate is right S_0 -linear, so this produces an injection

$$M/M_{\text{tor}} \longrightarrow S_0^r$$

for some finite r . The choice of left denominators is the one compatible with right modules.

Every submodule of S_0^r is free of finite rank. To prove this, induct on r and project to the first coordinate. The image is a right ideal. If it is zero, the induction hypothesis applies inside S_0^{r-1} . If it is nonzero, its principal generator is a nonzero element of the domain, so the image is free of rank one. The projection then splits, and its kernel lies in S_0^{r-1} . This proves the assertion in both cases. It follows that M/M_{tor} is finite free, and hence that the sequence with this quotient splits. The kernel of a finite free presentation of M is also finite free by the same argument. This gives a finite free resolution of length at most one.

Now view a dg module over $S_0[t^{\pm 1}]$ as a D_0 -periodic complex of right S_0 -modules. A finite-cell module has finite free modules in each period residue, and noetherianity of S_0 makes its homology finitely generated there. The same holds for a retract, hence for every perfect module P . For each $i \in \{0, \dots, D_0 - 1\}$ choose a resolution

$$0 \longrightarrow R_{1,i} \longrightarrow R_{0,i} \longrightarrow H_i(P) \longrightarrow 0$$

by finite free right modules. Place $R_{0,i}$ in degree i and $R_{1,i}$ in degree $i + 1$. Lift the free generators of $R_{0,i}$ to cycles in a representative complex for P . The images of the relations are boundaries. Since $R_{1,i}$ is free, lift them through the differential onto the boundaries in degree i . This gives a map of the two-term complex into P .

Extend the map periodically using multiplication by t . The positive even degree D_0 gives the usual periodization without changing the homological signs. This periodized complex has homology $H_i(P)$ in residue i and zero in the other residues. Sum the maps over $0 \leq i < D_0$. The sum is finite and induces an isomorphism in every homology degree, so it is a quasi-isomorphism. Finally split each $H_i(P)$ into its torsion and finite free parts. The periodizations of the free parts are finite sums of free shifts; those of the torsion parts are perfect because of their length-one finite free resolutions. This gives the asserted splitting. $\square$

The second puts every finite construction into a tower of free layers, with all retracts taken at the end.

Lemma 6.3 (Finite towers and retracts). *For an idempotent-complete triangulated category $\mathcal{C}$ and an object T_0 , every $P \in \text{Thick}_{\mathcal{C}}(T_0)$ is a retract of an object P_0 admitting a finite tower of triangles*

$$F_j \longrightarrow P_j \longrightarrow P_{j+1} \longrightarrow F_j[1], \quad 0 \leq j < s, \quad P_s = 0,$$

where each F_j is a finite sum of arbitrary integer shifts of T_0 .

Proof. Call an object a finite extension of free layers if it admits such a tower. Zero, finite sums of shifts of T_0 , and their shifts have this property. The octahedral axiom concatenates two towers. More explicitly, suppose $A \rightarrow E \rightarrow B$ is a triangle, and the first layer of a tower for A is $F_0 \rightarrow A \rightarrow A_1$. The cofiber E_1 of the composite $F_0 \rightarrow A \rightarrow E$ fits into triangles

$$F_0 \longrightarrow E \longrightarrow E_1, \quad A_1 \longrightarrow E_1 \longrightarrow B.$$

Induction on the number of layers in A , ending with the tower for B , gives a tower for E . In particular, if A and B have towers and $f : A \rightarrow B$ is a map, the triangle

$$B \longrightarrow \text{cofib}(f) \longrightarrow A[1]$$

gives a tower for $\text{cofib}(f)$. Finite sums of towers give the same closure for finite sums.

It remains to move retracts to the end. Let A and B be retracts of finite extensions $\overline{A}$ and $\overline{B}$. Idempotent completeness supplies complements

$$\overline{A} \simeq A \oplus A', \quad \overline{B} \simeq B \oplus B'.$$

Extend a map $f : A \rightarrow B$ to $\overline{f} : \overline{A} \rightarrow \overline{B}$ by setting all other matrix entries to zero. Its cone is isomorphic to

$$\text{cofib}(f) \oplus B' \oplus A'[1].$$

Thus $\text{cofib}(f)$ is a retract of a finite extension. Shifts, finite sums and successive retracts of such objects are also retracts of finite extensions. The resulting subcategory is thick and contains T_0 , which proves the lemma. $\square$

Proof of Theorem 6.1. Suppose that $U_0 \in \text{Thick}_{\mathcal{C}}(T_0)$. By Lemma 6.3, choose a retract insertion $g_0 : U_0 \rightarrow P_0$ and a tower as in that lemma. Let $g_j : U_0 \rightarrow P_j$ be the composite with the map to its j th tail, and put $Y_0 = \Psi(U_0)$. We prove backward from $j = s$ that

$$H(\Psi(g_j)) = 0, \quad H((\Psi(g_j))^*) = 0.$$

Both assertions hold for $P_s = 0$.

Write one extended step as

$$F \longrightarrow P'_1 \longrightarrow P'_2 \longrightarrow F[1], \quad g'_1 : Y_0 \rightarrow P'_1, \quad g'_2 : Y_0 \rightarrow P'_2,$$

where F is a finite sum of free shifts, and suppose that the two vanishing assertions hold for g'_2 . By Lemma 6.2, choose

$$P'_2 \simeq V' \oplus L',$$

with V' a finite sum of free shifts and $H(L')$ torsion. Each component of the projection $P'_2 \rightarrow V'$ is a class in the homology of $(P'_2)^*$. Dual-homology vanishing for g'_2 says that its composite with each of these classes is zero. Hence

$$Y_0 \xrightarrow{g'_2} P'_2 \longrightarrow V'$$

is zero as a derived map. The exact sequence of maps into the fiber triangle therefore lifts g'_1 to

$$Z' = \text{fib}(P'_1 \longrightarrow V').$$

The fiber construction, equivalently the octahedral axiom, gives a triangle

$$F \longrightarrow Z' \longrightarrow L' \longrightarrow F[1].$$

In its homology sequence, the map preceding $H(F) \rightarrow H(Z')$ has torsion source and free target in each degree. It is zero, so $H(F) \rightarrow H(Z')$ is injective. The lifted map $Y_0 \rightarrow Z'$ is zero on homology after composition with $Z' \rightarrow L'$, because that composite is the corresponding summand of g'_2 . Its homology therefore factors through the submodule $H(F) \subset H(Z')$. The source $H(Y_0)$ is torsion, so its map into the free module $H(F)$ is zero. It follows that $H(g'_1) = 0$.

The map g'_1 is the image of a map $U_0 \rightarrow P_j$ in $\mathcal{C}$, and this tower tail belongs to $\text{Thick}_{\mathcal{C}}(T_0)$. Hypothesis (6.1) gives $H((g'_1)^*) = 0$ as well. This completes the backward induction. In particular $\Psi(g_0)$ is zero on homology. It has a left inverse because g_0 is a retract insertion and every functor preserves its retraction equation. Thus $H(\Psi(g_0))$ is also a split injection, contradicting the nonzero homology of Y_0 . $\square$

The two tests have distinct roles in the induction. Dual homology removes the free quotient of a tower tail as an actual derived map. Ordinary homology then places the remaining map in a free submodule, where torsion forces it to vanish. The implication (6.1) restores the dual statement for the next step. There is no bound on the length of the proposed tower.

6.2 A flat coefficient extension

We now construct the coefficient ring required by the criterion. The extension must be flat on both sides: homology of right modules and homology of their left-module duals will both be tested after extending coefficients. It must also keep the cone of a nonzero while making its homology torsion.

There are two filtrations in what follows. The paired vectors in $W_i(Y) = \mathcal{Q} \oplus \mathcal{K}$ have Laurent pole bounds in $K_0 = k((x))$. The operation ring $\Delta = R[[H]]$, with $R = k[[x]]$, has instead the weight filtration of Proposition 2.8, with

$$\text{gr } \Delta = k[\sigma x, A, B, C], \quad |\sigma x| = 4, \quad |A| = |B| = 1, \quad |C| = 3.$$

The coefficient extension will be taken at the homogeneous prime (C) and used only to test homology of perfect modules. The later estimate concerns integral operators acting on Laurent functions: it compares the growth of their joint kernels with the larger exponent $\beta > 1/3$ supplied by Lemma 5.3.

We give the filtered algebra argument with explicit hypotheses. A decreasing filtration is indexed by integers and is multiplicative. Its initial-symbol map is denoted by σ . A filtration on a finitely generated module is called *good* if it is the quotient filtration from a finite direct sum of filtration shifts of the ring. The construction through truncated Rees rings is the algebraic approach to microlocalization developed by Asensio, Van den Bergh, and Van Oystaeyen [3, Sections 2–3]. We give the construction and flatness argument in the precise complete setting needed here.

Lemma 6.4 (Filtered microlocalization). *Let k be a finite field, and let $\mathcal{A}$ be a complete separated filtered k -algebra such that $F^n \mathcal{A} = \mathcal{A}$ for $n \leq 0$ and*

$$\text{gr } \mathcal{A} = k[z_1, \dots, z_d, c]$$

as a graded algebra, with all the displayed generators of positive integer degree. There is a filtered algebra $\mathcal{S}$ and a map $\mathcal{A} \rightarrow \mathcal{S}$ with the following properties:

1. *$\text{gr } \mathcal{S}$ is the homogeneous localization of $\text{gr } \mathcal{A}$ obtained by inverting all homogeneous elements outside (c) ;*
2. *$\mathcal{S}$ is separated and complete for its exhaustive filtration, which is allowed to have nonzero terms of negative degree;*
3. *$\mathcal{S}$ is flat as both a left and a right $\mathcal{A}$ -module.*

For a finitely generated good filtered right module M , the corresponding microlocalized module is $M \otimes_{\mathcal{A}} \mathcal{S}$ and its associated graded is the same homogeneous localization of $\text{gr } M$. The analogous assertion holds for left modules.

Proof. We first justify the finite-module filtration properties needed for the construction. If N is a submodule of a finite shifted filtered free module, choose homogeneous generators of $\text{gr } N$ over the noetherian ring $\text{gr } \mathcal{A}$ and lift them to N . For an element of N , subtract a linear combination of the lifts matching its initial symbol, and repeat. The remainder increases in filtration at each step. The coefficients of each chosen lift form a convergent series in $\mathcal{A}$, so the element is a finite linear combination of the lifts. This also proves that the resulting surjection from a finite shifted free module is strict for the induced filtration.

Each quotient $\mathcal{A}/F^n \mathcal{A}$ is finite: the graded pieces below n are finite-dimensional over the finite field k . Consequently $\mathcal{A}$ and its finite products are compact. The submodule just considered is therefore compact and closed, being the image of a compact finite free module. Notice that closedness was not assumed during the initial-symbol subtraction. These observations prove that $\mathcal{A}$ is noetherian on both sides and that every finitely generated module has a complete separated good filtration. Submodule and quotient filtrations are good, by using their preimages in a finite free presentation. A map of finite good filtered modules is bounded up to a filtration shift, as one sees on strict generators. In particular, any two good filtrations on one module differ by bounded shifts.

Form the Rees ring and Rees module

$$\tilde{\mathcal{A}} = \bigoplus_{n \in \mathbb{Z}} F^n \mathcal{A} s^{-n}, \quad \tilde{M} = \bigoplus_{n \in \mathbb{Z}} F^n M s^{-n},$$

where s is central of Rees degree -1 . For $j \geq 1$, work modulo s^j and invert the homogeneous elements whose reductions modulo s lie outside (c) . Such elements are regular, by induction through the s -layers and regularity of their nonzero reductions in the polynomial ring. They satisfy the Ore conditions on both sides. Indeed all commutators are divisible by s , so for a denominator d and an element e one has $\text{ad}_d^j(e) = 0$ modulo s^j . Expanding this identity shows that $d^j e$ has a right factor d , and that ed^j has a left factor d . Powers of a denominator are still denominators. These relations give the Ore conditions first for homogeneous elements and then for their finite sums. We may thus use ordinary two-sided Ore localization, retaining the Rees grading. Localization is exact on modules, by clearing common denominators.

Let M_j^μ denote this localization of $\tilde{M}/s^j \tilde{M}$. Denominators lift across truncation levels. The localizations therefore reduce compatibly, and localizing a module from a lower level at a higher level gives its own localization. Define

$$F^n \mu(M) = \lim_j (M_j^\mu)_n, \quad \mu(M) = \bigcup_{n \in \mathbb{Z}} F^n \mu(M). \quad (6.2)$$

The inclusions in this union come from multiplication by powers of s . More precisely, regularity of s on the untruncated Rees module gives, after exact localization,

$$0 \longrightarrow (M_j^\mu)_{n+l} \xrightarrow{s^l} (M_{j+l}^\mu)_n \longrightarrow (M_l^\mu)_n \longrightarrow 0 \quad (l, j \geq 1). \quad (6.3)$$

Transitions in each fixed Rees degree are surjective. Taking limits in (6.3) gives the asserted inclusions and

$$F^n \mu(M) / F^{n+l} \mu(M) = (M_l^\mu)_n.$$

The filtration is separated and complete. In fact compatible Cauchy data eventually lie in one fixed $F^n \mu(M)$, where both assertions follow from the inverse-limit description. Taking $l = 1$ identifies the associated graded with the homogeneous localization of $\text{gr } M$.

Products at the truncated levels give $\mathcal{S} = \mu(\mathcal{A})$ its ring structure and give $\mu(M)$ its module structure. The natural maps from $\mathcal{A}$ and M are obtained by taking their classes at each level. A bounded-shift map of good filtered modules gives a map on these unions; changing the bound does not change the map, by the transition inclusions. Thus the underlying module $\mu(M)$ is functorial and independent of the good filtration chosen.

This functor is exact on finite modules. Use induced and quotient filtrations in a short exact sequence to make it strict. Its Rees sequence remains exact modulo s^j , because the quotient Rees module is s -torsion-free. Ore localization preserves exactness. Inverse limits preserve it here because the transition maps in each module system are surjective, and taking the filtered union preserves it as well. The natural map

$$M \otimes_{\mathcal{A}} \mathcal{S} \longrightarrow \mu(M)$$

is an isomorphism for a free module and hence for every finite module by a finite presentation. This proves left flatness of $\mathcal{S}$; over a noetherian ring, exactness on finite modules suffices for this flatness test. The identical construction for left modules proves right flatness. $\square$

Apply Lemma 6.4 to $\mathcal{A} = \Delta$ and $c = C$, and write S for the resulting ring. The initial symbol of x is a unit in $\text{gr } S$. Lifting its inverse and correcting the successive errors in increasing filtration shows that x is a unit in S . Both left and right flatness hold also over $\Delta[x^{-1}]$: on a module over that ring, balancing by the invertible x identifies tensoring over Δ with tensoring over $\Delta[x^{-1}]$.

Recall from Proposition 3.3 that

$$\mathcal{E}[t^{-1}] = A_{\text{op}}[t^{\pm 1}], \quad A_{\text{op}} = (\Delta[x^{-1}])^{\Lambda}.$$

The tame character decomposition makes $\Delta[x^{-1}]$ finite free over A_{op} on either side. Indeed x has a faithful Λ -character and is a unit; multiplication on either side by its powers identifies every character space with the invariant space. We have proved the following coefficient-extension statement:

$$\mathcal{E} \longrightarrow A_{\text{op}}[t^{\pm 1}] \longrightarrow \Delta[x^{-1}][t^{\pm 1}] \longrightarrow \mathcal{B} := S[t^{\pm 1}] \quad (6.4)$$

is flat on both sides. Here t is central, of homological degree D ; S is in homological degree zero and has zero differential. The filtration on S is unrelated to this homological grading.

The following lemma verifies the remaining ring properties and shows that the cone of a retains nonzero torsion homology.

Lemma 6.5. *The ring S is a domain, every left or right ideal of S is principal, and S has a two-sided Ore division ring of fractions. The operator a constructed in Proposition 4.6 has no one-sided inverse in S . In particular $S/aS \neq 0$, and this right S -module is torsion.*

Proof. The graded ring of S is a domain, so multiplicativity of initial symbols proves that S is a domain. Every nonzero homogeneous element of $\text{gr } S$ is a power of C times a homogeneous unit. In a nonzero right ideal choose d_0 minimizing the nonnegative C -order of its initial symbol. For any member d of the ideal, its initial symbol is a multiple of $\sigma(d_0)$. Lift the multiplier and subtract the corresponding right multiple of d_0 . The remainder is still in the ideal and has strictly larger filtration, so its initial C -order is still at least the chosen minimum. Repeating, and summing the convergent multipliers, expresses d as a right multiple of d_0 . The successive multipliers tend to zero because their filtration degrees increase by the same amount as those of the remainders. This proves the right principal-ideal property; the left proof is identical.

The intersection argument in the proof of Lemma 6.2 now gives the two-sided Ore division ring of fractions.

The operator a still has initial symbol C in S . A one-sided inverse would make C invertible in $\text{gr } S$, which it is not. Thus $S/aS \neq 0$. Given a nonzero representative $b \in S$, the nonzero intersection $bS \cap aS$ supplies a nonzero d with $bd \in aS$. Every class in S/aS is therefore annihilated on the right by a nonzero element, proving torsion. $\square$

The instance supplied by the glued object. We now return to $\mathbf{D}(\mathcal{N})$ and the coefficient extension (6.4). Let

$$\mathcal{C}_{\text{comp}} = \{P \in \mathbf{D}(\mathcal{N}) : \widehat{P} \in \text{Thick}(\widehat{T}')\}$$

be the full subcategory with the displayed property. Completion is exact by Proposition 3.1, so the inverse image of $\text{Thick}(\widehat{T}')$ is thick. The ambient derived category is idempotent-complete, and the defining property is closed under retracts; hence $\mathcal{C}_{\text{comp}}$ is idempotent-complete as well. It contains T' , and it contains U because Proposition 5.4 identifies $\widehat{U}$ with the displayed finite construction Y . Compose completion, the inverse of the strict realization equivalence in Proposition 3.5, and derived extension of scalars to obtain an exact functor

$$\Psi : \mathcal{C}_{\text{comp}} \longrightarrow \text{Perf}(\mathcal{B}).$$

We write $\widehat{P}_{\mathcal{B}}$ for $\Psi(P)$ and use the same subscript for the image of a map. The functor sends T' to the free right module $\mathcal{B}$.

On a free right module, the map determined by $1 \mapsto a$ is left multiplication by a . It is injective over S because S is a domain, and its cokernel is the right module S/aS . Since Y is the sum of all D shifts of this cone,

$$H_i(\Psi(U)) \cong S/aS \quad \text{as right } S\text{-modules for every integer } i. \quad (6.5)$$

Lemma 6.5 says that this module is nonzero and torsion. This verifies the first hypothesis of Theorem 6.1. The remaining task is the implication from homology vanishing to dual-homology vanishing for every global map $g : U \rightarrow P$ with $P \in \text{Thick}_{\mathbf{D}(\mathcal{N})}(T')$.

6.3 A growth bound for joint operator kernels

To prove the remaining implication, we return to integral operators acting on $K_0 = k((x))$. For the remainder of this section let $\bar{b}$ denote the reduction of $b \in \Delta$ in $\Omega = \Delta/(x) = k[[H]]$. The notation $\sigma(\bar{b})$ means the subsequent initial symbol for the weight filtration on Ω ; reduction modulo x and passage to the initial symbol are distinct operations. The congruence subgroups H_l , their indices N_l , and the balanced sequence b_j^H are those used in Lemma 4.1 and Lemma 4.2. In particular, with $z_h = h(x)/x$, we have

$$r'_m = \sum_{l \geq 1} \lfloor m/N_l \rfloor \leq \kappa m, \quad \Gamma_s \subset H_{p^s}, \quad N_{p^s} \geq p^{3(s-1)} \quad (s \geq 1). \quad (6.6)$$

These are the inputs that will turn linear Laurent precision into cubic-root dimension growth.

Lemma 6.6 (Joint-kernel growth). *Suppose $a_1, b_1 \in \Delta$ satisfy*

$$(\sigma(\bar{a}_1)) = (C) \quad \text{in } \text{gr } \Omega = k[A, B, C], \quad \sigma(\bar{b}_1) \notin (C).$$

Fix an integer $L_0 \geq 0$. There is a constant c_0 , depending on the fixed chart, L_0 , a_1 , and b_1 , such that for every $n \geq 1$ the image of

$$\{f \in x^{-L_0}R : a_1(f) = b_1(f) = 0\} \longrightarrow K_0/x^n R$$

has k -dimension at most $c_0(1+n)^{1/3}$.

Proof. Choose any finite family $f_1, \dots, f_v$ in the indicated joint kernel which is independent modulo $x^n R$. We bound v uniformly. Consider the orbit functions

$$\varphi_i(h) = h(f_i), \quad h \in H,$$

as functions with values in K_0 . The proof first places them in a lattice of integral-valued functions whose reduction injects into functions on H . Interpolation then makes that reduction factor through a finite quotient, where the two annihilator relations give the required dimension bound.

We first bound the coefficients of integral-valued linear combinations of the φ_i . Put $F_i(y) = y^{L_0} f_i(y) \in k[[y]]$. For coefficients $c_i \in K_0$ with minimum x -valuation zero,

$$\sum_i c_i \varphi_i(h) = h(x)^{-L_0} \sum_i c_i F_i(h(x)).$$

The reduction modulo x of $\sum_i c_i F_i(y) \in R[[y]]$ is a nonzero series of order $d < n + L_0$. Indeed the residues of the c_i give a nontrivial k -linear combination, and the assumed independence of the f_i bounds its order after multiplication by y^{L_0} . Weierstrass preparation writes this series as a unit times a monic polynomial of degree d . Evaluation at $h(x) = xz_h$ preserves valuation of the polynomial because the evaluated unit is a unit.

As a polynomial in z_h , that monic polynomial evaluated at xz_h has leading coefficient x^d . Lagrange interpolation on the first $d+1$ balanced orbit points expresses x^d as the sum of its values divided by the interpolation denominators. Every such denominator has valuation at most r'_d . Therefore at least one of the values has valuation at most $d + r'_d$. By (6.6), some value of the normalized orbit combination has valuation $O(n)$, with constants independent of the chosen family and coefficients.

It follows first that the φ_i are independent over K_0 . It follows also that the coefficients of every integral-valued combination have valuations bounded below by $-O(n)$: factor out the smallest coefficient valuation and apply the preceding upper bound to the remaining normalized combination. Let $\mathcal{L}_{\text{orb}}$ be the R -module of integral-valued functions in the K_0 -span of the φ_i . It contains each $x^{L_0} \varphi_i$ and is contained in a fixed finite free lattice by the coefficient bound. Thus it is a full finite free lattice of rank v . Its reduction injects into the space of functions $H \rightarrow k$. In fact an integral-valued function whose values all lie in xR is divisible by x within $\mathcal{L}_{\text{orb}}$, by its definition.

Multiply an integral combination by $z_h^{L_0}$, which is congruent to one modulo x . Its residue is consequently the residue of

$$x^{-L_0} \sum_i c_i F_i(xz_h).$$

The uniform coefficient bound permits truncation to a polynomial in z_h of degree $J = O(n)$: all later terms vanish modulo x . The truncated polynomial is integral-valued on the whole orbit, since it differs from the preceding integral expression by an xR -valued function.

We next control the congruence level on which the residue of such a polynomial depends. The normalized Newton basis is the integral-valued-polynomial construction associated with P -orderings; see [9, §11, Theorem 24] for the general framework. The orbit-specific bounds and

residue estimates below are proved here. The integral-valued polynomials of degree at most J on the orbit points have the R -basis

$$x^{-r'_m} \prod_{i=0}^{m-1} (z - z_{b_i^H}), \quad 0 \leq m \leq J. \quad (6.7)$$

Balancedness gives at least $\lfloor m/N_l \rfloor$ factors congruent to zero modulo x^l at every orbit point. Hence each polynomial in (6.7) is integral-valued. At $z = z_{b_m^H}$ these counts are exact, so its value is a unit. Successively evaluating at $z_{b_0^H}, z_{b_1^H}, \dots$ now proves the basis assertion by triangular interpolation.

If two orbit points are congruent modulo x^l and $N_l > J$, the residues of all the polynomials in (6.7) agree at the two points. To see this, expand the difference of a product by replacing one factor at a time. Deleting that factor loses at most one of the required matches at each level below l . No matches at levels l and higher are needed, since $N_l > m$. The difference of the replaced factors contributes valuation at least l , exceeding the possible loss of $l - 1$. After division by $x^{r'_m}$ every difference is thus in xR .

Choose $s \geq 1$ with $N_{p^s} > J$ and $p^s = O(n^{1/3})$. Such a choice follows from the last inequality of (6.6), by taking the first s for which $p^{3(s-1)} > J$. All the residue functions coming from $\mathcal{L}_{\text{orb}}$ then factor through H/Γ_s .

We have placed the rank- v lattice, after reduction, inside functions on a finite quotient. The annihilator equations now impose two algebraic relations on that function space. For any $\varphi \in \mathcal{L}_{\text{orb}}$ and $\mu = a_1, b_1$, the original annihilator equations imply

$$\int_{g \in H} h(\mu(g)) \varphi(hg) = 0 \quad (h \in H). \quad (6.8)$$

For the functions φ_i this is obtained by applying h to $\mu(f_i) = 0$; field-linear combinations preserve the identity because their coefficients do not depend on h . All integrals can be computed at finite coefficient and group precision. On reduction, h acts trivially on the coefficient field k . The resulting functions on H/Γ_s therefore annihilate the left translates of $\bar{a}_1$ and $\bar{b}_1$. Their space has dimension at most

$$\dim_k \frac{\Omega}{\Omega \bar{a}_1 + \Omega \bar{b}_1 + \ker(\Omega \rightarrow k[H/\Gamma_s])}.$$

The associated graded of this quotient kills at least

$$C, \quad \sigma(\bar{b}_1), \quad A^{p^s}, \quad B^{p^s}, \quad C^{p^s},$$

since $g_i^{p^s} \in \Gamma_s$. The quotient filtration is finite, already on the finite group algebra. Modulo C , take a leading monomial $A^{u_0} B^{v_0}$ of the fixed nonzero polynomial $\sigma(\bar{b}_1) \bmod C$. Among the monomials $A^i B^j$ with $0 \leq i, j < p^s$, at most $(u_0 + v_0)p^s$ are not divisible by $A^{u_0} B^{v_0}$; the case of a nonzero constant gives a zero quotient. Thus the displayed dimension is $O(p^s) = O(n^{1/3})$. Because $\mathcal{L}_{\text{orb}}/x\mathcal{L}_{\text{orb}}$ has dimension v and injects into this function space, the required bound follows. $\square$

We now have a quantitative restriction on joint kernels of two integral operators. To apply it to a global map out of U , we must extract a second annihilating operator from a microlocal homology vanishing condition. The next lemma records both sides of this extraction; they will be used separately and must not be interchanged.

Lemma 6.7 (Removal of x -divisible remainders). *Let $a \in \Delta$ be the operator of Proposition 4.6, so that $\bar{a} = U_3$ and $\sigma(\bar{a}) = C$.*

1. Suppose a right ideal $J \subset \Delta$ contains $a\Delta$, is right x -saturated, meaning $bx \in J \Rightarrow b \in J$, and satisfies $JS = S$. Then J contains an element b such that $\sigma(\bar{b}) \notin (C)$.
2. Suppose $b_0 \in \Delta[x^{-1}]$ does not belong to Sa and annihilates a k -subspace $V \subset \ker(a : K_0 \rightarrow K_0)$. There is $b_1 \in \Delta$ annihilating V with $\sigma(\bar{b}_1) \notin (C)$.

Proof. For the first assertion, suppose that every member of J reduces into $\bar{a}\Omega$. Starting with $j \in J$, subtract a right multiple of a with the same reduction and write

$$j = ad_0 + j_1x.$$

Normality of x allows the right division, and right saturation puts j_1 back in J . Repetition gives

$$j = a(d_0 + d_1x + \cdots + d_{r-1}x^{r-1}) + j_rx^r.$$

The remainder tends to zero, and the coefficient series converges in Δ . Thus $j \in a\Delta$, and $J = a\Delta$. This contradicts $JS = S$, because $aS \neq S$ by Lemma 6.5.

Choose now $j \in J$ whose reduction is not in $\bar{a}\Omega$. If its reduction has initial symbol divisible by C , subtract a right multiple of $\bar{a}$ matching that symbol. Continue in the complete filtered ring Ω . The process must terminate with an initial symbol outside (C) : otherwise the convergent sum of multipliers would express the original reduction as a member of $\bar{a}\Omega$. Lift the finitely many subtractions to right multiples of a in Δ . They preserve membership in J and produce the asserted b .

For the second assertion, first multiply b_0 on the left by a power of x to obtain $b \in \Delta$. This preserves annihilation of V and nonmembership in Sa , since x is a unit on K_0 and in S . Whenever $\bar{b} \in \Omega\bar{a}$, subtract a left multiple of a and divide on the left by x . If this procedure never terminated, its equations would give

$$b = (d_0 + xd_1 + \cdots + x^{r-1}d_{r-1})a + x^rb_r, \quad b_r, d_r \in \Delta.$$

The limit would put b in Δa , a contradiction. Thus a reduction outside $\Omega\bar{a}$ is reached. As on the right, successive initial-symbol subtraction by left multiples of $\bar{a}$ then terminates with a symbol outside (C) . Throughout, subtracting a left multiple of a preserves annihilation of V , and division on the left by x does also. The final integral operator is the required b_1 . Neither of these two convergent expansions commutes x through a . $\square$

6.4 Dual detection for global maps

We now verify the second hypothesis of Theorem 6.1 for the functor $\Psi(P) = \hat{P}_{\mathcal{B}}$ defined above. Proposition 3.5 identifies each complete object in $\text{Thick}(\hat{T}')$ with a perfect right $\mathcal{E}$ -module. For such a module M , the two duals used below are

$$M^* = R\text{Hom}_{\mathcal{E}}(M, \mathcal{E}), \quad (M_{\mathcal{B}})^* = R\text{Hom}_{\mathcal{B}}(M_{\mathcal{B}}, \mathcal{B}).$$

They are left modules. The subscript $\mathcal{B}$ denotes the extension (6.4). A *homology ghost* here is a map inducing zero on module homology in every degree. We will show that a global map out of U which becomes a homology ghost also becomes zero on dual homology.

Lemma 6.8 (Adjoints on the Laurent field). *Every element $b \in \Delta$ has an adjoint $b^\dagger \in \Delta$ for the continuous pairing*

$$\langle u, f \rangle = \text{CT}_x(uf), \quad u, f \in K_0.$$

Here CT_x extracts the coefficient of x^0 in a Laurent series. Modulo x , taking the adjoint induces group inversion on $\Omega = k[[H]]$. In particular it preserves the conditions that an initial symbol generate (C) or lie outside (C) .

Proof. The pairing identifies K_0 with its continuous k -linear dual. Indeed a continuous functional kills some $x^j R$; its values on the monomials x^i are exactly the coefficients of a Laurent series bounded below, read in the reverse order. For a coefficient $c_0 \in R$ and a group element $g \in H$, formal residue substitution in dx/x gives

$$(c_0 \delta_g)^\dagger = g^{-1}(c_0) \frac{x(g^{-1}(x))'}{g^{-1}(x)} \delta_{g^{-1}}. \quad (6.9)$$

The Jacobian factor is an integral unit congruent to one modulo x , since $g(x)/x \in 1 + xR$. Residue substitution remains valid in characteristic p : for monomials, the requisite identities depend on only finitely many coefficients of an invertible formal series and hold integrally, as is seen by verifying them first over characteristic zero. They then specialize to k and extend to Laurent series.

Formula (6.9) extends to measures in Δ . At any fixed coefficient precision, moving the coefficients by g^{-1} and multiplying by the Jacobian require only a finite level of the group action; inversion is continuous in the group coordinates. Equivalently, integrate the displayed formula on sufficiently fine finite quotients, at each precision. The resulting compatible measures define $b^\dagger \in \Delta$ and satisfy the adjoint identity. Modulo x , this is group inversion on the completed group algebra. Its associated graded map sends the generator symbols A, B, C to their negatives. It therefore preserves the two asserted symbol conditions. $\square$

Proposition 6.9 (Dual detection of global ghosts). *For the object U of Proposition 5.4, let $g : U \rightarrow P$ be a map in $\mathbf{D}(\mathcal{N})$ with $P \in \text{Thick}(T')$. Then*

$$H(\widehat{g}_{\mathcal{B}}) = 0 \implies H((\widehat{g}_{\mathcal{B}})^*) = 0. \quad (6.10)$$

Both zero assertions mean that the corresponding induced homology maps vanish.

Proof. Assume the left side of (6.10). We first bound the images of the polar representatives belonging to the cokernel part of $W(Y)$, then use the paired lattice to obtain large killed subspaces of the paired span. Composites into the generator vanish on the cokernel part, so they annihilate the kernel projections of these subspaces. Lemma 6.6 will force the operators representing those composites into the left ideal Sa .

An integral second annihilator. Fix a residue r and let $\iota_r : \widehat{T}'[r] \rightarrow m[r]$ be the cofiber inclusion. Restrict $\widehat{g}$ to this summand of Y and put

$$h : \widehat{T}'[r] \xrightarrow{\iota_r} m[r] \longrightarrow Y \xrightarrow{\widehat{g}} \widehat{P}.$$

This map is a homology class of the perfect module representing $\widehat{P}$. Extend coefficients first to $\Delta[x^{-1}][t^{\pm 1}]$, but not yet to $\mathcal{B}$, and let $J_h \subset \Delta$ be the right annihilator of that class. The ring Δ acts in homological degree zero here. We have $a\Delta \subset J_h$ because $\iota_r a = 0$. The ideal is right x -saturated because x acts invertibly on the extended module.

Put $A_{\text{loc}} = \Delta[x^{-1}]$ and $N_h = H_r(\widehat{P}_{A_{\text{loc}}[t^{\pm 1}]})$, and let $I_h = \text{Ann}_{A_{\text{loc}}}(h)$ be the local right annihilator. Then $J_h = I_h \cap \Delta$ and $I_h = J_h A_{\text{loc}}$: every local coefficient is a right fraction dx^{-n} , and $hdx^{-n} = 0$ if and only if $hd = 0$, by multiplying on the right by the unit x^n . The cyclic map

$$A_{\text{loc}}/I_h \longrightarrow N_h, \quad \alpha + I_h \longmapsto h\alpha$$

is injective. Left flatness of S over A_{loc} preserves this injection, with source $(A_{\text{loc}}/I_h) \otimes_{A_{\text{loc}}} S \cong S/I_h S$. The same left flatness, after adjoining t , commutes extension with homology. It gives

$$0 \longrightarrow S/I_h S \longrightarrow N_h \otimes_{A_{\text{loc}}} S \cong H_r(\hat{P}_{\mathcal{B}}).$$

The generator maps to the extended class $h_{\mathcal{B}}$, which is zero because it factors through the homology ghost $\hat{g}_{\mathcal{B}}$. Injectivity forces $1 + I_h S = 0$, so $J_h S = I_h S = S$. Lemma 6.7 gives

$$b \in J_h, \quad \sigma(\bar{b}) \notin (C). \quad (6.11)$$

Crucially, b annihilates the homology class *before* microlocal extension; S was used to prove its existence, not to define a new action on functions.

From the annihilator to a Laurent rank bound. Extend the overlap calculation from Φ to K_0 by derived tensor with the left module $K_0[t^{\pm 1}]$ over $\Delta[x^{-1}][t^{\pm 1}]$. The integral skew action extends to Laurent functions by the same integration formula: for a fixed bounded input lattice and a fixed output precision, finite coefficient and group levels suffice. The relations $ha = hb = 0$ are zero homology classes of maps out of a free shift, hence zero derived maps before tensoring. Derived tensor therefore preserves them, without any flatness assumption on K_0 as a coefficient module.

To make the return to the original overlap explicit, put $D_{\text{loc}} = \Delta[x^{-1}][t^{\pm 1}]$ and let $\mathbf{P}$ be the perfect right $\mathcal{E}$ -module representing $\hat{P}$. Derived associativity gives the natural identification

$$(\mathbf{P} \otimes_{\mathcal{E}}^{\mathbf{L}} D_{\text{loc}}) \otimes_{D_{\text{loc}}}^{\mathbf{L}} K_0[t^{\pm 1}] \simeq \mathbf{P} \otimes_{\mathcal{E}}^{\mathbf{L}} K_0[t^{\pm 1}].$$

The same identification applies to the source free shift $\mathcal{E}[r]$ and is natural for h . The tame average

$$\text{pr} : K_0 \longrightarrow K_0^{\Lambda} = \Phi, \quad f \longmapsto \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} \lambda(f)$$

is $\mathcal{E}$ -linear, because the normalized operations in $\mathcal{E}$ are Λ -invariant. Tensoring this map over $\mathcal{E}$ projects the last displayed complex onto $\mathbf{P} \otimes_{\mathcal{E}}^{\mathbf{L}} \Phi[t^{\pm 1}]$, and similarly for the source. Naturality gives a commuting square for these two projections and the maps induced by h . Applying this projection *after* the preceding derived vanishing yields

$$W(h)(\text{pr}(a(f))) = W(h)(\text{pr}(b(f))) = 0 \quad (f \in K_0). \quad (6.12)$$

The operator b need not be invariant, and pr need not be linear over $\Delta[x^{-1}]$. The extended tensor complex is equally the original perfect $\mathcal{E}$ -module tensored with $K_0[t^{\pm 1}]$, so the $\mathcal{E}$ -linear projection is all that is required.

Choose a fixed compact open subspace C_P in the target overlap degree. Continuous functionals on that overlap which kill C_P pull back along $W(h)\text{pr}$ to functionals on K_0 . By Lemma 6.8, write them as $\text{CT}_x(uf)$. The map $W(h)$ is continuous by Proposition 3.8: both T' and P have degreewise finite sections homotopy, and the reduction maps give the intrinsic topology on their completions. Since the tame average is also continuous, $W(h)\text{pr}$ sends some $x^j R$ into C_P . All the pulled-back functionals therefore kill $x^j R$. In the constant-term pairing this means $u \in x^{1-j} R$, so all these u lie in one fixed $x^{-L_0} R$, with $L_0 = \max(0, j-1)$. Equation (6.12) and nondegeneracy of the pairing show that

$$a^{\dagger}(u) = b^{\dagger}(u) = 0.$$

The adjoint lemma and (6.11) verify the symbol hypotheses of Lemma 6.6.

It follows that, modulo C_P , the images under $W(h)$ of the selected polar representatives w^j , $1 \leq j \leq N$, have rank $O(N^{1/3})$. Indeed take independent functionals on their span in the discrete quotient and extend them to that quotient. Their pullback series u are independent modulo $x^{qN+1}R$: any linear combination lying in this subspace pairs to zero with every $w^j = x^{-qj}$, $j \leq N$, contradicting independence on the selected images. Lemma 6.6, with $n = qN + 1$, gives the rank bound. This argument applies in every period residue.

The paired lattice forces a large killed subspace. In a fixed degree identify $W_i(Y) = \mathcal{Q} \oplus \mathcal{K}$ as in Lemma 5.2. The images of $(q_j, 0)$ are the images just bounded, through the relevant map $W(h)$. All images of $(0, k_l)$ belong to one compact open subspace of the target. In fact the k_l lie in the compact subspace $\mathcal{K} \cap R$ of the source, the map is continuous, and any compact subset of the target lies in a sufficiently large compact open stage. Enlarge C_P to contain those images.

By Lemma 5.5, the sums (q_j, k_l) belong after applying $W(\hat{g})$ to the discrete lattice image of $\pi_i LP$. At scale N , Lemma 5.3 supplies at least cN^β simultaneous pairs. The rank of their images modulo C_P is $O(N^{1/3})$, while the intersection of the discrete lattice with C_P has fixed finite dimension. Therefore the span of these pairs contains a subspace killed by $W(\hat{g})$ of dimension at least

$$cN^\beta - O(N^{1/3}), \quad c > 0. \quad (6.13)$$

Its projection to $\mathcal{K}$ has the same dimension even modulo $x^{qN+1}R$, since the selected k_l are independent modulo that subspace. These projected vectors lie in $\mathcal{K} \cap R$.

Composites into a free shift. Take any map of perfect modules, before microlocal extension,

$$\theta : \hat{P} \longrightarrow \hat{T}'[\ell].$$

On the summand $m[r]$, the composite $\theta\hat{g}$ kills the cofiber inclusion ι_r . Indeed its composite c on that free shift satisfies $ca = 0$, and right multiplication by a on homogeneous operations is injective: the operation ring embeds in the domain $\Delta[x^{-1}][t^{\pm 1}]$. The cofiber triangle therefore factors this composite as

$$m[r] \xrightarrow{\partial_r} \hat{T}'[r+1] \xrightarrow{b'} \hat{T}'[\ell],$$

where b' is a homogeneous operation. Consequently all cokernel components of its overlap map vanish. In degree $i \equiv r+1 \pmod{D}$, the only kernel component is the one belonging to this summand, and ∂_r identifies it with $\mathcal{K}$. Thus b' annihilates the projected subspaces in (6.13). If the relevant operation degree is not a multiple of D , then $b' = 0$ already. Otherwise normalize by the corresponding power of t and regard its coefficient, still denoted by b' , as an element of $(\Delta[x^{-1}])^\Lambda$ acting on K_0 .

We claim that $b' \in Sa$. If not, apply the second part of Lemma 6.7 to the span of all the projected subspaces just found. It produces an integral operator b_1 annihilating them with $\sigma(\bar{b}_1) \notin (C)$. The vectors are also killed by a , lie in R , and at scale N contain a family independent modulo $x^{qN+1}R$ of dimension $cN^\beta - O(N^{1/3})$. Lemma 6.6 contradicts this because $\beta > 1/3$. The claim follows.

After extension to $\mathcal{B}$, a left multiple of a kills the cofiber connecting map, since $a[1]\partial_r = 0$. Hence every $\theta\hat{g}$ is zero after this extension, on every summand of Y . For a perfect right $\mathcal{E}$ -module M , evaluation gives a natural equivalence of left $\mathcal{B}$ -modules

$$\mathcal{B} \otimes_{\mathcal{E}}^{\mathbf{L}} M^* \xrightarrow{\sim} (M_{\mathcal{B}})^*.$$

It is an equivalence for the free module and its shifts. Both sides are exact contravariant functors, so the class of modules for which it is an equivalence is closed under finite sums, cones and retracts. Thus it holds for every perfect module. Right flatness of $\mathcal{B}$ over $\mathcal{E}$ in (6.4) then gives

$$H((M_{\mathcal{B}})^*) \cong \mathcal{B} \otimes_{\mathcal{E}} H(M^*).$$

For the module representing $\hat{P}$, each homogeneous class on the right is a finite sum of tensors $b_{\nu} \otimes [\theta_{\nu}]$. Here each $[\theta_{\nu}]$ is represented, under strict realization, by a pre-extension map to a shift of $\hat{T}'$ as above. By naturality, the dual of $\hat{g}_{\mathcal{B}}$ sends this tensor to $b_{\nu} \cdot [(\theta_{\nu})_{\mathcal{B}} \hat{g}_{\mathcal{B}}]$. Every such composite has just been shown to vanish. This proves the right side of (6.10). $\square$

6.5 Applying the criterion to the glued object

Proposition 6.9 supplies the constraint on global maps required by the finite-tower criterion. The nonzero torsion homology was established in (6.5), so we can now conclude the obstruction.

Theorem 6.10. *For the object $U \in \mathbf{D}(\mathcal{N})$ constructed in Proposition 5.4,*

$$U \notin \text{Thick}_{\mathbf{D}(\mathcal{N})}(T').$$

Proof. Take $S_0 = S$, $D_0 = D = 2(p-1)$ and $\mathcal{B}_0 = \mathcal{B}$ in Theorem 6.1. Lemma 6.5 and the positive even value of D verify its ring and period assumptions. Use the full thick subcategory $\mathcal{C}_{\text{comp}}$ and the exact functor Ψ defined above. The subcategory contains U and T' , and $\Psi(T') \simeq \mathcal{B}$. Since $\mathcal{C}_{\text{comp}}$ is full and thick in $\mathbf{D}(\mathcal{N})$ and contains T' , the ordinary thick closure of T' computed in either category is the same. Equation (6.5) and Lemma 6.5 verify that $H(\Psi(U))$ is torsion in every degree and nonzero. Proposition 6.9 verifies (6.1) for every map into that thick closure, including every tower tail. Theorem 6.1 therefore gives

$$U \notin \text{Thick}_{\mathcal{C}_{\text{comp}}}(T') = \text{Thick}_{\mathbf{D}(\mathcal{N})}(T').$$

$\square$

Theorem 6.10, together with Proposition 5.4, establishes all three conditions of the transfer criterion: degree-wise finite homotopy, a finite complete construction, and failure of a global finite construction. The remaining step is to transfer this object to the claimed connective spectrum and record its chromatic consequences.

7 Comparison with chromatic homotopy theory

The algebraic construction and its obstruction are now complete. To turn them into a spectrum, we use the large-prime comparison between $E(2)$ -local homotopy theory and the periodic derived category on $\mathcal{M}$. Two identifications make that comparison useful here: the localized sphere must correspond to the pure unit, and the localized truncated Brown–Peterson spectrum must correspond to its coefficient-chart sheaf. The first recovers ordinary homotopy groups from algebraic sections. The second carries a proposed finite construction into the algebraic setting, where the coefficient-change argument will compare it with the obstruction on $\mathcal{N}$.

7.1 The exact comparison

Let $E = E(2)$ be the p -local Johnson–Wilson theory, using the Hazewinkel coefficient chart from Proposition 2.1. In this section let $p > 8$ and $D = 2(p - 1)$; in particular $D > 12$. This bound concerns the comparison. The operator construction imposes the larger prime bounds already used in the algebraic argument.

Theorem 7.1 (Large-prime algebraicity). *For $p > 8$ there is a triangulated equivalence*

$$\mathcal{F} : h\mathrm{Sp}_E \longrightarrow \mathbf{D}(\mathcal{M})$$

*whose sheaf homology is naturally identified, under descent and the periodic regrading below, with the E_*E -comodule homology functor $E_*(-)$. Its suspension structure gives natural isomorphisms*

$$\mathcal{F}(\Sigma^j Z) \cong (\mathcal{F}Z)[j] \quad (j \in \mathbb{Z}).$$

We use the construction in the November 14, 2018 version (v2) of Pstrągowski’s preprint. Its Corollary 6.9 gives the range $p > n^2 + n + 2$, hence $p > 8$ at height two, by comparing homotopy 2-categories [27, Corollary 6.9]. Two inputs to that construction need explicit support here: the bound 6 for Ext between all graded comodules, and convergence of the topological and algebraic potential-stage towers. The quotient-closure inference in the proof of v2 Theorem 2.4 does not follow from the long exact Ext sequence, and Proposition 4.4 defers the needed convergence. We first identify the target, then prove these inputs for this fixed $E(2)$, and finally follow the stage-zero homology and triangulated structures through the same comparison.

Write $\mathcal{D}(E_*E)$ for the stable localization of differential E_*E -comodules at homology isomorphisms, as in v2 Definitions 3.1–3.2 and Proposition 3.3 [27]. Its homotopy category is our twisted periodic model by the following degree regrading. All coefficient and cooperation degrees are multiples of D . For a differential comodule M in the homological convention, so that its differential lowers total degree by one, define the weight- s part of C_j by

$$(C_j)_s = M_{j+sD} \quad (j, s \in \mathbb{Z}).$$

The graded comodule structure becomes the weight equivariance for flat descent to $\mathcal{M}$. The definition gives $(C_{j+D})_s = (C_j)_{s+1}$, hence $C_{j+D} = C_j(1)$, and the differential gives $C_j \rightarrow C_{j-1}$. Conversely, write each total degree uniquely as $n = r + sD$ with $0 \leq r < D$ and set $M_n = (C_r)_s$. For $r > 0$ the differential is the component of $C_r \rightarrow C_{r-1}$. At $r = 0$, the identity $C_{-1} = C_{D-1}(-1)$ identifies its target with $(C_{D-1})_{s-1}$, the component of total degree $n - 1$; this is the required twist at the residue boundary. The coefficient action and coaction are recovered from the weight-graded ones. Degree-zero maps restrict to families of sheaf maps commuting with differentials and periodicity, and the inverse formula recovers every such map uniquely. The two constructions identify homology in every degree and weight, so they identify quasi-isomorphisms. They also identify shifts and mapping cones: the suspension convention is $(C[h])_j = C_{j-h}$ with differential multiplied by $(-1)^h$. In particular $C[D] \cong C(-1)$ with its differential sign, since D is even. Thus regrading gives an exact equivalence $h\mathcal{D}(E_*E) \simeq \mathbf{D}(\mathcal{M})$. It applies to all differential comodules and periodic complexes, including arbitrary unbounded and non-finite-type objects.

Lemma 7.2 (The height-two comparison inputs). *Let $\mathcal{C}_E = \mathrm{Comod}_{E_*E}$ be the ordinary abelian category of all graded comodules. Then*

$$\mathrm{Ext}_{\mathcal{C}_E}^s(M, N) = 0 \quad (M, N \in \mathcal{C}_E, \ s > 6).$$

This includes every internal grading shift. Let Syn_E be the Grothendieck prestable infinity-category of spherical hypercomplete sheaves of spaces on the site $\mathrm{Sp}_E^{\mathrm{fp}}$ of finite spectra P with E_*P projective over E_* , with covers the single maps surjective on E -homology. Its objects are connective synthetic spectra, and each satisfies

$$X \simeq \lim_{i \geq 0} X_{\leq i} \quad (X \in \mathrm{Syn}_E).$$

The ordinary unbounded derived infinity-category $\mathcal{D}_{\mathrm{ord}}(\mathcal{C}_E)$ is left complete as well. For the topological potential stages of the unit shift algebra on this same Syn_E , and the algebraic potential stages of the periodicity algebra on $\mathcal{D}_{\mathrm{ord}}(\mathcal{C}_E)_{\geq 0}$, the natural functors are equivalences

$$\mathcal{M}_{\infty}^{\mathrm{top}} \simeq \lim_{\ell} \mathcal{M}_{\ell}^{\mathrm{top}}, \quad \mathcal{M}_{\infty}^{\mathrm{alg}} \simeq \lim_{\ell} \mathcal{M}_{\ell}^{\mathrm{alg}}.$$

These are the two convergence inputs ($\spadesuit 5$) of *v2 Propositions 4.4 and 4.8*.

Proof. Put

$$B = E_* = \mathbb{Z}_{(p)}[v_1^{\mathrm{Haz}}, (v_2^{\mathrm{Haz}})^{\pm 1}],$$

$$I_0 = (0), \quad I_1 = (p), \quad I_2 = (p, v_1^{\mathrm{Haz}}), \quad Q_h = B/I_h \quad (0 \leq h \leq 2).$$

The ideals I_h are invariant under the chart Hopf algebroid, by (2.2) and the triangular coordinate change. Because the coefficient and cooperation degrees lie in $D\mathbb{Z}$, the degree phases split $\mathcal{C}_E$ into a product of D abelian categories. After shifting a phase and reindexing its degrees as weights, flat descent on the chart of Proposition 2.1 identifies it with $\mathrm{QCoh}(\mathcal{M})$. This is abelian descent, independent of the differential regrading and of topological algebraicity.

Supported unit-source bounds. For a comodule V annihilated by I_h , the supported unit-source bounds are

$$\mathrm{Ext}_{\mathcal{C}_E}^a(B, V) = 0 \quad (a > 6 - h),$$

also after arbitrary internal shifts. For $h = 0$ this is the bound on $\mathcal{M}$ in Proposition 2.12. For $h = 1$, the closed-base-compatible coinduced resolution in Lemma 2.10 identifies the sections calculation with the characteristic- p one, where that proposition gives the bound 5.

For $h = 2$, regard the corresponding sheaf as a sheaf on $\mathcal{M}_0$ and extend it to $\mathcal{M}_k$, writing V_k for the result. The congruence proved in Remark 7.5 identifies v_1^{Haz} modulo p with the Cartier section t , so $tV_k = 0$. The central action is $\mathcal{O}$ -linear. Consequently both $r_*V_k = \ker(c - 1)$ and $R^1r_*V_k = \mathrm{coker}(c - 1)$ are still killed by t , and the higher derived invariants vanish, by Proposition 2.12. Since r^* is exact, r_* preserves injectives, and the derived-composite spectral sequence is

$$E_2^{a,b} = H^a(\mathcal{N}, R^b r_* V_k) \implies H^{a+b}(\mathcal{M}_k, V_k).$$

There are only rows $b = 0, 1$. The closed-base resolution and Lemma 2.11 apply to both rows on $\mathcal{N}/(t)$ and give vanishing for $a > 3$, for arbitrary sheaves. Thus the total degree is at most 4. Finite flat extension $\mathbb{F}_p \rightarrow k$ commutes with the coinduced sections resolution and detects vanishing. The closed-base compatibility of that resolution then returns the bound from $\mathcal{M}_0$ to $\mathcal{M}$. These three bounds are uniform under weight twists. They are bounds with unit source; the following argument is needed to obtain the all-pairs assertion.

From supported units to all pairs. The cooperation Hopf algebroid is flat and Adams. Indeed the empty regular sequence in Hovey's Proposition 1.4.8 gives topological flatness of BP; Theorem 1.4.9 carries it to its Landweber-exact module E , and Lemma 1.4.6 uses the commutative cooperation algebra to give the Adams property [16]. Propositions 1.4.4 and 1.4.1 there imply that $\mathcal{C}_E$ is Grothendieck, that its dualizable comodules generate, and that every comodule finitely generated over B is a quotient of a dualizable one. By Proposition 1.3.4 there, dualizability means that the underlying B -module is finitely generated projective.

The ring B is noetherian, and the regular sequence defining I_h gives Q_h a graded B -module Koszul resolution of length h . Starting with $K_0 = Q_h$, choose dualizable epimorphisms $P_j \rightarrow K_j$ and put $K_{j+1} = \ker(P_j \rightarrow K_j)$ for $0 \leq j < h$. Noetherianity keeps these kernels finitely generated over B . For every graded B -module W , dimension shifting gives

$$\mathrm{Ext}_B^1(K_h, W) \cong \mathrm{Ext}_B^{h+1}(Q_h, W) = 0.$$

Thus K_h is projective over B and is dualizable. For $h = 0$ the object $Q_0 = B$ already is dualizable. This constructs in every case a length- h resolution $P^\bullet \rightarrow Q_h$ by dualizable comodules, in ordinary cohomological degrees $-h, \dots, 0$.

For an arbitrary comodule V , the bounded complex $(P^\bullet)^\vee \otimes_B V$ computes the internal derived Hom from Q_h in $\mathcal{D}_{\mathrm{ord}}(\mathcal{C}_E)$. To see this within that category, each dualizable term has evaluation and coevaluation, and its tensor functor is exact; bounded totalization therefore preserves quasi-isomorphisms and the duality descends to the ordinary derived category. The underlying complex is the graded $\mathrm{Hom}_B(P^\bullet, V)$. Its cohomology in degree b is $\mathrm{Ext}_B^b(Q_h, V)$, including its internal grading, and vanishes outside $0 \leq b \leq h$. It is annihilated by I_h : the B -action on this Ext can be taken through its source Q_h , or, equivalently, multiplication by an element of I_h lifts to a null-homotopic map on a projective resolution. Since I_h is invariant, these cohomology objects are supported comodules. The bounded hypercohomology spectral sequence is consequently

$$E_2^{a,b} = \mathrm{Ext}_{\mathcal{C}_E}^a(B, H^b((P^\bullet)^\vee \otimes_B V)) \implies \mathrm{Ext}_{\mathcal{C}_E}^{a+b}(Q_h, V).$$

Its entries have $a \leq 6 - h$ by the supported unit bounds and $0 \leq b \leq h$ by the resolution. Hence $\mathrm{Ext}_{\mathcal{C}_E}^{>6}(Q_h, V) = 0$ for arbitrary V , uniformly in internal shifts. This calculation uses a bounded dualizable comodule resolution in the ordinary derived category; it requires no comparison with a different derived category of sheaves.

By the Landweber filtration theorem [18, Theorem 5.7], every comodule finitely presented over B has a finite filtration by internal shifts of Q_0, Q_1, Q_2 . The long exact Ext sequence therefore gives the same bound 6 for every such source and every target. Let $\mathcal{U}$ be this class of finite sources. It is quotient closed: a quotient remains finitely generated over the noetherian ring B , hence finitely presented, and comodule quotients have the underlying module quotients. It also detects every nonzero comodule, because the primitive theorem [18, Theorem 5.1] supplies a nonzero map from an internal shift of B to each one.

We use the partial-section argument underlying Franke's injective-truncation criterion [12, Section 3.4.3, Lemma 2]. In a Grothendieck abelian category, let a quotient-closed class $\mathcal{U}$ detect nonzero objects. If K is injective and $\pi : K \rightarrow L$ induces surjections $\mathrm{Hom}(U, K) \rightarrow \mathrm{Hom}(U, L)$ for every $U \in \mathcal{U}$, then π admits a section. Indeed partial sections on subobjects of L form a set, by well-poweredness and local smallness, and exact filtered unions give upper bounds for chains. Choose a maximal partial section with domain L_0 . If $L_0 \neq L$, choose a nonzero map from an object of $\mathcal{U}$ to L/L_0 . Its image V still belongs to $\mathcal{U}$ by quotient closure; let L_1 be its preimage in L . Injectivity extends the partial section to a map $\tilde{s} : L_1 \rightarrow K$. The discrepancy $\pi\tilde{s} - (L_1 \hookrightarrow L)$

vanishes on L_0 , so it factors through a map $V \rightarrow L$. Lift that map to K by the Hom surjection and subtract its composite with $L_1 \rightarrow V$. This extends the section to L_1 , contradicting maximality.

For any target N , take an injective resolution $0 \rightarrow N \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots$ and set $Z^7 = \ker(I^7 \rightarrow I^8)$. Dimension shifting and the finite-source bound show that $I^6 \rightarrow Z^7$ induces surjections on $\text{Hom}(U, -)$ for every $U \in \mathcal{U}$: their cokernels are $\text{Ext}^7(U, N) = 0$. The criterion splits this epimorphism. Its kernel is an injective direct summand of I^6 , so the resolution truncates at degree 6. Thus every target has injective dimension at most 6, proving the stated Ext bound for all sources and targets.

Postnikov completeness. It remains to prove the two convergence assertions. We apply Pstrągowski–VanKoughnett’s general synthetic construction and the argument of their Theorem 7.4, with locators in arXiv v4 [28]. The theory $E(2)$ is Adams-type by their Example 6.2. The site in the statement consists of the unlocalized finite spectra P and its stated cover maps. Taking all its objects and their identity covers in their Lemma 7.2 gives the colimit-generating family

$$G_P = \nu(L_E P) \quad \text{in } \text{Syn}_E.$$

For $X \in \text{Syn}_E$, put $X[i] = \text{fib}(X_{\leq i} \rightarrow X_{\leq i-1}) \simeq \Sigma^i \pi_i X$. This layer is canonically a module over the unit truncation $\mathbf{1}_{\leq 0}$. Their Theorem 3.11 together with Proposition 6.4 identifies that module category with $\mathcal{D}_{\text{ord}}(\mathcal{C}_E)_{\geq 0}$, and identifies the reduction of G_P with $E_* P$. By the module adjunction, for $j \geq 0$ we therefore have

$$\pi_j \text{Map}_{\text{Syn}_E}(G_P, X[i]) \cong \text{Ext}_{\mathcal{C}_E}^{i-j}(E_* P, \pi_i X).$$

This vanishes when $i > 6 + j$, uniformly in P and in the arbitrary layer comodule $\pi_i X$. The fiber sequences of the Postnikov tower therefore stabilize these mapping-space homotopy groups.

For precision, the completeness criterion in the proof of their Theorem 7.4 asks that, for every $m \geq 0$, the projection $\lim_i X_{\leq i} \rightarrow X_{\leq k}$ become an m -equivalence for large k , meaning an equivalence after m -truncation. It is enough that the induced maps on mapping spaces out of the colimit generators be $(m+1)$ -equivalences. Representable mapping spaces preserve limits, so the preceding stabilization and the Milnor exact sequence give this condition. Comparing with $X \rightarrow X_{\leq k}$ and using separatedness of Syn_E proves $X \simeq \lim_i X_{\leq i}$. The literal Theorem 7.4 is stated for Morava E -theory; this is its proof applied on the stated $E(2)$ site, using Lemma 7.2 and the local uniform Ext bound in place of its Morava-specific finite-source input.

Potential-stage convergence. This objectwise Postnikov completeness supplies the separate formal tower step. Proposition 3.2 of the same source recovers unit-module categories as the limit of the categories of modules over the unit truncations; Definition 4.1 and Remark 4.4 restrict this equivalence to the potential stages. Proposition 6.4 identifies the periodic unit modules in this synthetic category with Sp_E . On the algebraic side the finite homological dimension just proved gives left completeness of $\mathcal{D}_{\text{ord}}(\mathcal{C}_E)$ by the argument recorded in Remark 7.5. The same formal module and potential-stage step applies to the algebraic periodicity algebra of v2 Proposition 4.8. This proves both displayed tower equivalences in the exact categories used by the comparison. $\square$

Proof of Theorem 7.1. We now follow the comparison in Pstrągowski v2 [27]. Propositions 4.4 and 4.8, item ($\spadesuit 6$), identify stage-zero reduction with E -homology on the topological side and differential-comodule homology on the algebraic side. Lemma 5.3 identifies the bottom functor on finitely projective comodules with their inclusion into the standard comodule heart, and its

cocontinuous extension gives the heart identification. Lemmas 5.6 and 5.8 identify the heart and the homotopy objects used by the stages. Corollary 6.5 uses the preferred unit/free-algebra map and commutes with reduction to stage zero; Corollary 6.7 restricts that comparison to potential stages.

Lemma 7.2 supplies the dimension $d = 6$ and the two convergence inputs used by v2 Theorem 4.10. Its stabilization stage is $\ell = d + k - 1$ at homotopy level k , hence $\ell = 7$ at level two. Theorem 6.8, in the stated Corollary 6.9 range $p > 8$, then gives the zigzag of homotopy 2-category equivalences

$$h_2\mathrm{Sp}_E \xrightarrow{\sim} h_2\mathcal{M}_7^{\mathrm{top}} \simeq h_2\mathcal{M}_7^{\mathrm{alg}} \xleftarrow{\sim} h_2\mathcal{D}(E_*E).$$

Choose an inverse to the final leg. Compose from left to right and then apply the regrading above on homotopy categories. Call the resulting equivalence $\mathcal{F}$. The stage-zero compatibility throughout the restriction and assembly gives its natural identification of sheaf homology with $E_*(-)$.

The recovery of triangles uses Appendix B of v2, with one local correction: the proof of Lemma B.3 overstates the local connectivity of the projection from a stable infinity-category to its homotopy 2-category. Its needed uniqueness follows directly from the cone mapping space. Write $\mathcal{H} = h_2\mathcal{T}$ for a stable infinity-category $\mathcal{T}$, and fix a span $\sigma = (B \leftarrow A \rightarrow C)$. The space of cones with vertex W is

$$\mathrm{Cone}_\sigma(W) \simeq \mathrm{Map}_{\mathcal{H}}(B, W) \times_{\mathrm{Map}_{\mathcal{H}}(A, W)}^h \mathrm{Map}_{\mathcal{H}}(C, W),$$

where the homotopy pullback retains the cone's 2-cell. If q_c is a cone with vertex V and q is any target cone with vertex W , Lemma A.5 identifies the mapping space of cones as

$$\mathrm{Map}_{\mathcal{H}_{\sigma_j}}(q_c, q) \simeq \mathrm{hofib}_q(\mathrm{Map}_{\mathcal{H}}(V, W) \longrightarrow \mathrm{Cone}_\sigma(W)).$$

The arrow postcomposes the entire cone q_c with a vertex map. When q_c is 1-cocartesian, Definition B.1 says that the displayed comparison is 1-connected. Its fiber over the arbitrary cone q is therefore nonempty and connected, so it has exactly one π_0 class. Existence follows by lifting the span to $\mathcal{T}$, forming a pushout, and applying Lemma B.2 to its image. This proves the existence and full uniqueness needed in Lemma B.3. Corollary B.4 then identifies any two such cones. The arbitrary-target form also gives suspension maps and their naturality: for $f : X \rightarrow Y$, precompose the chosen cone under $0 \leftarrow Y \rightarrow 0$ to obtain a cone under $0 \leftarrow X \rightarrow 0$ with vertex ΣY , and apply that uniqueness. The two-dimensional lifting in Theorem B.6, together with Lemma B.2 and Corollary B.4, consequently recovers distinguished triangles, their boundary classes, and the natural suspension comparison; Corollary B.7 makes the induced equivalence triangulated.

Apply this recovery to the chosen equivalence of homotopy 2-categories and then use the exact regrading. It equips the same $\mathcal{F}$ with its triangulated structure. Iterating its natural suspension isomorphism and its inverse gives the displayed isomorphisms for every integer j . $\square$

Franke introduced the differential-object model and realization construction [12, Section 2.2 and Theorem 10 of Section 3.5]. Patchkoria identified a gap in the general equivalence argument [25, Remark 6.3.1]. The comparison just justified is Pstrągowski's later Goerss–Hopkins construction; these results do not identify it with Franke's particular functor. Patchkoria–Pstrągowski give a homology-theoretic formulation of the comparison [26, Theorems 7.56 and 8.13], and Barkan proves a tensor-triangulated strengthening in its stated prime range [4, Theorem 1.1 and Corollary 3.42]. We use the chosen exact comparison, its homology identification, and distinguished triangles and retracts. The inverse limits, completions, and gluing used to construct the algebraic object were formed in the algebraic homotopy theory before realization; no monoidal or stable infinity-category comparison between the topological and algebraic theories is used.

7.2 The Brown–Peterson quotient and its comodule

Recall the rings $P = \mathrm{BP}_*$ and $B_2 = \mathbb{Z}_{(p)}[v_1, v_2]$, the chart $f : \mathcal{X} \rightarrow \mathcal{M}$, and the sheaves

$$T_{\mathbb{Z}} = f_* \mathcal{O}_{\mathcal{X}}, \quad T_{\mathbb{Z}_p} = f_*(\mathcal{O}_{\mathcal{X}} \otimes_{\mathbb{Z}_{(p)}} \mathbb{Z}_p).$$

The comparison theorem identifies homology, but this alone does not identify the image of a chosen spectrum with a pure sheaf. We first calculate the comodule of a standard Brown–Peterson quotient; the sparsity argument in the next subsection will then identify the object itself.

We use the regular-quotient form b of $\mathrm{BP}\langle 2 \rangle$. The existence of an $\mathbb{E}_4$ multiplication on BP [7, Theorem 1.1 and Corollary 1.2] gives a monoidal module category in which to form the cofibers of the scalar maps v_i , $i > 2$. Taking these regular quotients successively and then their sequential colimit gives a BP -module with coefficients B_2 . Indeed the scalar action on the homotopy of a module is multiplication by the corresponding coefficient: homotopy classes are maps out of shifts of the monoidal unit, and tensor naturality identifies the scalar action with composition at that unit.

We explain why the enhanced multiplication causes no change in the cooperation formulas used below. An $\mathbb{E}_4$ ring with underlying spectrum BP has even homotopy and admits a complex orientation. The orientation determines a unital multiplicative map from $MU_{(p)}$ in the homotopy category: its Thom classes multiply under Whitney sum. Composing with the multiplicative inclusion of the Quillen summand gives a homotopy ring map from the usual BP to this enhancement; see [29, Theorem 4.1.12 and Lemma 4.1.13]. Both $MU_{(p)}$ and $MU_{(p)} \wedge MU_{(p)}$ have filtrations by finite even-cell skeleta. Maps from their suspended finite skeleta to the even target vanish by cellular induction. The corresponding Milnor $\lim^1$ terms therefore vanish, so the compatible Thom maps and their multiplicativity identities pass to the whole spectra without phantom ambiguities. This is the even-cell case of the no-even-phantoms criterion in [17, Lemma 2.11, Corollary 2.15, and Proposition 2.18].

This map is an equivalence. It induces a unit on degree-zero mod- p cohomology, since the unit of $\pi_0 \cong \mathbb{Z}_{(p)}$ is not p -divisible. The Steenrod module $H^*(\mathrm{BP}; \mathbb{F}_p)$ is cyclic on this class, and source and target have the same finite dimensions in each degree. The induced map is therefore an isomorphism on mod- p cohomology. Its cofiber is connective and of finite type over $\mathbb{Z}_{(p)}$; lowest-degree Hurewicz and Nakayama show that this cofiber vanishes. We may accordingly choose the usual coefficient generators and cooperation formulas for the multiplication acting on these modules. Naturality of multiplication and of the unit insertions identifies the units and coproduct on cooperations under this homotopy ring equivalence. All module constructions remain in the module category of the chosen $\mathbb{E}_4$ enhancement; this argument requires no refinement of the comparison map to a map of structured rings.

Proposition 7.3 (The quotient comodule). *For the regular quotient b just constructed,*

$$\mathrm{BP}_* b \cong \Gamma_P \otimes_{\eta_R, P} B_2.$$

Under Landweber base change to E and descent to $\mathcal{M}$, this comodule is the pure sheaf $T_{\mathbb{Z}}$. The corresponding comodule for $b_p^\wedge$ is $T_{\mathbb{Z}_p}$. Both sheaves are p -monic, and their reductions modulo p equal T_0 .

Proof. There is a relative smash identification

$$\mathrm{BP} \wedge b \simeq (\mathrm{BP} \wedge \mathrm{BP}) \otimes_{\mathrm{BP}} b.$$

As a right BP-module, $\mathrm{BP} \wedge \mathrm{BP}$ is a wedge of free shifts: lift a right-module basis of its homotopy to obtain the equivalence. Thus the flat Künneth calculation gives the asserted tensor product. Under that identification, insertion of units, which defines the coaction, agrees with the cooperation coproduct. Only the homotopy cooperation formulas, not a further multiplicative enhancement, enter this calculation.

Landweber exactness now gives, on the Hazewinkel chart,

$$A_{\mathrm{Haz}} \otimes_{P, \eta_L} \Gamma_P \otimes_{\eta_R, P} B_2.$$

These are the functions on strict arrows with the chosen chart law at the left and a $\mathcal{X}$ -law at the right, with grading frames identified. Descent is composition of arrows. This is precisely the pullback of $f_* \mathcal{O}_{\mathcal{X}}$. All degrees lie in $D\mathbb{Z}$, so its regraded sheaf is pure. Since f is affine and $\mathcal{O}_{\mathcal{X}}$ is p -monic, so is $T_{\mathbb{Z}}$.

The same calculation also identifies the completed generator. The completion $b_p^\wedge$ is the homotopy limit of b/p^a . The tower is surjective on homotopy, so its coefficients are $B_2 \otimes_{\mathbb{Z}_{(p)}} \mathbb{Z}_p$, with ordinary completion in each graded degree. The limit is complete by its Postnikov description and has the same multiplication-by- p cofiber as b . It remains a BP-module, and the same free-right-module calculation gives $T_{\mathbb{Z}_p}$. Its p -monicity and reduction follow on the affine chart and then by affine pushforward. $\square$

We have now matched the quotient coefficients and coaction to the chart. To ensure that this particular regular quotient is a standard truncated Brown–Peterson spectrum, we check its Steenrod module.

Proposition 7.4 (Standard mod- p cohomology). *At an odd prime, the quotient b has the standard Steenrod module*

$$H^*(b; \mathbb{F}_p) \cong \mathcal{A} // E(Q_0, Q_1, Q_2),$$

where Q_i are the Milnor primitives and $E(Q_0, Q_1, Q_2)$ is their exterior subalgebra. In particular its completion is a standard form of $\mathrm{BP}\langle 2 \rangle_p^\wedge$.

The displayed Steenrod module is classical [32, Proposition 1.7]; we verify it for the module quotient chosen here.

Proof. Connective BP-module truncation and reduction give a map $b \rightarrow H\mathbb{F}_p$. For $N = b, H\mathbb{F}_p$, consider the connective Künneth spectral sequence for

$$H\mathbb{F}_p \otimes_{\mathrm{BP}} (\mathrm{BP} \wedge N),$$

where BP acts on the first factor of $\mathrm{BP} \wedge N$. The free-right-module calculation in Proposition 7.3 applies to both choices of N . The map on Tor terms is therefore

$$\mathbb{F}_p[t_1, t_2, \dots] \otimes \bigwedge (\epsilon_i : i > 2) \longrightarrow \mathbb{F}_p[t_1, t_2, \dots] \otimes \bigwedge (\epsilon_i : i \geq 0), \quad |\epsilon_i| = 2p^i - 1,$$

and is the inclusion. To compute it, use the regular Koszul resolutions of the right-base-change quotients of Γ_P , on $\eta_R v_i$ for $i > 2$, and on $p, \eta_R v_i$ for $i \geq 1$, respectively. These resolutions are free over the left base. After tensoring with the left $\mathbb{F}_p$, every differential vanishes.

The target spectral sequence collapses by the dimensions in Milnor’s dual Steenrod algebra calculation, recalled in [29, Theorem 3.1.1(b)]. The injective map and naturality then force the source to collapse page by page. Connectivity gives convergence, and the resulting map on mod- p homology is injective. Dually $\mathcal{A}$ surjects onto $H^*(b; \mathbb{F}_p)$. There are no classes in the odd degrees of Q_0, Q_1, Q_2 , so these primitives kill the degree-zero generator. The resulting quotient map from $\mathcal{A} // E(Q_0, Q_1, Q_2)$ is an isomorphism by the same dimension calculation. $\square$

Remark 7.5 (Choice of generators and forms). The quotient comparison can equally be made with the Hazewinkel chart. Its characteristic- p reduction agrees with the Araki chart used here. The congruence between the Araki and Hazewinkel generators modulo p is classical [29, Theorem A2.2.3]; we give the calculation in our notation. If l_i are the logarithm coefficients, subtraction of the two logarithm recursions gives

$$v_n - v_n^{\text{Haz}} = -p^n l_n + \sum_{0 < i < n} l_i ((v_{n-i}^{\text{Haz}})^{p^i} - v_{n-i}^{p^i}).$$

The facts that $p^i l_i \in P$ and that raising to p^i improves a congruence modulo p to one modulo p^{i+1} prove inductively that $v_n \equiv v_n^{\text{Haz}} \pmod{p}$. Thus the subsequent reduction argument does not depend on this coordinate choice.

The quotient b is connective and each $\pi_i b$ is finitely generated over $\mathbb{Z}_{(p)}$. Together with the Steenrod-module calculation above, the completed uniqueness theorem of Angeltveit–Lind [2], with Lee’s corrected proof, identifies $b_p^\wedge$ with the standard underlying spectrum $\text{BP}\langle 2 \rangle_p^\wedge$ [20, Theorem 1.1 and Remark 1.4]; see also [21, Note 2]. For independence of the quotient generators already before completion, see [20, Corollary 1.9]. That statement concerns underlying spectra; the BP-module structures may differ. For a refutation of the universal thick generation assertion, the specified standard quotient form already suffices.

7.3 Identifying the pure representatives

We have identified the homology of the comparison objects. The remaining step is to rule out nontrivial attachments between copies of that homology in different periods. The cohomological bounds from Section 2 do this: the next lemma makes precise how a gap in homology forces a periodic complex to be pure.

Lemma 7.6 (Sparse representatives). *In a twisted periodic derived category of period D , suppose a complex has homology only in one residue modulo D , with sheaf W in a chosen representative of that residue. It is isomorphic to the corresponding pure object in either of the following cases:*

1. W has projective dimension at most a in the Ext sense and $D > a + 1$;
2. W has injective dimension at most b and $D > b + 1$.

The assertion applies to every shift of the complex.

Proof. Shift the chosen residue to degree zero. For the first assertion, take a fibrant representative C with injective terms and write B_j, Z_j for its boundaries and cycles. Lifting the isomorphism $W \rightarrow H_0 C$ to $Z_0 C$ is obstructed by $\text{Ext}^1(W, B_0 C)$. In the intervening positive degrees the homology vanishes, so the short exact sequences

$$0 \longrightarrow B_j C \longrightarrow C_j \longrightarrow B_{j-1} C \longrightarrow 0$$

dimension-shift this obstruction through injective terms. Before reaching the next nonzero homology residue it becomes an Ext group of degree greater than a , and hence vanishes. A lift gives a map from the pure W into C . Repeat it by the period-twist identifications. It induces the given isomorphism on every homology sheaf and is a quasi-isomorphism.

For the second assertion take an injective resolution of W in degrees $0, -1, \dots, -b$, and periodize it. Map C into this resolution. The isomorphism on $Z_0 C/B_0 C$ extends to $C_0/B_0 C$ with values in

the first injective. Its composite with the next differential vanishes on cycles, so it prescribes a map on $B_{-1}C$; extend that map to C_{-1} by injectivity. Continue in this way. At every intermediate step, vanishing of homology identifies the cycles with the boundaries on which the chain-map equation has already been prescribed. The finite resolution ends before the next homology residue, so its periodically repeated copies do not interfere. The resulting map is again a quasi-isomorphism. The same arguments begin in any chosen residue, proving the assertion for all shifts. $\square$

Proposition 7.7 (The comparison generators). *For every integer j ,*

$$\begin{aligned}\mathcal{F}(\Sigma^j L_E S^0) &\cong \mathcal{O}[j], \\ \mathcal{F}(\Sigma^j L_E b) &\cong T_{\mathbb{Z}}[j], \\ \mathcal{F}(\Sigma^j L_E b_p^\wedge) &\cong T_{\mathbb{Z}_p}[j].\end{aligned}$$

Proof. The sphere has pure comodule homology $\mathcal{O}$, and Proposition 2.12 gives $\text{Ext}^{>6}(\mathcal{O}, -) = 0$. Thus the first part of Lemma 7.6 identifies its image, since $D > 12 > 6 + 1$.

For the chart objects we claim injective dimension at most 5. Resolve $\mathcal{O}_{\mathcal{X}}$ by its sheaf Čech resolution for the three principal opens, of length 2. On each quotient-affine chart the localized regular polynomial ring has global dimension at most 3. Hence each term has injective dimension at most 3 after pushforward. This assertion applies to arbitrary graded modules: a graded free resolution is also an ungraded free resolution, and in its Hom complex the projection of every homogeneous input to the output of matching degree splits off the grading-preserving Hom complex. Its Ext groups are therefore summands of the ordinary Ext groups. Flat-affine adjunction for f preserves the injective-dimension bound. Combining the length-2 resolution with the bound 3 gives the bound 5 for $T_{\mathbb{Z}}$.

The same argument applies after tensoring the chart sheaf with $\mathbb{Z}_p$: global dimension bounds hold for arbitrary modules over the localized regular rings, so no finite-generation assertion over $\mathbb{Z}_{(p)}$ is needed. It gives injective dimension at most 5 for $T_{\mathbb{Z}_p}$. Proposition 7.3, homology compatibility, and the second part of Lemma 7.6 now give the remaining identifications, since $D > 5 + 1$. The bounds are unchanged by the weight twists, so the sparsity proof applies in every residue. Equivalently, once the degree-zero objects have been identified, the suspension structure of $\mathcal{F}$ gives the displayed isomorphisms for every j . $\square$

Proposition 7.7 completes the two identifications needed for transfer. For every realized algebraic object, maps from the pure unit compute ordinary homotopy groups. Moreover, exactness of $\mathcal{F}$ sends every finite construction from $L_E b_p^\wedge$ to a finite construction from $T_{\mathbb{Z}_p}$: it carries shifts, distinguished triangles, and retracts to the corresponding operations. The transfer argument will use precisely these two consequences.

The main counterexample proof continues in Section 8. The following subsection gives an independent categorical extension.

7.4 A transport principle without exactness

There is a useful weaker principle for comparisons that are known only as equivalences of underlying categories. In that setting the functor need not identify suspension, so all shifts must appear in the membership condition. This principle is independent of the exact comparison chosen above.

Lemma 7.8 (Transport of thick membership). *Let $F : \mathcal{C} \rightarrow \mathcal{D}$ be an equivalence of the underlying categories of triangulated categories, and let $\mathcal{D}_0$ be a thick subcategory of $\mathcal{D}$. Then*

$$\{X \in \mathcal{C} : F(\Sigma^j X) \in \mathcal{D}_0 \text{ for every } j \in \mathbb{Z}\}$$

is a thick subcategory of $\mathcal{C}$.

Proof. Equivalences preserve zero objects, finite biproducts, and retracts. Closure under shifts is built into the definition. To prove closure under triangles, consider

$$A \longrightarrow B \longrightarrow C \longrightarrow \Sigma A$$

and first suppose only that $F(B), F(\Sigma A) \in \mathcal{D}_0$. Complete the map $F(B) \rightarrow F(C)$ to a triangle

$$F(B) \longrightarrow F(C) \xrightarrow{s} D_1 \longrightarrow F(B)[1].$$

The map s kills $F(B) \rightarrow F(C)$. Full faithfulness and essential surjectivity transport the exactness of represented sequences in $\mathcal{C}$, so s factors through the image of the next map:

$$s = hF(C \rightarrow \Sigma A), \quad h : F(\Sigma A) \longrightarrow D_1.$$

Form a weak homotopy pullback P of s along h . Its triangle is

$$F(B) \longrightarrow P \longrightarrow F(\Sigma A) \longrightarrow F(B)[1],$$

so $P \in \mathcal{D}_0$. The pair consisting of the identity of $F(C)$ and $F(C \rightarrow \Sigma A)$ has equal composites to D_1 . The exact sequence defining the weak pullback therefore gives a map $F(C) \rightarrow P$ whose composite with $P \rightarrow F(C)$ is the identity. Thus $F(C)$ is a retract of P and belongs to $\mathcal{D}_0$. Weak pullbacks here can be constructed from fiber triangles and the octahedral axiom.

Now suppose A and B belong to the subcategory in the statement. Apply the preceding argument to each shifted triangle: its two inputs are $F(\Sigma^j B)$ and $F(\Sigma^{j+1} A)$, both in $\mathcal{D}_0$. It follows that $F(\Sigma^j C) \in \mathcal{D}_0$ for every j . Rotating triangles gives the other two-out-of-three implications. $\square$

Even under this weaker hypothesis, the equivalence preserves addition on Hom sets: addition is determined by biproduct diagonals and folds. Consequently an identification of maps out of the sphere obtained from such a comparison is an identification of abelian groups, as well as of their underlying sets.

8 Transfer to a topological counterexample

Theorem 6.10 completes the algebraic construction on $\mathcal{N}$. We now turn that object into a topological counterexample. The comparison of Section 7 takes place over the integral stack $\mathcal{M}$. Recall the coefficient maps

$$\mathcal{N} \xleftarrow{r} \mathcal{M}_k \xrightarrow{e} \mathcal{M}_0 \xrightarrow{i} \mathcal{M},$$

where r removes the central automorphisms, e is the finite field extension, and i is the characteristic- p inclusion. The main algebraic step is to reverse a hypothetical finite construction through these maps. The comparison triangles for reduction and removal of the center may be nonsplit; their boundary maps will instead be nilpotent.

Proposition 8.1 (Transfer criterion). *Let $p > 8$, with $D = 2(p - 1) > 12$, and choose the comparison of Theorem 7.1. Suppose $U \in \mathbf{D}(\mathcal{N})$ satisfies*

$$\pi_i U \text{ is finite for every } i \in \mathbb{Z}, \quad \widehat{U} \in \text{Thick}(\widehat{T}'), \quad U \notin \text{Thick}(T'). \quad (8.1)$$

Choose an $E(2)$ -local spectrum Z_{sp} whose image under $\mathcal{F}$ is

$$\widetilde{U} = i_* e_* r^* U \in \mathbf{D}(\mathcal{M}).$$

Then

$$X_{\text{sp}} = \tau_{\geq 0} Z_{\text{sp}}$$

is a connective p -complete fp-spectrum of fp-type at most 2, and

$$L_{E(2)} X_{\text{sp}} \simeq Z_{\text{sp}}, \quad X_{\text{sp}} \notin \text{Thick}_{\text{Sp}_p^\wedge}(b_p^\wedge).$$

Here b is the standard regular quotient of Proposition 7.4.

The proposition is stated for any U with these three properties; the object already constructed will be its final application. The proof has two parts. First we construct derived reduction $Li^* : \mathbf{D}(\mathcal{M}) \rightarrow \mathbf{D}(\mathcal{M}_0)$ and, writing $F = e_* r^* U$, establish the implications

$$\begin{aligned} Li^* i_* F \in \text{Thick}(T_0) &\implies F \in \text{Thick}(T_0) \\ &\implies r^* U \in \text{Thick}(T_k) \implies U \in \text{Thick}(T'). \end{aligned}$$

The last conclusion contradicts the third hypothesis. We then use the comparison of the pure unit to prove that the connective cover is p -complete and satisfies the *total* finiteness test for fp-type.

8.1 Derived reduction and its obstruction

Call a sheaf, or a term of a complex, *p-monic* if multiplication by p is injective on it. Derived reduction can be constructed using such objects even though the characteristic- p inclusion is not flat. The following triangle records exactly how reducing an object already in characteristic p differs from returning that object unchanged.

Lemma 8.2 (Reduction triangle). *There is an exact functor*

$$Li^* : \mathbf{D}(\mathcal{M}) \longrightarrow \mathbf{D}(\mathcal{M}_0)$$

which on a termwise p -monic complex is ordinary reduction modulo p . For $F \in \mathbf{D}(\mathcal{M}_0)$ it has a natural comparison triangle

$$F[1] \longrightarrow Li^* i_* F \longrightarrow F \xrightarrow{\delta_F} F[2]. \quad (8.2)$$

Proof. We first construct enough p -monic epimorphisms of sheaves. On the whole graded BP stack, the cofree-on-the-base comodule Γ_P is a union of graded dualizable subcomodules which are finite free over the left base. Specifically, take the left spans of the arrow monomials of bounded weight. There are finitely many such monomials at each bound. The coproduct preserves these spans because it is homogeneous, all base and arrow coefficients have nonnegative weight, and left scalars remain in the outer factor of the coaction.

For a graded comodule M , the diagonal tensor $\Gamma_P \otimes_P M$ is cofree on the underlying graded module of M . This is also the projection formula for the affine chart $[\text{Spec } P/\mathbb{G}_m] \rightarrow \mathcal{M}_{\text{all}}$ of

the whole stack. A homogeneous element of M therefore corresponds to a map from a grading shift of the unit into this diagonal tensor. It factors through $N \otimes_P M$ for one of the finite-free subcomodules N : primitives commute with the filtered union, or equivalently the given coaction expression has finite support. The inclusions of these spans are split on underlying left modules, so tensoring with M preserves the injections and the factorization respects coactions. Duality turns it into a map from a shifted $N^\vee$ to M . Evaluation at the counit restricted to N shows that the original element is in its image. Summing these maps gives an epimorphism from a sum of dualizables to every graded comodule.

To apply this on $\mathcal{M}$, extend a quasi-coherent sheaf by pushforward from the height-open into the whole stack. This pushforward is quasi-coherent: on the flat affine charts the open is quasi-compact, and its pushforward commutes with flat base change. Restrict the epimorphism just constructed. The restricted dualizables are still p -monic, so $\mathcal{M}$ has enough p -monic epimorphisms.

For a twisted periodic complex C , use disk sums on these generators to choose a functorial degreewise epimorphism $P_C \rightarrow C$. Its kernel K_C is termwise p -monic, as is P_C . The natural map

$$\mathrm{cofib}(K_C \rightarrow P_C) \rightarrow C$$

is a quasi-isomorphism, and its source is termwise p -monic. The same construction works within the full subcategory of termwise p -monic complexes, so that subcategory has the same quasi-isomorphism localization as all complexes. An acyclic complex in the subcategory stays acyclic after reduction: apply the short exact multiplication-by- p sequence. Since the subcategory is closed under shifts and cones, ordinary reduction on it induces an exact functor on its localized homotopy category. This constructs Li^* .

For $C = i_*F$, the snake sequence for multiplication by p is

$$0 \rightarrow F \rightarrow K_C/p \rightarrow P_C/p \rightarrow F \rightarrow 0$$

as a sequence of complexes over $\mathcal{M}_0$. The cone on the middle map is a representative of Li^*i_*F . It extends the cokernel F by the shifted kernel $F[1]$, giving (8.2). The iterated boundary is natural for chain maps and hence descends naturally to the localization. $\square$

This supplies the first boundary map whose nilpotence will reverse a coefficient change. The other comparison triangle was constructed in Proposition 2.13:

$$U \rightarrow Qr^*U \rightarrow U[-1] \rightarrow U[1], \quad Q = Rr_*. \quad (8.3)$$

After a shift, its last map is a natural transformation $\epsilon_U : U \rightarrow U[2]$. We will use δ_F for reduction and ϵ_U for removal of the center. The next lemma explains why nilpotence of these transformations suffices to recover thick membership from the middle terms.

Lemma 8.3 (Nilpotent comparison obstructions). *Let $\delta : \mathrm{id} \rightarrow [2]$ be a natural transformation on a triangulated category. For an object A , consider the property that there is an integer $N \geq 1$ such that*

$$\delta_{A[j]}^N : A[j] \rightarrow A[j + 2N] \quad \text{is zero for every } j \in \mathbb{Z},$$

where the power denotes the successive natural transformations on the intervening shifts. No compatibility between $\delta_{A[j]}$ and $\delta_A[j]$ is assumed. This property is thick.

Suppose in addition that the cones of the maps $\delta_{A[j]}$ are shifts of $B(A[j])$, where B is an exact functor. If A has the property, then $A \in \mathrm{Thick}(B(A))$.

Proof. Shifts and retracts preserve the stated property. Consider a triangle $A \rightarrow B_1 \rightarrow C$ and exponents a, c which work on every shift of A, C , respectively. Naturality makes the composite of the c -fold power on $B_1[j]$ with the map to $C[j + 2c]$ vanish. Exactness of represented sequences factors that power through $A[j + 2c]$. Another a applications of δ kill the factorization, again by naturality. Thus $a + c$ works uniformly on all shifts of B_1 .

For the second assertion, the cone of each factor in δ_A^N belongs to $\text{Thick}(B(A))$: exactness identifies $B(A[j])$ with a shift of $B(A)$. Repeated octahedra show that the cone of their composite also belongs there. If the composite is zero, that cone is $A[2N] \oplus A[1]$. Taking a retract and then a shift proves the claim. $\square$

For (8.2) the cone of δ_F is $(Li^*i_*F)[1]$. For the shifted boundary ϵ_U of (8.3), it is $(Qr^*U)[2]$. Both middle-object functors are exact, so the lemma applies to each.

8.2 Nilpotence on the open and complete pieces

We next prove nilpotence for the particular objects in the transfer criterion. Fracture reduces this to the t -open and the completion. On an object pushed forward from the t -open, both obstructions vanish. Indeed Lemma 3.2 gives a finite decomposition into pure period-residue summands. For $\mathcal{N}$ these are vector spaces; for $\mathcal{M}_0$ they are smooth representations of the pro-cyclic center. Each pure summand has no self-map to its 2-shift. Naturality against the inclusions of the summands therefore kills the obstruction on their direct sum, and the same reasoning applies to every shift. The vector spaces or representations themselves need not have finite dimension: there are only finitely many period residues.

By Corollary 3.4, the complete generators $\widehat{T}'$, $\widehat{T}_k$, and $\widehat{T}_0$ likewise have no self-maps to their 2-shifts, and the same holds after every shift. Hence the appropriate obstruction is nilpotent, uniformly over all shifts, on each of their thick subcategories. To use this for the reduction comparison we must also control the completed pullback through the center.

Lemma 8.4 (Completed central pullback). *Suppose $U \in \mathbf{D}(\mathcal{N})$ and $\widehat{U} \in \text{Thick}(\widehat{T}')$. Set $F = e_*r^*U \in \mathbf{D}(\mathcal{M}_0)$. Then*

$$\widehat{F} \in \text{Thick}(\widehat{T}_0).$$

The comparison obstructions of (8.3) and (8.2) are uniformly nilpotent over all shifts of U and F , respectively.

Proof. The fiber of $U \rightarrow \widehat{U}$ is t -invertible by Proposition 3.1. Pullback preserves this property, and completion kills such objects. Therefore

$$\widehat{r^*U} \simeq \widehat{r^*(\widehat{U})}.$$

The exact functor obtained by composing r^* with completion reduces the assertion to the generator. Its value on $\widehat{T}'$ is its value on T' , by the same invertible-fiber argument. The finite reductions are $r^*T'_d$, where $T'_d = T'/t^d$.

At each level the full coinduction description of Proposition 2.6 gives an exact sequence

$$0 \longrightarrow r^*T'_d \longrightarrow T_{k,d} \xrightarrow{c-1} T_{k,d} \longrightarrow 0, \quad T_{k,d} = T_k/t^d. \quad (8.4)$$

Here c is a chosen topological generator of the center $Z = 1 + p\mathbb{Z}_p$. We include the surjectivity check, which uses characteristic p . A locally constant function on G_{up}/Λ factors through a finite quotient. Refine this quotient until each central period has increased by a multiple of p . This is

possible because the center embeds in the profinite group and its central p -power images meet the tame Λ -images trivially. On each refined central cycle, the sum of the repeated values is zero in characteristic p . The function can therefore be integrated as a discrete difference. This proves surjectivity of $c - 1$; its kernel is the rigidified coinduction object.

Take homotopy limits of (8.4). The result is a triangle expressing $\widehat{r^*T'}$ using two copies of $\widehat{T}_k$. Thus $\widehat{r^*U} \in \text{Thick}(\widehat{T}_k)$. This argument uses compatible finite reductions, not commutation of r^* with arbitrary homotopy limits. Restriction along the finite field extension commutes with the limits and sends $\widehat{T}_k$ to a finite sum of copies of $\widehat{T}_0$. Hence it gives the claimed membership for $\widehat{F}$.

We have already proved nilpotence on the open pieces and on the thick subcategories of the complete generators. Apply it to the fracture triangles

$$U \longrightarrow \widehat{U} \oplus LU \longrightarrow L\widehat{U}, \quad F \longrightarrow \widehat{F} \oplus LF \longrightarrow L\widehat{F}.$$

The open terms have zero obstruction, and the complete terms have nilpotent obstruction. Lemma 8.3 supplies a uniform exponent over all shifts of U and F . $\square$

The completed central pullback is now controlled by the original completed chart, and both comparison boundaries have uniform nilpotence exponents. We can therefore reverse the coefficient changes at the level of thick membership.

Lemma 8.5 (Excluding the reduced construction). *If U satisfies (8.1), then*

$$Li^*(i_*e_*r^*U) \notin \text{Thick}(T_0).$$

Proof. Write $F = e_*r^*U$. If the displayed membership held, the reduction comparison triangle, Lemma 8.4, and Lemma 8.3 would give $F \in \text{Thick}(T_0)$. Exact scalar extension then gives

$$e^*F \in \text{Thick}(e^*T_0) = \text{Thick}(T_k).$$

The object r^*U is a retract of $e^*e_*r^*U$ by separability of the finite field extension. Hence $r^*U \in \text{Thick}(T_k)$. Apply the exact functor Q and use $QT_k = T'$ from Proposition 2.13. It follows that $Qr^*U \in \text{Thick}(T')$. Nilpotence of the central comparison obstruction and Lemma 8.3 now imply $U \in \text{Thick}(T')$, contrary to (8.1). $\square$

8.3 The connective cover and the finite test

Proof of Proposition 8.1. Choose Z_{sp} as in the statement, using essential surjectivity of Theorem 7.1. By Proposition 7.7, its ordinary homotopy groups identify with sections homotopy of $\widetilde{U}$:

$$\begin{aligned} \pi_i Z_{\text{sp}} &\cong [\Sigma^i L_E S^0, Z_{\text{sp}}]_{h\text{Sp}_E} \\ &\cong [\mathcal{O}[i], i_*e_*r^*U]_{\mathbf{D}(\mathcal{M})} \\ &\cong \pi_i Qr^*U. \end{aligned}$$

These are identifications of abelian groups, since a triangulated functor is additive. The last identification uses the acyclic resolution and closed-base change of Section 2. For example, i_*e_* sends termwise injectives to complexes of sections-acyclic objects, and the finite sections bounds justify the computation on periodic complexes. The comparison triangle (8.3) and the first condition in (8.1) show that these groups are finite p -groups in every degree.

Set $X_{\text{sp}} = \tau_{\geq 0} Z_{\text{sp}}$. It is connective, and its homotopy groups are finite p -groups. Its finite Postnikov stages are p -complete, and p -complete spectra are closed under limits. The Postnikov limit consequently proves that X_{sp} is p -complete.

Next, $H\mathbb{F}_p$ is $E(2)$ -acyclic. One can see this directly in the same module calculation as Proposition 7.3: connective ring truncation gives its BP-module structure and

$$\mathrm{BP}_* H\mathbb{F}_p = \Gamma_P \otimes_{\eta_R, P} \mathbb{F}_p.$$

The left positive coefficient actions vanish successively, using $\eta_R v_i \equiv v_i \pmod{(p, v_1, \dots, v_{i-1})}$. Landweber base change to the chart with v_2 invertible therefore gives zero. The negative tail $\tau_{<0} Z_{\mathrm{sp}}$ is a filtered colimit of finite Postnikov ranges with finite p -group homotopy. A composition series of each finite p -group expresses its Eilenberg–Mac Lane spectrum as a finite construction from $H\mathbb{F}_p$. Consequently each range is a finite construction from shifts of $H\mathbb{F}_p$ and is $E(2)$ -acyclic. Thus the whole tail is acyclic and

$$L_{E(2)} X_{\mathrm{sp}} \simeq Z_{\mathrm{sp}}.$$

Let V be any finite type-three spectrum. Such spectra exist by the periodicity theorem, taking successive self-map cofibers [30, Theorem 1.5.4 and Corollary 2.3.5]. The chromatic Bousfield-class calculation [30, Theorem 7.3.2(d)]

$$\langle E(2) \rangle = \langle K(0) \vee K(1) \vee K(2) \rangle$$

shows that both V and its Spanier–Whitehead dual are $E(2)$ -acyclic. Indeed finite duality over each Morava K -theory field preserves the vanishing that defines type three. If DV is the Spanier–Whitehead dual and W is any spectrum, finite duality gives

$$[W, V \wedge Z_{\mathrm{sp}}] \cong [DV \wedge W, Z_{\mathrm{sp}}] = 0,$$

because $DV \wedge W$ is $E(2)$ -acyclic and Z_{sp} is local. Hence $V \wedge Z_{\mathrm{sp}} = 0$. Consequently

$$V \wedge X_{\mathrm{sp}} \simeq \Sigma^{-1}(V \wedge \tau_{<0} Z_{\mathrm{sp}}).$$

The left side is bounded below and has finite homotopy in each degree, since X_{sp} is connective with finite homotopy and V has finitely many cells. The right side is bounded above, since its tail is bounded above and V has finitely many cells. It follows that $\pi_*(V \wedge X_{\mathrm{sp}})$ is finite in total. Since X_{sp} is connective and p -complete, Mahowald–Rezk’s criterion [22, Proposition 3.2] now proves fp-ness as well as fp-type at most 2; fp-ness is not an additional assumption in that criterion. This connective-cover step is also related to their general formulation in Proposition 3.5 and the chromatic construction in Section 7.2(2).

Finally suppose that X_{sp} belonged to $\mathrm{Thick}_{\mathrm{Sp}_p^\wedge}(b_p^\wedge)$. The full category of p -complete spectra is stable, so its finite constructions remain finite constructions of spectra. Exact localization gives

$$Z_{\mathrm{sp}} \in \mathrm{Thick}(L_E b_p^\wedge).$$

Apply the triangulated equivalence $\mathcal{F}$. It preserves shifts, distinguished triangles, and retracts, while Proposition 7.7 identifies the generator with $T_{\mathbb{Z}_p}$. Hence

$$\tilde{U} \in \mathrm{Thick}(T_{\mathbb{Z}_p}).$$

The pure sheaf $T_{\mathbb{Z}_p}$ is p -monic and has reduction T_0 by Proposition 7.3. Exactness of Li^* therefore gives $Li^* \tilde{U} \in \mathrm{Thick}(T_0)$. This contradicts Lemma 8.5 and completes the proof. $\square$

For the glued object, Proposition 5.4 supplies the first two conditions in (8.1), and Theorem 6.10 supplies the third. The transfer criterion therefore produces the spectrum used in Theorem 1.1. We next establish its exact fp-type and its localization properties.

9 The counterexample and its localizations

The algebraic construction and obstruction now give all three inputs to Proposition 8.1. We first determine the exact fp-type of its connective spectrum and prove both localization comparisons for that same spectrum. These conclusions finish the main theorem. We then explain why its positive local classification is compatible with the failure of global thick generation, and isolate the separate obstruction for the connective cover of Morava E -theory.

9.1 Exact type and telescope comparisons

Fix a prime satisfying the operator bounds in (4.8). Let U be the object of Proposition 5.4, and write

$$Z_{\text{sp}} \in \text{Sp}_{E(2)}, \quad X_{\text{sp}} = \tau_{\geq 0} Z_{\text{sp}}$$

for the spectra obtained from U by Proposition 8.1. That proposition gives finite p -groups $\pi_i Z_{\text{sp}}$ in every degree, identifies $L_{E(2)} X_{\text{sp}}$ with Z_{sp} , and shows that X_{sp} is p -complete of fp-type at most two with

$$X_{\text{sp}} \notin \text{Thick}(b_p^\wedge).$$

Here b is the regular BP-module quotient representing $\text{BP}\langle 2 \rangle$ in Proposition 7.7. We first strengthen the upper bound on fp-type to an equality.

Proposition 9.1. *The spectrum X_{sp} has fp-type exactly two.*

Proof. Suppose that X_{sp} had fp-type at most one. The height-one theorem of Lee–Pstrągowski [21, Theorems 1.3 and 5.4] would give

$$X_{\text{sp}} \in \text{Thick}(\text{BP}\langle 1 \rangle_p^\wedge).$$

We check that the generator on the right is already in $\text{Thick}(b_p^\wedge)$.

In the same monoidal category of BP-modules used to construct b , form the further scalar cofiber

$$b_1 = \text{cofib}(\Sigma^{2(p^2-1)} b \xrightarrow{v_2} b).$$

Multiplication by v_2 is regular on $\pi_* b = \mathbb{Z}_{(p)}[v_1, v_2]$, so $\pi_* b_1 = \mathbb{Z}_{(p)}[v_1]$. The mod- p comparison in Proposition 7.4, with the cutoff changed from two to one, identifies

$$H^*(b_1; \mathbb{F}_p) \cong \mathcal{A}/E(Q_0, Q_1).$$

Indeed the same Koszul calculation has exterior generators ϵ_i for $i > 1$; its comparison with the calculation for $H\mathbb{F}_p$ is injective, so the spectral sequence collapses and the Milnor-basis dimension comparison identifies the displayed Steenrod module. Here $E(Q_0, Q_1)$ is the exterior subalgebra generated by the first two Milnor primitives. The quotient is connective and of finite type over $\mathbb{Z}_{(p)}$. Lee’s odd-prime uniqueness theorem therefore identifies $(b_1)_p^\wedge$ with $\text{BP}\langle 1 \rangle_p^\wedge$ as a spectrum [20, Theorem 1.1]; see also [21, Note 2].

Since p -completion is exact, completing the defining cofiber sequence of b_1 gives

$$\text{BP}\langle 1 \rangle_p^\wedge \in \text{Thick}(b_p^\wedge).$$

The assumed height-one membership would consequently put X_{sp} in $\text{Thick}(b_p^\wedge)$, contradicting the transfer criterion. Its fp-type is at most two and not at most one, as required. $\square$

Write $L_2 = L_{E(2)}$, and let L_2^f denote finite chromatic localization away from finite spectra of type greater than two. For $j > 0$, let $T(j)$ be the telescope of a positive-degree v_j -self-map on a finite type- j spectrum, and put $T(0) = H\mathbb{Q}$. These localizations occur in the restricted telescope conjecture of Mahowald–Rezk [22, Conjecture 7.3], and in the proposed reconstruction route of [21, Conjecture 1.2 and Remark 5.1]. The following comparisons are proved directly for the spectrum constructed here.

Proposition 9.2. *There are equivalences*

$$L_2^f X_{\text{sp}} \simeq L_2 X_{\text{sp}} \simeq Z_{\text{sp}}, \quad L_{T(2)} X_{\text{sp}} \simeq L_{K(2)} X_{\text{sp}}.$$

Proof. Put $N = \tau_{<0} Z_{\text{sp}}$, so that

$$X_{\text{sp}} \longrightarrow Z_{\text{sp}} \longrightarrow N$$

is a cofiber sequence. The spectrum N is the filtered colimit of its finite Postnikov ranges $\tau_{\geq -m} N$. Every homotopy group of each such range is a finite p -group. Filtering those groups by their composition series expresses each range as a finite extension of shifts of $H\mathbb{F}_p$.

For $j > 0$, the ordinary mod- p homology of the finite spectrum on which a v_j -self-map is chosen is bounded. Iterating a positive-degree self-map therefore gives the zero telescope on that homology, and hence $H\mathbb{F}_p \wedge T(j) = 0$. The same conclusion for $j = 0$ is rational acyclicity. It follows that N is $T(j)$ -acyclic for $0 \leq j \leq 2$. Finite localization is smashing and its acyclics are generated by a finite type-three spectrum [31, author-preprint Corollary 1.19(i) and Theorem 2.7(ii)]. Together with the Boolean algebra theorem [30, Theorem 7.2.9], this identifies $\langle L_2^f S^0 \rangle$ with $\langle T(0) \vee T(1) \vee T(2) \rangle$. Thus N is L_2^f -acyclic.

Every finite spectrum of type greater than two is $E(2)$ -acyclic. Thus an $E(2)$ -local spectrum is also L_2^f -local: it is right orthogonal to the localizing subcategory generated by those finite spectra. In particular Z_{sp} is L_2^f -local, and the preceding cofiber sequence proves $L_2^f X_{\text{sp}} \simeq Z_{\text{sp}}$. The equality with $L_2 X_{\text{sp}}$ was established in the transfer criterion. This argument is consistent with the finite-localization property of the connective-cover examples in [22, Section 7.2(2)].

For the second comparison we may work with Z_{sp} , since N is $T(2)$ -acyclic and hence also $K(2)$ -acyclic. The latter implication follows from the field property: $K(2)_* T(2)$ is a nonzero graded vector space over $K(2)_*$, and the Künneth isomorphism shows that a $T(2)$ -acyclic spectrum has zero $K(2)$ -homology. The same implication shows that every $K(2)$ -local spectrum is $T(2)$ -local.

Consider the fiber

$$G = \text{fib}(Z_{\text{sp}} \longrightarrow L_{K(2)} Z_{\text{sp}}).$$

It is $K(2)$ -acyclic and $E(2)$ -local. Here the target is $E(2)$ -local because every $E(2)$ -acyclic spectrum is $K(2)$ -acyclic. The Hopkins–Ravenel smashing theorem implies that $T(2) \wedge G$ remains $E(2)$ -local [30, Theorem 7.5.6]. Its $K(0)$ - and $K(1)$ -homology vanish because they vanish on $T(2)$, while its $K(2)$ -homology vanishes because it vanishes on G . The Bousfield-class identity

$$\langle E(2) \rangle = \langle K(0) \vee K(1) \vee K(2) \rangle$$

therefore makes $T(2) \wedge G$ also $E(2)$ -acyclic [30, Theorem 7.3.2(d)]. Being both local and acyclic, it is zero. Thus the map from Z_{sp} to its $K(2)$ -localization is a $T(2)$ -equivalence with $T(2)$ -local target. This proves the claimed equivalence of the two localizations, and hence the assertion for X_{sp} . $\square$

Proof of Theorem 1.1. The parameter choices for Proposition 4.6 are available at every sufficiently large prime. For any such prime, Proposition 5.4 provides an object U with finite $\pi_i U$ in every degree

and $\widehat{U} \in \text{Thick}(\widehat{T}')$. Theorem 6.10 gives $U \notin \text{Thick}(T')$. Applying Proposition 8.1 produces the connective, p -complete spectrum X_{sp} of fp-type at most two outside $\text{Thick}(\text{BP}\langle 2 \rangle_p^\wedge)$. Proposition 9.1 gives exact fp-type two, and Proposition 9.2 gives both localization comparisons. $\square$

This gives the asserted failure at sufficiently large primes, without a claim at the remaining primes. The telescope comparisons are properties of this example; they neither assume nor settle the general restricted telescope conjecture for fp-spectra.

9.2 The local classification and its hypotheses

We now explain the positive local conclusion for this same example. For general height n , write E_n for Morava E -theory; this notation is distinct from the Johnson–Wilson theory $E(n)$. A $K(n)$ -local spectrum Y' is *locally fp* if $\pi_i(F \wedge Y')$ is finite for every integer i and every finite type- n spectrum F . This condition asks for degreewise finiteness. It does not ask that the direct sum over all degrees be finite, as in the finite-spectrum test for fp-spectra.

Corollary 9.3. *The spectrum $Y' = L_{K(2)}X_{\text{sp}}$ is locally fp. In the $K(2)$ -local category it belongs to*

$$\text{Thick}(E_2) = \text{Thick}(L_{K(2)}\text{BP}\langle 2 \rangle).$$

*Its Morava E -cohomology E_2^*Y' is finitely generated over the ring of operations $E_2^*E_2$, and it is reflexive in the $K(2)$ -local category.*

Proof. The exact fp-type from Proposition 9.1 and the $T(2)$ -to- $K(2)$ comparison from Proposition 9.2 verify the hypotheses of [21, Lemma 4.1]. That lemma gives local fp-ness. The equivalent characterizations in [21, Theorems 1.7, 1.8, and 4.3] then give thick membership, finite generation over Morava operations, and local reflexivity. $\square$

By Corollary 9.3, the spectrum $L_{K(2)}X_{\text{sp}}$ satisfies the hypotheses of the local reduction and resolution results discussed below. Their operations are Morava operations, and their finite constructions take place in the local category. They do not provide integral resolutions over global $\text{BP}\langle 2 \rangle$ -operations or lifts of maps through the height-open fracture square. We spell out the relevant hypotheses, since they specify exactly what finite constructions the local theorem provides.

Remark 9.4 (Periodic reductions in the local theorem). Theorem 3.9 of Lee–Pstrągowski, stated as Theorem 1.9 in their introduction, chooses a generalized type- n Moore spectrum M and an ideal I generated by suitable powers of the generators of the height ideal, so that

$$E_{n*}M \cong E_{n*}/I.$$

The powers are chosen to permit the required multiplicative structure. They also choose a sufficiently small normal open subgroup U_{gp} of the extended Morava stabilizer group. After extension to the finite $K(n)$ -local Galois extension $E_n^{hU_{\text{gp}}}$, the reduction object

$$V' = E_n^{hU_{\text{gp}}} \wedge M$$

has homotopy

$$\pi_* V' \cong \bigwedge_{E_{n*}/I} (\bar{x}_1, \dots, \bar{x}_{n^2}), \quad |\bar{x}_i| = -1.$$

Their lower p -series and trivial-action hypotheses ensure that the subgroup acts trivially on the coefficient quotient, that its cohomology has the required exterior form, and that the Adams–Novikov spectral sequence collapses with the stated extensions. The assertion concerns these choices of powers and subgroup; arbitrary Moore exponents and arbitrary open subgroups need not have these properties [21, Section 3 and Theorem 3.9].

The object V' is compact in the $K(n)$ -local category and is a retract of the localization of a finite type- n spectrum, as used in [21, Notation 4.6]. Thus, if Y' is locally fp, the groups $\pi_i L_{K(n)}(V' \wedge Y')$ are finite for all i . The finite-smash version of this assertion needs no extra localization: if F is finite and Y' is local, finite duality makes $F \wedge Y'$ local.

In fact $V' \wedge Y'$ itself is $K(n)$ -local here. Both $E_n^{hU_{\text{gp}}}$ and Y' are $E(n)$ -local, so

$$A' = E_n^{hU_{\text{gp}}} \wedge Y'$$

is $E(n)$ -local by smashing localization. For a $K(n)$ -acyclic spectrum W , finite duality gives

$$[W, M \wedge A']_* \cong [M^\vee \wedge W, A']_*.$$

The spectrum on the left of the second mapping group is $E(n)$ -acyclic. Its lower Morava K -theories vanish because $M^\vee$ has type n , and its $K(n)$ -homology vanishes because that of W vanishes. The mapping group is consequently zero, proving the claim. It follows that $\pi_*(V' \wedge Y')$ is degreewise finite. Periodicity and the Artin coefficient quotient then make it a finitely generated module over the displayed graded exterior algebra.

Remark 9.5 (Operations and finite local resolutions). The reductions of Remark 9.4 are used in a universal-coefficient spectral sequence for V' -linear cohomology with target $E_n \wedge M$. This spectral sequence is a module over the universal-coefficient spectral sequence for the V' -linear endomorphisms of that target. At the coefficient level the relevant self-Ext algebra is

$$\text{Ext}_{\pi_* V'}^*(E_{n*}/I, E_{n*}/I),$$

a polynomial algebra on the exterior generators' Ext classes, all even in total degree. The even-degree endomorphism spectral sequence collapses. The accompanying noetherianity and Ext-finiteness arguments give finite-page collapse and strong convergence for the module spectral sequence, together with finite generation over operations at the quotient level. Derived completeness then permits passage back from these reductions. These are the steps in [21, Lemmas 4.7–4.9 and Proposition 4.11] that produce finite generation of $E_n^* Y'$ over $E_n^* E_n$.

For the converse direction, the twisted group algebra description gives noetherian operations and global dimension $n^2 + n$ for suitable uniform open subgroups [21, Proposition 2.12 and Corollary 2.13]. Taking the subgroup to be pro- p and contained in the kernel of the Galois quotient also makes finitely generated projectives free [21, Proposition 2.14 and Corollary 2.15]. The finite resolution and cofiber argument tracks projective dimension over this small-subgroup algebra. After a suitable $E_n^{hU_{\text{gp}}}$ -extension it reduces to a free case and then descends thick membership in the $K(n)$ -local category [21, Lemmas 4.4–4.5 and the proof of Theorem 4.3].

The subgroup filtrations and completed operations used in this paper provide related control of the complete part of a gluing diagram. After the field extension, characteristic reduction, and removal of central inertia, the digit filtration, compact Koszul resolution, and complete endomorphism calculation give the bounds and models proved above. The discrete lattice and the obstruction to global maps impose additional conditions. Theorem 6.10 uses those conditions to exclude a global finite construction even though the corresponding local membership holds.

9.3 Connective covers and tensor-ideal classifications

There is also a direct obstruction to carrying the height-one connective-cover step of [21, Lemma 5.3] over to Morava E -theory at height two.

Proposition 9.6. *For height-two Morava E -theory,*

$$\tau_{\geq 0}E_2 \notin \text{Thick}(b_p^\wedge).$$

Proof. For a spectrum P , write $P/p = \text{cofib}(p : P \rightarrow P)$. The property that $\pi_i(P/p)$ is finite for every i is thick: the multiplication-by- p cofiber functor is exact, and degreewise finiteness is preserved by shifts, finite sums, cofiber sequences, and retracts. The property holds for $b_p^\wedge$, since

$$\pi_*(b_p^\wedge/p) = \mathbb{F}_p[v_1, v_2]$$

has finite-dimensional homogeneous pieces.

On the other hand,

$$\pi_0E_2 = W(\mathbb{F}_{p^2})[[u_1]],$$

where $W(\mathbb{F}_{p^2})$ is the Witt-vector ring. Since $\tau_{\geq 0}E_2$ is connective, its multiplication-by- p cofiber has

$$\pi_0((\tau_{\geq 0}E_2)/p) \cong \mathbb{F}_{p^2}[[u_1]],$$

which is infinite. It cannot lie in the thick subcategory generated by $b_p^\wedge$. $\square$

Consequently, taking connective covers of the pieces in a finite $K(2)$ -local construction and correcting by Postnikov sections does not by itself reproduce the height-one argument. In the construction here, the connective cover is taken only after an $E(2)$ -local object with degreewise finite homotopy has been glued. The transfer criterion proves fp-ness for that connective cover, while the global obstruction determines its failure of thick generation.

Remark 9.7 (Local tensor ideals and comodule realization). Barthel–Heard–Naumann study a different classification problem in *On conjectures of Hovey–Strickland and Chai*. Their Definition 3.2 and Conjecture 3.4 concern thick tensor ideals of $K(n)$ -locally dualizable spectra. For $0 \leq j \leq n$, their subcategory $\mathcal{D}_j$ consists of retracts of $A \otimes F$, where A is locally dualizable, F is finite of type at least j , and the tensor product is taken in the $K(n)$ -local category; they set $\mathcal{D}_{n+1} = (0)$. Theorem 4.9 deduces the proposed classification from the Chai-type condition on invariant radical ideals. Theorem 4.13 gives the converse implication under $2p - 2 > n^2 + n$, and Theorem 4.15 establishes the height-two classification at every prime [5].

The corresponding local Hovey–Strickland classification is established unconditionally for every prime and every height $n \geq 1$ in [24, Theorem 1.2]: the category of dualizable $K(n)$ -local spectra has exactly $n + 2$ thick tensor ideals.

The telescope-versus-Morava vanishing comparison used in Barthel–Heard–Naumann’s discussion of Proposition 3.3 also applies to our $E(2)$ -local calculation, as the proof of Proposition 9.2 shows. The classification itself has different hypotheses and closure operations: its objects are locally dualizable, with the corresponding finiteness of their Morava modules, and its subcategories are closed under tensoring with locally dualizable objects. The ordinary global thick subcategory of fp-spectra considered in Theorem 1.1 is not such a tensor-ideal category.

A further point of contact is large-prime comodule realization. [5, Lemma 3.11] recalls Pstrągowski’s realization functor from $E_{n*}E_n$ -comodules to the E_n -local homotopy category under $2p - 2 > n^2 + n$. Barthel–Heard–Naumann apply it to invariant-ideal reductions and reassemble

them to prove their Theorem 3.7, realizing finitely generated Morava modules by local dualizable spectra. Our prime range satisfies this inequality at $n = 2$. However, the object constructed here includes a lattice map and is glued in a derived category over heights at most two; thick membership depends on maps between such objects, in addition to the realization of their individual homology modules.

For this reason the transfer criterion uses the homology-compatible triangulated comparison for $E(2)$ in Theorem 7.1, together with the checks on pure representatives in Proposition 7.7; see [27, Corollary 6.9 in the cited preprint version]. The comparison transports the derived object and the finite constructions needed in the proof. The reduced group action and orbit control for its particular height-two charts were established directly in the preceding algebraic sections.

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